

Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences

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Abstract

In the field of artificial intelligence, a combination of scale in data and model capacity enabled by unsupervised learning has led to major advances in representation learning and statistical generation. In the life sciences, the anticipated growth of sequencing promises unprecedented data on natural sequence diversity. Protein language modeling at the scale of evolution is a logical step toward predictive and generative artificial intelligence for biology. To this end we use unsupervised learning to train a deep contextual language model on 86 billion amino acids across 250 million protein sequences spanning evolutionary diversity. The resulting model contains information about biological properties in its representations. The representations are learned from sequence data alone. The learned representation space has a multi-scale organization reflecting structure from the level of biochemical properties of amino acids to remote homology of proteins. Information about secondary and tertiary structure is encoded in the representations and can be identified by linear projections. Representation learning produces features that generalize across a range of applications, enabling state-of-the-art supervised prediction of mutational effect and secondary structure, and improving state-of-the-art features for long-range contact prediction.

1. Introduction

Growth in the number of protein sequences in public databases has followed an exponential trend over decades,

creating a deep view into the breadth and diversity of protein sequences across life. This data is a promising ground for studying predictive and generative models for biology using artificial intelligence. Our focus here will be to fit a single model to many diverse sequences from across evolution. Accordingly we study high-capacity neural networks, investigating what can be learned about the biology of proteins from modeling evolutionary data at scale.

The idea that biological function and structure are recorded in the statistics of protein sequences selected through evolution has a long history (Yanofsky et al., 1964; Altschuh et al., 1987; 1988). Out of the possible random perturbations to a sequence, evolution is biased toward selecting those that are consistent with fitness (Göbel et al., 1994). The unobserved variables that determine a protein’s fitness, such as structure, function, and stability, leave a record in the distribution of observed natural sequences (Göbel et al., 1994).

Unlocking the information encoded in protein sequence variation is a longstanding problem in biology. An analogous problem in the field of artificial intelligence is natural language understanding, where the distributional hypothesis posits that a word’s semantics can be derived from the contexts in which it appears (Harris, 1954).

Recently, techniques based on self-supervision, a form of unsupervised learning in which context within the text is used to predict missing words, have been shown to materialize representations of word meaning that can generalize across natural language tasks (Collobert & Weston, 2008; Dai & Le, 2015; Peters et al., 2018; Devlin et al., 2018). The ability to learn such representations improves significantly with larger training datasets (Baevski et al., 2019; Radford et al., 2019).

Protein sequences result from a process greatly dissimilar to natural language. It is uncertain whether the models and objective functions effective for natural language transfer across differences between the domains. We explore this question by training high-capacity Transformer language models on evolutionary data. We investigate the resulting unsupervised representations for the presence of biological organizing principles, and information about intrinsic

生物结构和功能源于将无监督学习扩展到2.5亿个蛋白质序列

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摘要

在人工智能领域, 数据规模与模型能力的结合, 加上无监督学习的支持, 推动了表示学习和统计生成的重大进展。在生命科学中, 预计测序的增长将为自然序列的多样性提供前所未有的数据。以进化为规模的蛋白质语言建模是朝着生物学领域的预测性和生成性人工智能迈出的合理一步。为此, 我们使用无监督学习来训练一个深度上下文语言模型, 该模型在超过860亿个氨基酸中, 跨越250万个蛋白质序列, 涵盖了进化多样性。所产生的模型在其表示中包含有关生物特性的信息。表示是仅通过序列数据学习而来。学习到的表示空间具有多尺度的组织, 反映了从氨基酸的生化特性到蛋白质的远缘同源性结构的信息。有关二级和三级结构的信息被编码在表示中, 并可以通过线性投影进行识别。表示学习产生可在各种应用中泛化的特征, 使得对突变效应和二级结构的最先进监督预测成为可能, 并改善了长程接触预测的最先进特征。

为研究生命中蛋白质序列的广度和多样性提供了深刻的视角。这些数据为使用人工智能研究生物学的预测和生成模型提供了良好的基础。我们这里的重点是将单个模型适配于来自进化过程中许多不同的序列。因此, 我们研究高容量的神经网络, 探讨通过大规模建模进化数据可以学习到的关于蛋白质生物学的知识。

生物功能和结构在通过进化选择的蛋白质序列的统计中记录的这一想法历史悠久 (Yanofsky等, 1964; Altschuh等, 1987; 1988)。在对序列可能的随机扰动中, 进化倾向于选择与适应度一致的扰动 (Göbel等, 1994)。决定蛋白质适应度的未观测变量, 如结构、功能和稳定性, 在观察到的自然序列分布中留下了记录 (Göbel等, 1994)。

解锁编码在蛋白质序列变异中的信息是生物学中的一个长期问题。在人工智能领域, 类似的问题是自然语言理解, 其中分布假设认为一个词的语义可以从其出现的上下文中推导出来 (Harris, 1954)。

最近, 基于自我监督的技术, 即一种在文本中使用上下文来预测缺失词的无监督学习形式, 已被证明可以实现跨自然语言任务的词义表示 (Collobert和Weston, 2008; Dai和Le, 2015; Peters等, 2018; Devlin等, 2018)。学习此类表示的能力随着更大训练数据集的使用显著提高 (Baevski等, 2019; Radford等, 2019)。

蛋白质序列源于一个与自然语言大相径庭的过程。目前尚不确定对自然语言有效的模型和目标函数是否可以跨越这些领域之间的差异。我们通过在进化数据上训练高容量的变换器语言模型来探讨这个问题。我们调查了所产生的无监督表示中生物组织原则的存在, 以及关于内在的

1. 引言

公共数据库中蛋白质序列的数量增长在数十年中遵循指数趋势,

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biological properties. We find metric structure in the representation space that accords with organizing principles at scales from physicochemical to remote homology. We also find that secondary and tertiary protein structure can be identified in representations. The structural properties captured by the representations generalize across folds. We apply the representations to a range of prediction tasks and find that they improve state-of-art features across the applications.

2. Background

Sequence alignment and search is a longstanding basis for comparative and statistical analysis of biological sequence data. (Altschul et al., 1990; Altschul & Koonin, 1998; Eddy, 1998; Remmert et al., 2011). Search across large databases containing evolutionary diversity assembles related sequences into a multiple sequence alignment (MSA). Within sequence families, mutational patterns convey information about functional sites, stability, tertiary contacts, binding, and other properties (Altschuh et al., 1987; 1988; Göbel et al., 1994). Conserved sites correlate with functional and structural importance (Altschuh et al., 1987). Local biochemical and structural contexts are reflected in preferences for distinct classes of amino acids (Levitt, 1978). Covarying mutations have been associated with function, tertiary contacts, and binding (Göbel et al., 1994).

The prospect of inferring biological structure and function from evolutionary statistics has motivated development of machine learning on individual sequence families. Direct coupling analysis (Lapedes et al., 1999; Thomas et al., 2008; Weigt et al., 2009) infers constraints on the structure of a protein by fitting a generative model in the form of a Markov Random Field (MRF) to the sequences in the protein’s MSA. Various methods have been developed to fit the MRF (Morcos et al., 2011; Jones et al., 2011; Balakrishnan et al., 2011; Ekeberg et al., 2013b). The approach can also be used to infer functional constraints (Figliuzzi et al., 2016; Hopf et al., 2017), and the generative picture can be extended to include latent variables (Riesselman et al., 2018).

Recently, self-supervision has emerged as a core direction in artificial intelligence research. Unlike supervised learning which requires manual annotation of each datapoint, self-supervised methods use unlabeled datasets and thus can exploit far larger amounts of data. Self-supervised learning uses proxy tasks for training, such as predicting the next word in a sentence given all previous words (Bengio et al., 2003; Dai & Le, 2015; Peters et al., 2018; Radford et al., 2018; 2019) or predicting words that have been masked from their context (Devlin et al., 2018; Mikolov et al., 2013).

Increasing the dataset size and the model capacity has shown improvements in the learned representations. In recent work, self-supervision methods used in conjunction with large data

and high-capacity models produced new state-of-the-art results approaching human performance on various question answering and semantic reasoning benchmarks (Devlin et al., 2018), and coherent natural text generation (Radford et al., 2019).

This paper explores self-supervised language modeling approaches that have demonstrated state-of-the-art performance on a range of natural language processing tasks, applying them to protein data in the form of unlabeled amino acid sequences. Since protein sequences use a small vocabulary of twenty canonical elements, the modeling problem is more similar to character-level language models (Mikolov et al., 2012; Kim et al., 2016) than word-level models. Like natural language, protein sequences also contain long-range dependencies, motivating use of architectures that detect and model distant context (Vaswani et al., 2017).

3. Scaling language models to 250 million diverse protein sequences

Large protein sequence databases contain diverse sequences sampled across life. In our experiments we explore datasets with up to 250 million sequences of the Uniparc database (The UniProt Consortium, 2007) which has 86 billion amino acids. This data is comparable in size to large text datasets that are being used to train high-capacity neural network architectures on natural language (Devlin et al., 2018; Radford et al., 2019). To model the data of evolution with fidelity, neural network architectures must have capacity and inductive biases to represent its breadth and diversity.

We investigate the Transformer (Vaswani et al., 2017), which has emerged as a powerful general-purpose model architecture for representation learning and generative modeling, outperforming recurrent and convolutional architectures in natural language settings. We use a deep Transformer (Devlin et al., 2018), taking as input amino acid character sequences.

The Transformer processes inputs through a series of blocks that alternate self-attention with feed-forward connections. Self-attention allows the network to build up complex representations that incorporate context from across the sequence. Since self-attention explicitly constructs pairwise interactions between all positions in the sequence, the Transformer architecture directly represents residue-residue interactions.

We train models using the masked language modeling objective (Devlin et al., 2018). Each input sequence is corrupted by replacing a fraction of the amino acids with a special mask token. The network is trained to predict the missing tokens from the corrupted sequence:

生物特性。我们在表示空间中发现度量结构，这与从物理化学到远程同源的组织原则相符合。我们还发现，二级和三级蛋白质结构可以在表示中被识别。表示所捕捉的结构特性在折叠之间具有普遍性。我们将这些表示应用于一系列预测任务，发现它们在各个应用中提高了最先进的特征。

2. 背景 序列比对和搜索是比较和统计分析生物序列数据的长期基础。(阿尔茨舒尔等, 1990年; 阿尔茨舒尔与库宁, 1998年; Eddy, 1998年; Remmert等, 2011年)。在包含进化多样性的巨大数据库中进行搜索，将相关序列组装成多个序列比对 (MSA)。在序列家族中，突变模式传达有关功能位点、稳定性、三级接触、结合和其他特性的信息 (阿尔茨休等, 1987年; 1988年; Göbel等, 1994年)。保守位点与功能和结构重要性相关 (阿尔茨休等, 1987年)。局部生化和结构背景反映在对不同类别氨基酸的偏好中 (Levitt, 1978)。协变突变与功能、三级接触和结合有关 (Göbel等, 1994年)。从进化统计推断生物结构和功能的前景，促使对单个序列家族的机器学习的发展。直接耦合分析 (Lapedes等, 1999年; 托马斯等, 2008年; Weigt等, 2009 年) 通过将生成功能形式为马尔可夫随机场 (MRF) 的模型拟合到蛋白质的MSA中的序列，从而推断出蛋白质结构的约束。已经开发了多种方法来拟合 MRF (Morcos等, 2011年; 琼斯等, 2011年; 巴拉克里希南等, 2011年; 埃克伯格等, 2013年b)。该方法还可用于推断功能约束 (菲利乌齐等, 2016年; 霍普等, 2017年)，并可以扩展生成图，包括潜在变量 (里瑟尔曼等, 2018年)。最近，自监督已成为人工智能研究的核心方向。与需要手动标注每个数据点的监督学习不同，自监督方法使用未标记的数据集，因此可以利用更大数量的数据。自监督学习使用代理任务进行训练，例如在给定所有先前单词的情况下，预测句子中的下一个单词 (本吉奥等, 2003年; Dai和Le, 2015 年; Peters等, 2018年; Radford等, 2018年; 2019年) 或预测已从上下文中屏蔽的单词 (Devlin等, 2018年; 米科洛夫等, 2013年)。增加数据集大小和模型容量已显示出在学习表示方面的改进。在最近的工作中，自监督方法与大数据结合使用。

并且高容量模型在各种问题回答和语义推理基准测试上产生了接近人类表现的新最先进结果 (Devlin等人, 2018)，以及连贯的自然文本生成 (Radford等人, 2019)。

本文探讨了自监督语言建模方法，这些方法在一系列自然语言处理任务中展示了最先进的性能，将其应用于以无标签氨基酸序列形式的蛋白质数据。由于蛋白质序列使用了二十个标准元素的小词汇，因此建模问题与字符级语言模型 (米科洛夫等, 2012; 金等, 2016) 更为相似。与自然语言一样，蛋白质序列也包含长距离依赖，这促使我们使用能够检测和建模远程上下文的架构 (Vaswani等, 2017)。

3. 将语言模型扩展到2.5亿个多样化的蛋白质序列 大型蛋白质序列数据库包含跨生活采样的多样序列。在我们的实验中，我们探索了最多包含2.5亿个 UniParc数据库序列的datasets (UniProt联盟, 2007)，其中包含860亿个氨基酸。这些数据在规模上可与用于训练高容量神经网络架构的的大型文本数据集相媲美 (Devlin等人, 2018; Radford等人, 2019)。为了忠实地建模进化的数据，神经网络架构必须具备代表其广度和多样性的容量和归纳偏差。我们研究了变换器 (Vaswani等人, 2017)，它作为一种强大的通用模型架构，在表示学习和生成建模中脱颖而出，在自然语言设置中优于递归和卷积架构。我们使用深层变换器 (Devlin等人, 2018)，以氨基酸字符序列作为输入。变换器通过一系列交替使用自注意力和前馈连接的块处理输入。自注意力使网络能够构建复杂的表示，结合序列中的上下文。由于自注意力显式构造了序列中所有位置之间的成对交互，因此变换器架构直接表示残基-残基交互。我们使用掩蔽语言建模目标 (Devlin等人, 2018)。每个输入序列通过用特殊掩码标记替换部分氨基酸而受到损坏。网络被训练以从损坏的序列中预测缺失的标记：

	Model	Params	Training	ECE
(a)	Oracle			1
	Uniform Random			25
(b)	n-gram	4-gram	UR50/S	17.18
(c)	LSTM	Small	UR50/S	14.42
	LSTM	Large	UR50/S	13.54
(d)	Transformer	6-layer	UR50/S	11.79
	Transformer	12-layer	UR50/S	10.45
(e)	Transformer	34-layer	UR100	10.32
	Transformer	34-layer	UR50/S	8.54
	Transformer	34-layer	UR50/D	8.46
(f)	Transformer	10% data	UR50/S	10.99
	Transformer	1% data	UR50/S	15.01
	Transformer	0.1% data	UR50/S	17.50

Table 1. Evaluation of language models for generalization to held-out UniRef50 clusters. (a) Exponentiated cross-entropy (ECE) ranges from 25 for a random model to 1 for a perfect model. (b) Best n-gram model across range of context sizes and Laplace-smoothing settings. (c) State-of-the-art LSTM bidirectional language models (Peters et al., 2018). (d) Transformer model baselines with 6 and 12 layers. Small Transformer models have better performance than LSTMs despite having fewer parameters. (e) 34-layer Transformer models trained on datasets of differing sequence diversity. Increasing the diversity of the training set improves generalization. High-capacity Transformer models outperform LSTMs and smaller Transformers. (f) 34-layer Transformer models trained on reduced fractions of data. Increasing training data improves generalization.

$$\mathcal{L}_{\text{MLM}} = \mathbb{E}_{x \sim X} \mathbb{E}_M \sum_{i \in M} -\log p(x_i | x_{/M}) \quad (1)$$

For each sequence x we sample a set of indices M to mask, replacing the true token at each index i with the mask token. For each masked token, we independently minimize the negative log likelihood of the true amino acid x_i given the masked sequence $x_{/M}$ as context. Intuitively, to make a prediction for a masked position, the model must identify dependencies between the masked site and the unmasked parts of the sequence.

Evaluation of language models We begin by training a series of Transformers on all the sequences in UniParc (The UniProt Consortium, 2007), holding out a random sample of 1M sequences for validation. We use these models throughout to investigate properties of the representations and the information learned during pre-training.

To comparatively evaluate generalization performance of different language models we use UniRef50 (Suzek et al., 2015), a clustering of UniParc at 50% sequence identity. For evaluation, a held-out set of 10% of the UniRef50 clusters is randomly sampled. The evaluation dataset consists of the representative sequences of these clusters. All sequences belonging to the held-out clusters are removed from the pre-training datasets.

We explore the effect of the underlying sequence diversity in the pre-training data. Clustering UniParc shows a power-law

distribution of cluster sizes (Suzek et al., 2007), implying the majority of sequences belong to a small fraction of clusters. Training using a clustering of the sequences results in a re-weighting of the masked language modeling loss toward a more diverse set of sequences. We use UniRef (Suzek et al., 2015) to create three pre-training datasets with differing levels of diversity: (i) the low-diversity dataset (UR100) uses the UniRef100 representative sequences; (ii) the high-diversity sparse dataset (UR50/S) uses the UniRef50 representative sequences; (iii) the high-diversity dense dataset (UR50/D) samples the UniRef100 sequences evenly across the UniRef50 clusters.

Table 1 presents modeling performance on the held-out UniRef50 sequences across a series of experiments exploring different model classes, number of parameters, and pre-training datasets. Models are compared using the exponentiated cross entropy (ECE) metric, which is the exponential of the model’s loss averaged per token. In the case of the Transformer this is $2^{\mathcal{L}_{\text{MLM}}}$. ECE describes the mean uncertainty of the model among its set of options for every prediction: ranging from 1 for an ideal model to 25 (the number of unique amino acid tokens in the data) for a completely random prediction. To measure the difficulty of generalization to the evaluation set, we train a series of n-gram models across a range of context lengths and settings of Laplace smoothing on UR50/S. The best n-gram model has an ECE of 17.18 with context size of 4.

As a baseline we train recurrent LSTM bidirectional language models (Peters et al., 2018), which are state-of-the-art

	模型	参数	训练	ECE
(a)	Oracle			1
	均匀随机			25
(b)	n-gram	4-gram	UR50/S	17.18
(c)	长短期记忆网络	小型	UR50/S	14.42
	长短期记忆网络	大型	UR50/S	13.54
(d)	变换器	6层	UR50/S	11.79
	变换器	12层	UR50/S	10.45
(e)	变换器	34层	UR100	10.32
	变换器	34层	UR50/S	8.54
	变换器	34层	UR50/D	8.46
(f)	变换器	10% 数据	UR50/S	10.99
	变换器	1% 数据	UR50/S	15.01
	变换器	0.1% 数据	UR50/S	17.50

表1。语言模型在对未纳入的数据集UniRef50集群的泛化评估。(a) 指数交叉熵 (ECE) 范围从随机模型的25到完美模型的1。(b) 不同上下文大小和拉普拉斯平滑设置下表现最佳的n-gram模型。(c) 目前最先进的LSTM双向语言模型 (Peters et al., 2018)。(d) 拥有6层和12层的Transformer模型基线。小型Transformer模型的表现优于LSTM，尽管参数更少。(e) 对不同序列多样性的数据集进行训练的34层Transformer模型。增加训练集的多样性可以提高泛化能力。高容量的Transformer模型优于LSTM和较小的Transformer。(f) 对减少数据分数进行训练的34层Transformer模型。增加训练数据可改善泛化能力。

$$\mathcal{L}_{\text{MLM}} = \mathbb{E}_{x \sim X} \mathbb{E}_M \sum_{i \in M} -\log p(x_i | x_{/M}) \quad (1)$$

对于每个序列 x ，我们采样一组索引 M 进行掩码，替换每个索引 i 处的真实标记为掩码标记。对于每个掩码标记，我们独立最小化真实氨基酸 x_i 在掩码序列 $x_{/M}$ 作为上下文的负对数似然。直观地说，要对掩码位置进行预测，模型必须识别掩码位置与未掩码位置之间的依赖关系。

序列的部分。**语言模型的评估** 我们开始在所有UniParc中的序列上训练一系列变换器（UniProt联盟，2007年），保留1M序列的随机样本用于验证。我们在整个过程中使用这些模型来研究表示的属性以及在预训练期间学习到的信息。为了比较不同语言模型的泛化性能，我们使用UniRef50（Suzek等，2015年），这是在50%序列同一性下对UniParc的聚类。为了评估，随机抽取了10%的UniRef50聚类作为保留集。评估数据集由这些聚类的代表性序列组成。所有属于保留聚类的序列都被从预训练数据集中移除。我们探讨了预训练数据中基础序列多样性对结果的影响。对UniParc进行聚类显示出幂律

簇大小的分布（Suzek等，2007），这意味着大多数序列属于一小部分簇。使用对序列的聚类进行训练会导致掩码语言建模损失的重新加权，向

一个更具多样性的序列集。我们使用UniRef（Suzek et al., 2015）创建了三个具有不同多样性水平的预训练数据集：(i) 低多样性数据集（UR100）使用UniRef100代表序列；(ii) 高多样性稀疏数据集（UR50/S）使用UniRef50代表序列；(iii) 高多样性密集数据集（UR50/D）则从UniRef50集群中均匀采样UniRef100序列。表1展示了在不同模型类别、参数数量和预训练数据集上的一系列实验中对保留的UniRef50序列的建模性能。使用指数交叉熵（ECE）度量来比较模型，该度量是每个令牌的模型损失的指数平均值。在变换器的情况下，这是 $2^{\mathcal{L}_{\text{MLM}}}$ 。ECE描述了模型在每个预测选项集合中的平均不确定性：从1（理想模型）到25（数据中独特氨基酸令牌的数量）（完全随机预测）。为了衡量对评估集的泛化难度，我们在UR50/S上训练了一系列n-gram模型，范围覆盖不同的上下文长度和拉普拉斯平滑设置。表现最佳的n-gram模型的ECE为17.18，上下文大小为4。作为基线，我们训练递归LSTM双向语言模型（Peters et al., 2018），这是目前最先进的。

for recurrent models in the text domain. Unlike standard left-to-right autoregressive LSTMs, these models use the whole sequence context, making them comparable to the Transformers we study. We evaluate a small model with approximately 25M parameters, and a large model with approximately 113M parameters. Trained on the UR50/S dataset, the small and large LSTM models have an ECE of 14.4 and 13.5 respectively.

We also train two small Transformers, a 12-layer (85.1M parameters) and 6-layer Transformer (42.6M parameters) on the UR50/S dataset. Both Transformer models have better ECE values (10.45, and 11.79 respectively) than the small and large LSTM models, despite the large LSTM having more parameters. These results show the Transformer enables higher fidelity modeling of protein sequences for a comparable number of parameters.

We train high-capacity 34-layer Transformers (approx 670M parameters) across the three datasets of differing diversity. The high-capacity Transformer model trained on the UR50/S dataset outperforms the smaller Transformers indicating an improvement in language modeling with increasing model capacity. Transformers trained on the two high-diversity datasets, UR50/S and UR50/D, improve generalization over the UR100 low-diversity dataset. The best Transformer trained on the most diverse and dense dataset reaches an ECE of 8.46, meaning that intuitively the is model choosing among approximately 8.46 amino acids for each prediction.

We also train a series of 34-layer Transformer models on 0.1%, 1%, and 10% of the UR50/S dataset, seeing the expected relationship between increased data and improved generalization performance. Underfitting is observed even for the largest models trained on 100% of UR50/S suggesting potential for additional improvements with higher capacity models.

ESM-1b Transformer Finally we perform a systematic optimization of model hyperparameters on 100M parameter models to identify a robust set of hyperparameters. The hyperparameter search is described in detail in [Appendix B](#). We scale the hyperparameters identified by this search to train a model with approximately 650M parameters (33 layers) on the UR50/S dataset, resulting in the ESM-1b Transformer.

4. Multi-scale organization in sequence representations

The variation observed in large protein sequence datasets is influenced by processes at many scales, including properties that affect fitness directly, such as activity, stability, structure, binding, and other properties under selection ([Hor-](#)

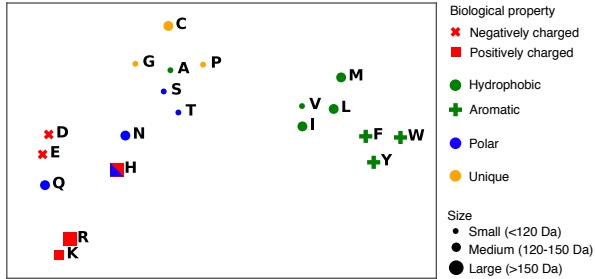


Figure 1. Biochemical properties of amino acids are represented in the Transformer model’s output embeddings, visualized here with t-SNE. Through unsupervised learning, residues are clustered into hydrophobic, polar, and aromatic groups, and reflect overall organization by molecular weight and charge. Visualization of 36-layer Transformer trained on UniParc.

[moz, 2013](#); [Hopf et al., 2017](#)) as well as by contributions from phylogenetic bias ([Gabaldon, 2007](#)), experimental and selection biases ([Wang et al., 2019](#); [Overbaugh & Bangham, 2001](#)), and sources of noise such as random genetic drift ([Kondrashov et al., 2003](#)).

Unsupervised learning may encode underlying factors that, while unobserved, are useful for explaining the variation in sequences seen by the model during pre-training. We investigate the representation space of the network at multiple scales from biochemical to evolutionary homology to look for signatures of biological organization.

Neural networks contain inductive biases that impart structure to representations. Randomly initialized networks can produce features that perform well without any learning ([Jarrett et al., 2009](#)). To understand how the process of learning shapes the representations, it is necessary to compare representations before and after they have been trained. Furthermore, a basic level of intrinsic organization is expected in the sequence data itself as a result of biases in amino acid composition. To disentangle the role of frequency bias in the data we also compare against a baseline that maps each sequence to a vector of normalized amino acid counts.

Learning encodes biochemical properties The Transformer neural network represents the identity of each amino acid in its input and output embeddings. The input embeddings project the input amino acid tokens into the first Transformer block. The output embeddings project the final hidden representations back to logarithmic probabilities. The interchangeability of amino acids within a given structural or functional context in a protein depends on their biochemical properties ([Hormoz, 2013](#)). Self-supervision can be expected to capture these patterns to build a representation space that reflects biochemical knowledge.

在文本领域的递归模型中。与标准的从左到右自回归长短期记忆网络不同, 这些模型使用整个序列上下文, 使它们可以与我们的研究中的变换器相比。我们评估了一个大约有2500万个参数的小型模型, 以及一个大约有1.13亿个参数的大型模型。经过UR50/S数据集训练后, 小型和大型长短期记忆网络模型的ECE分别为14.4和13.5。

我们还在UR50/S数据集上训练了两个小型变换器, 一个是12层(85.1M参数) 和一个6层变换器(42.6M参数)。这两个变换器模型的ECE值(分别为10.45和11.79) 均优于小型和大型长短期记忆网络模型, 尽管大型长短期记忆网络的参数更多。这些结果表明, 变换器能够在相似参数数量下更高保真度地建模蛋白质序列。

我们在三个不同多样性的数据集上训练了高容量的34层变换器(大约670M参数)。在UR50/S数据集上训练的高容量变换器模型的表现优于较小的变换器, 表明随着模型容量的增加, 语言建模得到了改善。在两个高多样性数据集UR50/S和UR50/D上训练的变换器在通用性上优于UR100低多样性数据集。最好的变换器在最具多样性和密集的数据集上达到了8.46的ECE, 这意味着直观上该模型在每次预测时大约选择8.46种氨基酸。

我们还在UR50/S数据集的0.1%、1%和10%上训练了一系列34层的变换器模型, 观察到数据增加与泛化性能提升之间的预期关系。即使对于在100% UR50/S上训练的最大模型, 也观察到了欠拟合, 暗示使用更高容量模型的潜在改进空间。

ESM-1b 变换器最后, 我们对100亿参数模型的超参数进行系统优化, 以识别出一组稳健的超参数。超参数搜索的详细描述见[附录B](#)。我们将此搜索中识别的超参数进行扩展, 用以训练一个约650亿参数(33层)的模型在UR50/S数据集上, 最终得到ESM-1b变换器。

4. 序列表示中的多尺度组织

在大型蛋白质序列数据集中观察到的变异受到多种尺度过程的影响, 包括

影响适应度的属性, 例如活动性、稳定性、结构、结合能力, 以及其他受选择影响的属性([霍尔-](#)

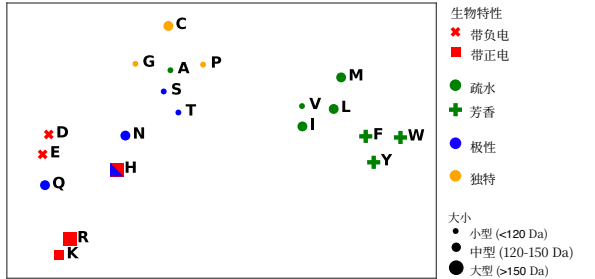


图1. 氨基酸的生化特性在变换器模型的输出嵌入中表示, 使用t-SNE可视化。通过无监督学习, 残基被聚类为疏水、极性和芳香组, 并反映出按分子量和电荷的整体组织。可视化经过UniParc训练的36层变换器。

[莫兹, 2013年](#); [霍普等, 2017年](#)) 以及来自系统发育偏差的贡献([加巴尔登, 2007年](#)), 实验和选择偏差([王等, 2019年](#); [欧弗博与班汉, 2001年](#)), 以及随机基因漂移等噪声源([孔德拉肖夫等, 2003年](#))。

无监督学习可以编码潜在因素, 尽管未被观察到, 但对解释模型在预训练期间看到的序列变异是有用的。我们研究网络在多个尺度上的表示空间, 从生化到进化同源性, 以寻找生物组织的特征。

神经网络包含赋予表示结构的归纳偏差。随机初始化的网络可以在没有任何学习的情况下产生表现良好的特征([Jarrett et al., 2009](#))。为了理解学习过程如何塑造表示, 有必要比较训练前后的表示。此外, 由于氨基酸组成的偏差, 序列数据本身也预计会有基本的内在组织水平。为了理清频率偏差在数据中的作用, 我们还将其与将每个序列映射到规范化氨基酸计数向量的基线进行比较。

学习编码生化属性变换器神经网络在其输入和输出嵌入中表示每个氨基酸的身份。输入嵌入将输入氨基酸标记投影到第一个变换器块中。输出嵌入将最终的隐藏表示投影回对数概率。氨基酸在蛋白质中给定结构或功能上下文中的可互换性依赖于其生化属性([霍尔莫兹, 2013年](#))。可以期望自我监督捕捉这些模式, 以构建反映生化知识的表示空间。

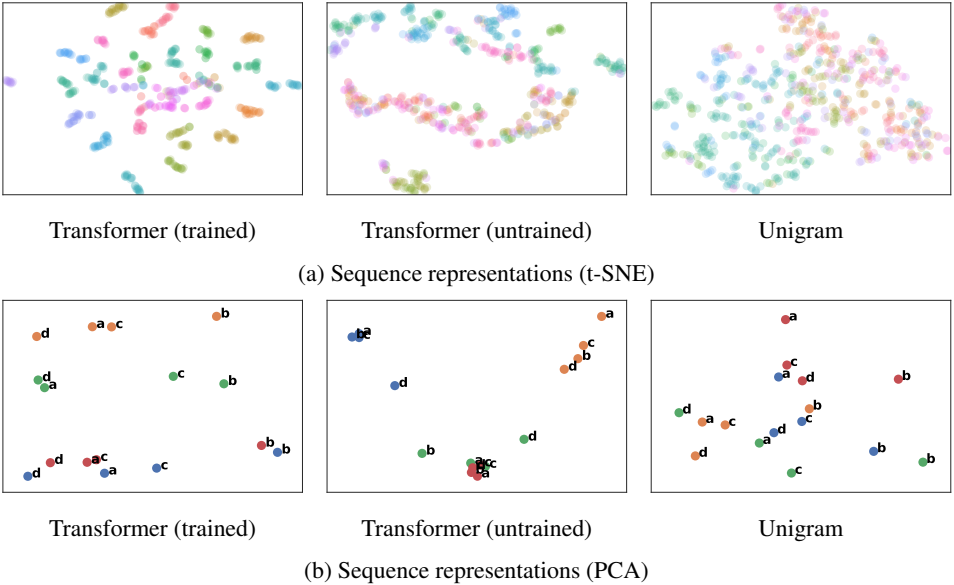


Figure 2. Protein sequence representations encode and organize biological variations. (a) Each point represents a gene, and each gene is colored by the orthologous group it belongs to (dimensionality is reduced by t-SNE). Orthologous groups of genes are densely clustered in the trained representation space. By contrast, the untrained representation space and unigram representations do not reflect strong organization by evolutionary relationships. (b) Genes corresponding to a common biological variation are related linearly in the trained representation space. Genes are colored by their orthologous group, and their species is indicated by a character label. PCA recovers a species axis (horizontal) and orthology axis (vertical) in the trained representation space, but not in the untrained or unigram spaces. Representations are from the 36-layer Transformer model trained on UniParc.

To investigate if the network has learned to encode physico-chemical properties in representations, we project the weight matrix of the final embedding layer of the network into two dimensions with t-SNE (Maaten & Hinton, 2008). In Figure 1 the structure of the embedding space reflects biochemical interchangeability with distinct clustering of hydrophobic and polar residues, aromatic amino acids, and organization by molecular weight and charge.

Biological variations are encoded in representation space Each protein can be represented as a single vector by averaging across the hidden representation at each position in its sequence. Protein embeddings represent sequences as points in a high dimensional space. Each sequence is represented as a single point and sequences assigned to similar representations by the network are mapped to nearby points. We investigate how homologous genes are represented in this space.

The structure and function of orthologous genes is likely to be retained despite divergence of their sequences (Huerta-Cepas et al., 2018). We find in Figure 2a that training shapes the representation space so that orthologous genes are clustered. Figure 2a shows a two-dimensional projection of the model's representation space using t-SNE. Prior to training the organization of orthologous proteins in the model's representation space is diffuse. Orthologous genes are clustered

in the learned representation space.

We examine whether unsupervised learning encodes biological variations into the structure of the representation space. We apply principal component analysis (PCA), to recover principal directions of variation in the representations, selecting 4 orthologous genes across 4 species to look for directions of variation. Figure 2b indicates that linear dimensionality reduction recovers species and orthology as primary axes of variation in the representation space after training. This form of structure is absent from the representations prior to training.

To quantitatively investigate the structure of the representation space, we assess nearest neighbor recovery under vector similarity queries. If biological properties are encoded along independent directions in the representation space, then proteins corresponding with a unique biological variation are related by linear vector arithmetic. In Figure S1 we find that learning improves recovery of target proteins under queries encoded as linear transformations along the species or gene axes.

Learning encodes remote homology Remotely homologous proteins have underlying structural similarity despite divergence of their sequences. If structural homology is encoded in the metric structure of the representation space,

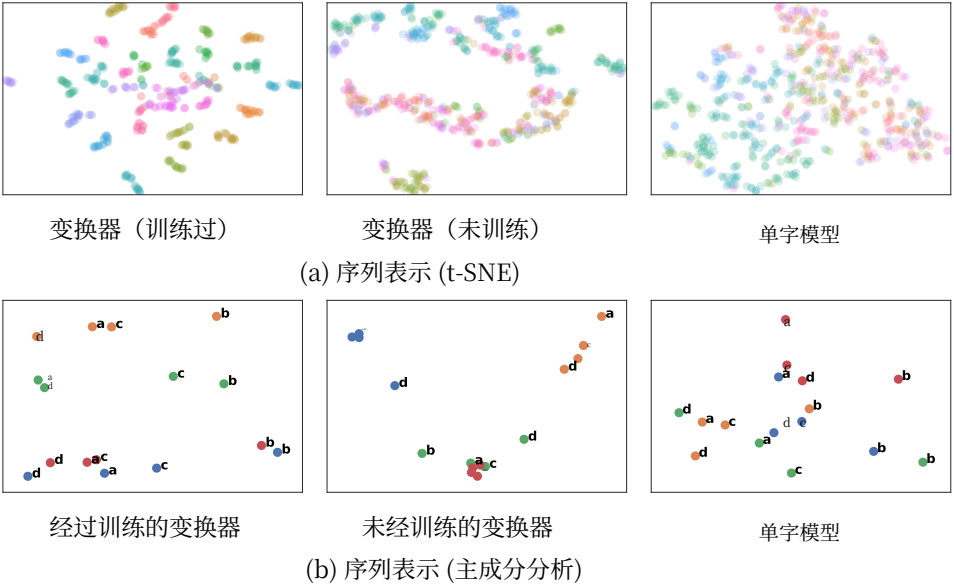


图 2. 蛋白质序列表示编码并组织生物变异。(a) 每个点代表一个基因, 且每个基因的颜色由其所属的同源组决定 (维度通过 t-SNE 降维)。经过训练的表示空间中, 同源基因组聚集得很密集。相反, 未经训练的表示空间和单字模型表示并未显示出由进化关系组成的强组织。(b) 在经过训练的表示空间中, 对应于共同生物变异的基因线性相关。基因的颜色由其同源组决定, 物种由字符标签表示。主成分分析在经过训练的表示空间中恢复了物种轴 (水平) 和同源轴 (垂直), 但在未经训练或单字模型空间中未能恢复。表示来自于在 UniParc 上训练的 36 层 Transformer 模型。

为了调查网络是否学会了在表示中编码物理化学属性, 我们使用 t-SNE 将网络最终嵌入层的权重矩阵投影到二维空间中 (Maaten & Hinton, 2008 年)。在

图 1 嵌入空间的结构反映了生物化学的可互换性, 其中疏水性和极性残基、芳香氨基酸的聚类是明显的, 并按分子量和电荷进行组织。

生物变异以表示空间进行编码 每个蛋白质可以通过对其序列中每个位置的隐藏表示进行平均来表示为一个单一向量。蛋白质嵌入将序列表示为高维空间中的点。每个序列表示为一个单独的点, 网络将赋予类似表示的序列映射到附近的点。我们研究同源基因在这个空间中的表示方式。

同源基因的结构和功能可能在它们的序列发生分歧时仍然被保留 (Huerta-Cepas et al., 2018)。我们在图 2a 中发现, 训练塑造了表示空间, 使得同源基因被聚类。图 2a 展示了模型表示空间的二维投影, 使用了 t-SNE。在训练之前, 模型表示空间中同源蛋白的组织是分散的。同源基因被聚类

在学习到的表示空间中。

我们研究无监督学习是否将生物变异编码到表示空间的结构中。我们应用主成分分析 (PCA), 以恢复表示中的主要变异方向, 选择 4 个同源基因跨 4 个物种寻找变异方向。图 2b 表明线性降维在训练后恢复了物种和同源性作为表示空间中的主要变异轴。这种结构在训练之前的表示中是不存在的。

为了定量研究表示空间的结构, 我们在向量相似性查询下评估最近邻恢复。如果生物属性沿表示空间中的独立方向进行编码, 那么与独特生物变异相关的蛋白质通过线性向量算术相联系。在图 S1 中, 我们发现学习改善了在按物种或基因轴的线性变换编码的查询下目标蛋白质的恢复。

学习编码远程同源性 远程同源蛋白质尽管序列发生了分歧, 但仍具有基本的结构相似性。如果结构同源性在表示空间的度量结构中被编码,

Pre-training		Hit-10		AUC	
		Fold	SF	Fold	SF
HHblits†		.584	.965	.831	.951
LSTM (S)	UR50/S	.558	.760	.801	.863
LSTM (L)	UR50/S	.574	.813	.805	.880
Transf-6	UR50/S	.653	.878	.768	.901
Transf-12	UR50/S	.639	.915	.778	.942
Transf-34	None	.481	.527	.755	.807
Transf-34	UR100	.599	.841	.753	.876
Transf-34	UR50/D	.617	.932	.822	.932
Transf-34	UR50/S	.639	.931	.825	.933
ESM-1b	UR50/S	.532	.913	.770	.880

Table 2. Remote homology detection. Structural homology at the fold and superfamily (SF) level is encoded in the metric structure of the representation space. Results for unsupervised classifier based on distance between vector sequence embeddings. Hit-10 reports the probability that a remote homolog is included in the ten nearest neighbors of the query sequence. Area under the ROC curve (AUC) is reported for classification by distance from the query in representation space. Transformer models have higher performance than LSTMs and similar performance to HMMs at the fold level. Best neural models are indicated in bold. † HH-blits (Remmert et al., 2011), a state-of-the-art HMM-based method for remote homology detection, using 3 iterations of sequence search.

then the distance between proteins reflects their degree of structural relatedness.

We investigate whether the representation space enables detection of remote homology at the superfamily (proteins that belong to different families but are in the same superfamily) and fold (proteins that belong to different superfamilies but have the same fold) level. We construct a dataset to evaluate remote homology detection using SCOPe (Fox et al., 2014). Following standard practices (Söding & Remmert, 2011) we exclude Rossman-like folds (c.2-c.5, c.27 and 28, c.30 and 31) and four- to eight-bladed β -propellers (b.66-b.70).

An unsupervised classifier on distance from the query measures the density of homologous proteins in the neighborhood of a query sequence. For each domain, a vector similarity query is performed against all other domains, ranking them by distance to the query domain. For evaluation at the fold level, any domain with the same fold is a positive; any domain with a different fold is a negative; and domains belonging to the same superfamily are excluded. For evaluation at the superfamily level, any domain with the same superfamily is a positive; any domain with a different superfamily is a negative; and domains belonging to the same family are excluded. We report AUC for the classifier, and Hit-10 (Ma et al., 2014) which gives the probability

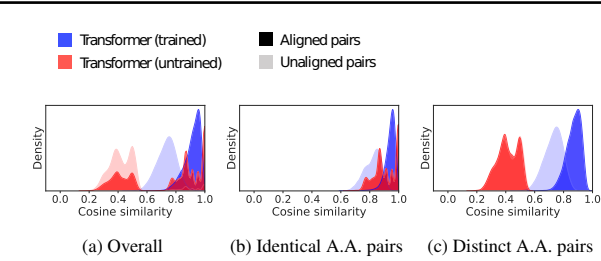


Figure 3. Final representations from trained models implicitly align sequences. Cosine similarity distributions are depicted for the final representations of residues from sequences within PFAM family PF01010. The differences between the aligned (dark blue) and unaligned (light blue) distributions imply that the trained Transformer representations are a powerful discriminator between aligned and unaligned positions in the sequences. In contrast representations prior to training do not separate the aligned (dark red) and unaligned positions (light red). AUCs across 128 PFAM families are reported in Table S1.

of recovering a remote homolog in the ten highest ranked results.

Table 2 indicates that vector nearest neighbor queries using the representations can detect remote homologs that are distant at the fold level with similar performance to HH-blits (Remmert et al., 2011) a state-of-the-art HMM-HMM alignment-based method. At the superfamily level, where sequence similarity is higher, HMM performance is better, but Transformer embeddings are close. Fast vector nearest neighbor finding methods allow billions of sequences to be searched for similarity to a query protein within milliseconds (Johnson et al., 2017).

Learning encodes alignment within a protein family
An MSA identifies corresponding sites across a family of related sequences (Ekeberg et al., 2013a). These correspondences give a picture of evolutionary variation at different sites within the sequence family. The model receives as input individual sequences and is given no access to the family of related sequences except via learning. We investigate whether the final hidden representations of a sequence encode information about the family it belongs to.

Family information could appear in the network through assignment of similar representations to positions in different sequences that are aligned in the family’s MSA. Using the collection of MSAs of structurally related sequences in Pfam (Bateman et al., 2013), we compare the distribution of cosine similarities of representations between pairs of residues that are aligned in the family’s MSA to a background distribution of cosine similarities between unaligned pairs of residues. A large difference between the aligned and unaligned distributions implies that the representations use shared features for related sites within all the sequences of the family.

前-训练		Hit-10		AUC	
		折叠	SF	折叠	SF
HHblits†		.584	.965	.831	.951
LSTM (S)	UR50/S	.558	.760	.801	.863
LSTM (L)	UR50/S	.574	.813	.805	.880
Transf-6	UR50/S	.653	.878	.768	.901
Transf-12	UR50/S	.639	.915	.778	.942
Transf-34	无	.481	.527	.755	.807
Transf-34	UR100	.599	.841	.753	.876
Transf-34	UR50/D	.617	.932	.822	.932
Transf-34	UR50/S	.639	.931	.825	.933
ESM-1b	UR50/S	.532	.913	.770	.880

表 2.远程同源检测。在折叠和超家族 (SF) 级别的结构同源性被编码在表示空间的度量结构中。基于向量序列嵌入之间距离的无监督分类器的结果。Hit-10

报告了查询序列的十个最近邻中包含远程同源物的概率。ROC曲线下的面积
在表示空间中, 曲线 (AUC) 是通过与查询的距离来报告分类结果。相比于长短期记忆网络, Transformer模型的性能更高, 而在折叠级别上与隐马尔可夫模型的表现相似。最佳神经模型以粗体字表示。

† HH-blits(Remmert等, 2011), 一种最先进的基于HMM的方法
用于远程同源检测, 进行3次序列搜索迭代。

那么, 蛋白质之间的距离反映了它们的结构相关性程度。我们研究表示空间是否能够在超家族(属于不同家族但在同一超家族中的蛋白质)和折叠(属于不同超家族但具有相同折叠的蛋白质)级别上检测远程同源。我们构建了一个数据集来评估使用SCOPe进行远程同源检测(Fox等, 2014)。遵循标准做法(Söding & Remmert, 2011)我们排除了Rossman样折叠(c.2-c.5, c.27和28, c.30

和31)以及四到八叶 β 螺旋桨(b.66-b.70)。一个无监督分类器根据查询的距离来测量同源蛋白质在查询序列附近的密度。对于每个结构域, 都要针对其他所有结构域执行一个向量相似性查询, 并按与查询结构域的距离进行排序。为了在折叠水平进行评估, 任何具有相同折叠的结构域都是正例; 任何具有不同折叠的结构域都是负例; 并且

属于相同超家族的结构域被排除。在超家族水平进行评估时, 任何具有相同超家族的结构域都是正例; 任何具有不同超家族的结构域都是负例; 并且属于该

相同家族被排除。我们报告分类器的AUC和Hit-10 (Ma et al., 2014), 这给出了

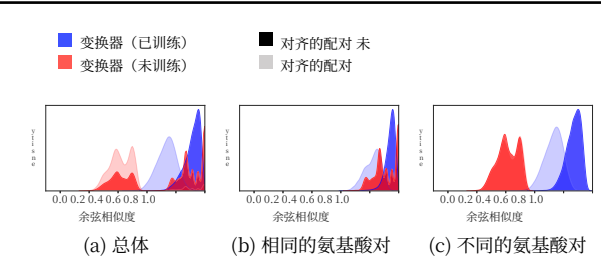


图 3. 经过训练的模型产生的最终表示隐式对序列进行对齐。余弦相似度分布显示了来自PFAM家族PF01010中的序列的残基的最终表示。对齐(深蓝色)和未对齐(浅蓝色)分布之间的差异暗示训练后的

变换器表示在序列中对齐和未对齐位置之间是一种强大的区分器。相比之下, 在训练前的表示并未将对齐(深红色)和未对齐位置(浅红色)区分开来。128个PFAM家族的AUC值见表 S1。

在排名前十的最高得分中恢复远程同源物的概率
结果。表2 表明, 使用表示的向量最近邻查询能够检测到在折叠层级上距离较远的远程同源物, 其性能与HH-blits相似(Remmert et al., 2011)是一种最先进的HMM-HMM

基于对齐的方法。在超家族层级, 序列相似性更高, HMM的表现更好, 但变换器嵌入的表现也相近。快速向量最近邻查找方法能够在毫秒内搜索数十亿个序列与查询蛋白质的相似性(Johnson et al., 2017)。

学习在蛋白质家族中编码对齐
一个MSA识别相关序列家族中的对应位点(Ekeberget al., 2013a)。这些对应关系展示了不同位点在序列家族中的进化变异。模型接收个别序列作为输入, 但除了通过学习之外, 不会接触到相关序列的家族信息。我们研究一个序列的最终隐藏表示是否编码了其所属家族的信息。家族信息可能通过将相似的表示分配给不同序列中在家族的MSA中对齐的位置而出现在网络中。利用PFAM中结构相关序列的MSA集合(Bateman et al., 2013), 我们对在家族的MSA中对齐的残基对之间表示的余弦相似性分布与未对齐的余弦相似性背景分布进行比较。

残基的配对。在对齐的部分之间存在较大差异, 未对齐的分布意味着表示在家族的所有序列中使用相关位点的共享特征。

Model	Pre-training	SSP			Contact		
		Family	Superfamily	Fold	Family	Superfamily	Fold
HMM Profile	-	47.9 ± 1.1	47.8 ± 1.3	48.0 ± 1.6	14.8 ± 0.6	14.5 ± 1.4	14.6 ± 1.6
CCMpred	-	-	-	-	32.7 ± 1.0	32.5 ± 2.3	32.6 ± 0.4
Transformer-6	UR50/S	62.4 ± 0.7	61.8 ± 0.7	61.8 ± 1.0	24.4 ± 2.2	19.4 ± 3.1	18.6 ± 0.5
Transformer-12	UR50/S	65.5 ± 0.7	65.0 ± 1.0	65.2 ± 1.0	32.8 ± 2.9	25.8 ± 3.7	25.2 ± 0.9
Transformer-34	(None)	41.5 ± 0.9	41.3 ± 0.8	41.3 ± 1.3	8.7 ± 0.3	8.4 ± 0.7	8.4 ± 0.3
Transformer-34	UR100	65.4 ± 0.7	64.9 ± 0.9	64.9 ± 0.9	28.3 ± 2.5	22.5 ± 3.3	22.0 ± 0.8
Transformer-34	UR50/S	69.4 ± 0.6	68.8 ± 0.9	68.9 ± 0.9	43.9 ± 2.8	36.4 ± 4.2	35.3 ± 1.7
Transformer-34	UR50/D	69.5 ± 0.6	68.9 ± 0.9	69.0 ± 0.9	43.9 ± 2.8	37.1 ± 4.6	36.3 ± 2.0
ESM-1b	UR50/S	71.1 ± 0.5	70.5 ± 0.9	70.6 ± 0.9	49.2 ± 2.5	43.5 ± 4.8	42.8 ± 2.3

Table 3. Linear projections. Five-fold cross validation experiment for generalization at the family, superfamily, and fold level. 8-class accuracy (secondary structure), Top-L long-range precision (contacts), mean and standard deviation across test sets for the five partitions. Minimal information about structure is present in representations prior to training. Information about secondary and tertiary structure emerges in representations as a result of unsupervised learning on sequences with the language modeling objective. Increasing diversity of sequences improves learning of structure. (Higher diversity UR50 datasets improve over UR100). Learned representations enable linear projections to generalize to held-out folds, outperforming projections of the sequence profile, and contacts identified by the CCMpred (Seemayer et al., 2014) implementation of direct coupling analysis.

Figure 3a depicts the distribution of cosine similarity values between aligned and unaligned positions within a representative family for the trained model and baselines. Unsupervised learning produces a marked shift between the distributions of aligned and unaligned pairs. Figure 3b and Figure 3c indicate that these trends hold under the constraints that the residue pairs (1) share the same amino acid identity or (2) have different amino acid identities.

We estimate differences between the aligned and unaligned distributions across 128 Pfam families using the area under the ROC curve (AUC) as a metric of discriminative power between aligned and unaligned pairs. Table S1 shows a quantitative improvement in average AUC after unsupervised training, supporting the idea that self-supervision encodes information about the MSA of a sequence into its representation of the sequence.

5. Prediction of secondary structure and tertiary contacts

There is reason to believe that unsupervised learning will cause the model’s representations to contain structural information. The underlying structure of a protein is a hidden variable that influences the patterns observed in sequence data. For example local sequence variation depends on secondary structure (Levitt, 1978); and tertiary structure introduces higher order dependencies in the choices of amino acids at different sites within a protein (Marks et al., 2011; Anishchenko et al., 2017). While the model cannot observe protein structure directly, it observes patterns in the sequences of its training data that are determined by struc-

ture. In principle, the network could compress sequence variations by capturing commonality in structural elements across the data, thereby encoding structural information into the representations.

Linear projections We begin by identifying information about protein structure that is linearly encoded within the representations. The use of linear projections ensures that the information originates in the Transformer representations, enabling a direct inspection of the structural content of representations. By comparing representations of the Transformer before and after pre-training, we can identify the information that emerges as a result of the unsupervised learning.

We perform a five-fold cross validation experiment to study generalization of structural information at the family, superfamily, and fold level. For each of the three levels, we construct a dataset of 15,297 protein structures using the SCOPe database. We partition the structures into five parts, splitting by family, superfamily, and fold accordingly. Five-fold cross validation is performed independently for each of the levels of structural hold-out.

To detect information about secondary structure we fit a logistic regression to the hidden representations using the 8-class secondary structure labels. To detect information about tertiary structure, we fit two separate linear projections to the hidden representations of pairs of positions in the sequence, taking their dot product to regress a binary variable indicating whether the positions are in contact in the protein’s 3-dimensional structure. The neural representations are compared to (i) projections of the sequence profile;

模型	前-训练	SSP			联系		
		家族	超家族	折叠	家族	超家族	折叠
HMM配置文件	-	47.9 ± 1.1	47.8 ± 1.3	48.0 ± 1.6	14.8 ± 0.6	14.5 ± 1.4	14.6 ± 1.6
CCMpred	-	-	-	-	32.7 ± 1.0	32.5 ± 2.3	32.6 ± 0.4
变换器-6	UR50/S	62.4 ± 0.7	61.8 ± 0.7	61.8 ± 1.0	24.4 ± 2.2	19.4 ± 3.1	18.6 ± 0.5
变换器-12	UR50/S	65.5 ± 0.7	65.0 ± 1.0	65.2 ± 1.0	32.8 ± 2.9	25.8 ± 3.7	25.2 ± 0.9
变换器-34	(无)	41.5 ± 0.9	41.3 ± 0.8	41.3 ± 1.3	8.7 ± 0.3	8.4 ± 0.7	8.4 ± 0.3
变换器-34	UR100	65.4 ± 0.7	64.9 ± 0.9	64.9 ± 0.9	28.3 ± 2.5	22.5 ± 3.3	22.0 ± 0.8
变换器-34	UR50/S	69.4 ± 0.6	68.8 ± 0.9	68.9 ± 0.9	43.9 ± 2.8	36.4 ± 4.2	35.3 ± 1.7
变换器-34	UR50/D	69.5 ± 0.6	68.9 ± 0.9	69.0 ± 0.9	43.9 ± 2.8	37.1 ± 4.6	36.3 ± 2.0
ESM-1b	UR50/S	71.1 ± 0.5	70.5 ± 0.9	70.6 ± 0.9	49.2 ± 2.5	43.5 ± 4.8	42.8 ± 2.3

表3。线性投影。关于家族、超家族和折叠层级的一般化的五折交叉验证实验。八类准确率（一级结构），Top-L 长距离精度（接触），在五个分区的测试集中均值和标准偏差。训练前，表示中包含的结构信息有限。经过无监督学习后，关于二级和三级结构的信息在表示中出现，其目标为语言建模。序列的多样性增加有助于结构学习。（更多多样性的 UR50 数据集优于 UR100）。学习到的表示使线性投影能够推广到保留的折叠，超越序列轮廓的投影，以及通过 CCMpred 实施的直接耦合分析所识别的接触（Seemayer et al., 2014）。

图 3a 描绘了经过训练的模型和基线中，代表性家族内对齐和未对齐位置之间余弦相似度值的分布。无监督学习在对齐和未对齐对之间的分布中产生了明显的变化。图 3b和

图3c表明这些趋势在以下约束下成立：残基对 (1) 共享相同氨基酸的身份或 (2) 具有不同的氨基酸身份。我们估计对齐和

未对齐之间的差异

使用ROC曲线下面积(AUC)作为对齐和未对齐对之间区分能力的指标，128个Pfam家族的分布。表S1展示了经过无监督训练后平均AUC的定量改善，支持自我监督将序列的MSA信息编码到序列表示中的观点。

5. 二级结构预测和三级接触

有理由相信无监督学习将

导致模型的表示包含结构信息。蛋白质的基本结构是一个隐藏变量，影响序列数据中观察到的模式。例如，局部序列变异依赖于二级结构（Levitt, 1978）；而三级结构引入了不同位点氨基酸选择中的高阶依赖性（Marks et al., 2011; Anishchenko et al., 2017）。虽然模型不能直接观察蛋白质结构，但它观察到训练数据序列中的模式，这些模式是由结构决定的。

ture. 从原则上讲，网络可以通过捕捉数据中结构元素的共性来压缩序列变体，从而将结构信息编码到表示中。**线性投影** 我们首先识别在表示中线性编码的蛋白质结构信息。线性投影的使用确保信息源自变换器表示，使我们能够直接检查表示的结构内容。通过比较变换器在预训练之前和之后的表示，我们可以识别出作为无监督学习结果而出现的信息。

学习。我们进行五折交叉验证实验，以研究在家族、超家族和折叠水平上结构信息的泛化。对于三个层级，我们使用SCOPe数据库构建了15,297个蛋白质结构的数据集。我们将结构划分为五个部分，按家族、超家族和折叠相应地拆分。对每个结构的保留水平独立执行五折交叉验证。为了检测二级结构信息，我们对隐藏表示拟合逻辑回归，使用8类次级结构标签。为了检测三级结构信息，我们对序列中位置对的隐藏表示拟合两个独立的线性投影，通过它们的点积回归一个二元变量，指示位置在蛋白质的三维结构中是否接触。神经表示与(i)序列概况的投影进行比较；

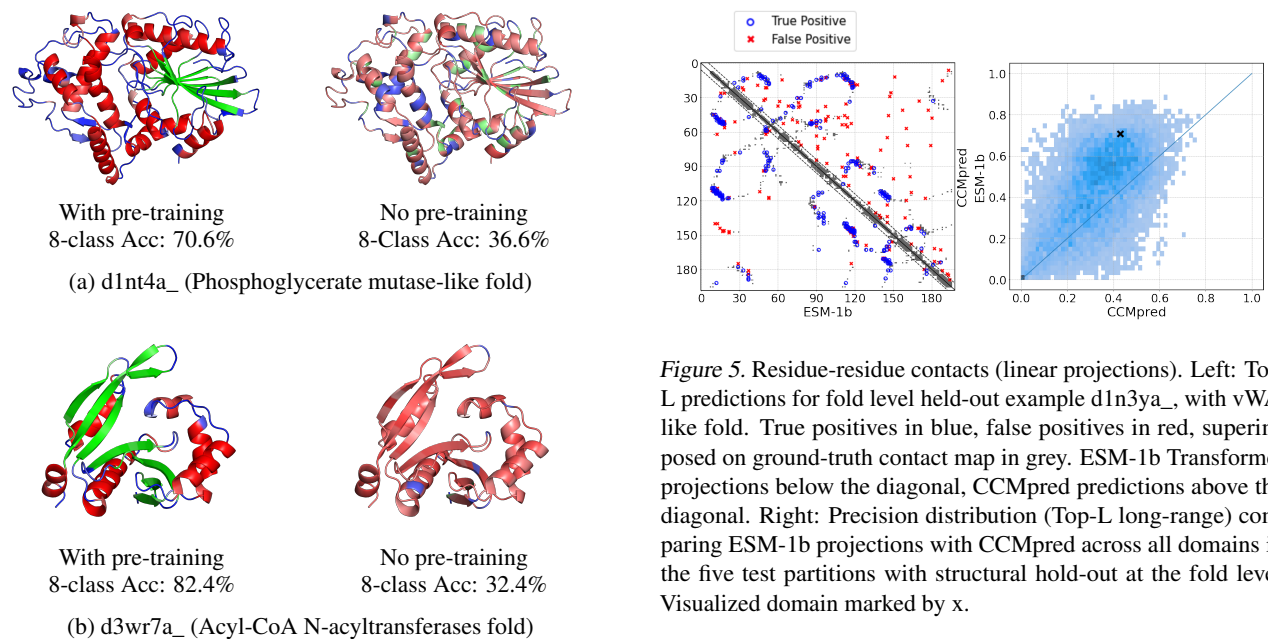


Figure 4. Secondary structure (linear projections). Example predictions for held-out folds. Unsupervised pre-training encodes secondary structure into representations. Following pre-training, linear projections recover secondary structure (left column). Without pre-training little information is recovered (right column). Colors indicate secondary structure class identified by the projection: helix (red), strand (green), and coil (blue). Color intensities indicate confidence. Representations from ESM-1b Transformer are used.

(ii) unsupervised contacts predicted by the CCMpred implementation (Seemayer et al., 2014) of direct coupling analysis. MSAs for the baselines are generated from the UniClust30 (Mirdita et al., 2017) database using 3 iterations of search by Hhblits. For secondary structure, we report 8-class accuracies. For contact precision we report Top-L long-range precision, i.e. the precision of the L (length of the protein) highest ranked predictions for contacts with sequence separation of at least 24 residues.

Table 3 shows results of the cross validation experiments. Prior to pre-training, minimal information about secondary structure and contacts can be detected. After pre-training, projections recover information about secondary structure and long-range contacts that generalizes across families, superfamilies, and folds. Secondary structure prediction 8-class accuracy distributions (Figure S2), and long-range contact prediction Top-L precision distributions (Figure S3) demonstrate that pre-training produces an increase in structural information across the entire distribution of test domains. Table 3 shows that projections of the Transformer representations recover more structure than projections of the sequence profile. For long-range contacts, projections of the best Transformer models have higher precision than con-

Figure 5. Residue-residue contacts (linear projections). Left: Top-L predictions for fold level held-out example d1n3ya_, with vWA-like fold. True positives in blue, false positives in red, superimposed on ground-truth contact map in grey. ESM-1b Transformer projections below the diagonal, CCMpred predictions above the diagonal. Right: Precision distribution (Top-L long-range) comparing ESM-1b projections with CCMpred across all domains in the five test partitions with structural hold-out at the fold level. Visualized domain marked by x.

tacts predicted by CCMpred across all levels of structural generalization. As the level of structural split becomes more remote, there is little degradation for secondary structure, with performance at the family level similar to the fold level. For long-range contacts, while generalization is reduced at the fold level in comparison to the family level, the best models still capture more structure than the unsupervised baseline. Training with higher-diversity sequences (UR50 datasets) improves learning of both secondary structure and long-range contacts, with a more pronounced effect on long-range contacts.

Figure 4 visualizes three-class secondary structure projections for two domains belonging to held-out folds. Prior to pre-training, projections produce an incoherent prediction of secondary structure. After pre-training projections recover a coherent prediction with most errors occurring at the boundaries of secondary structure regions. Figure 5 compares a projected contact map to predictions from CCMpred. Transformer projections recover complex contact patterns, including long-range contacts. Further visualizations of projected contacts for eight randomly selected test proteins are shown in Figure S7.

Deep neural network We train deep neural networks to predict secondary structure and contacts from the representations. We choose state-of-the-art neural architectures for both tasks. These downstream models are trained with a supervised loss to predict either the secondary structure or contact map from the pre-trained representations. The architecture of the downstream model is kept fixed across experiments with different representations and baselines to enable comparison.

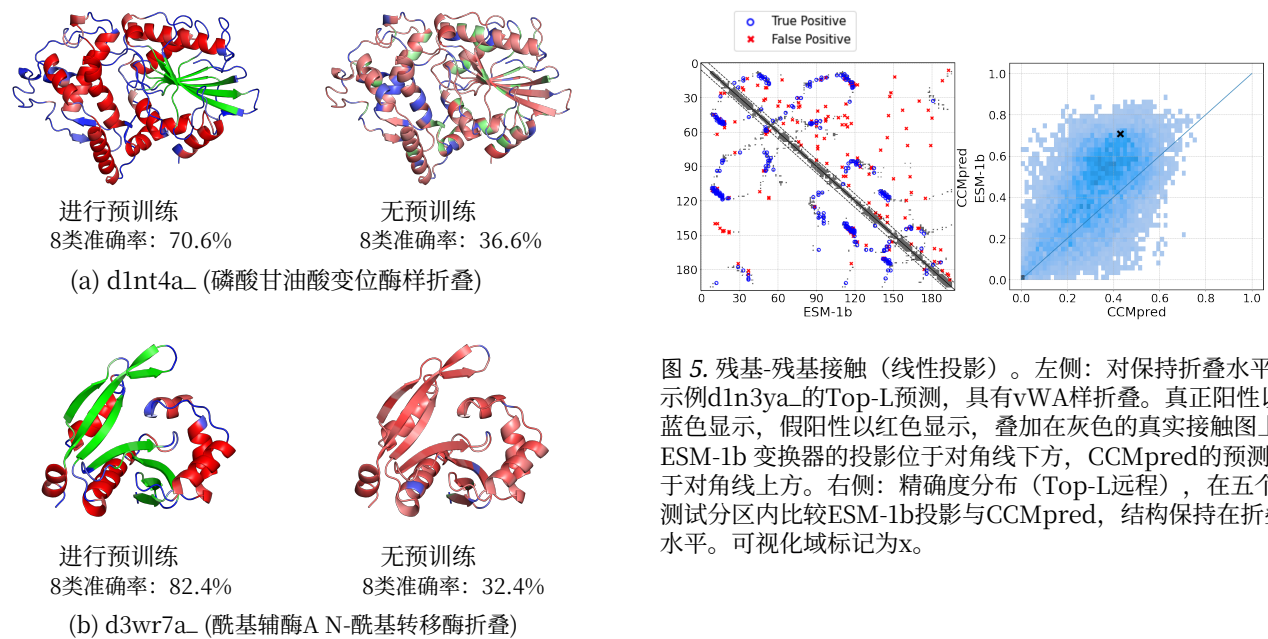


图 4. 二级结构（线性投影）。为保持折叠的预测示例。无监督预训练将二级结构编码为表示。经过预训练后，线性投影恢复二级结构（左侧列）。没有预训练时几乎没有恢复的信息（右侧列）。颜色表示通过投影识别的二级结构类别：螺旋（红色）、链（绿色）和卷曲（蓝色）。颜色强度表示置信度。使用来自 ESM-1b 变换器的表示。

(ii) 由CCMpred实现预测的无监督接触（Seemayer等，2014）的直接耦合分析。基准的 MSAs是从UniClust30 (Mirdita等，2017) 数据库生成的，使用Hhblits进行3次搜索迭代。对于二级结构，我们报告8类准确率。对于接触精度，我们报告 Top-L长距离精度，即与序列分隔至少24个残基的接触的L（蛋白质长度）个最高排名预测的精度。

表3显示了交叉验证实验的结果。在预训练之前，关于二级结构和接触的最小信息可以被检测到。预训练之后，投影恢复了有关二级结构和长距离接触的信息，这些信息在不同的家族、超家族和折叠之间具有泛化性。二级结构预测8类准确率分布（图 S2）和长距离接触预测Top-L精准度分布（图 S3）证明了预训练在测试领域的整个分布中产生了结构信息的增加。表3显示了Transformer表示的投影恢复的结构比序列轮廓的投影更多。对于长距离接触，最佳Transformer模型的投影具有比接触预测更高的精准度。

图 5. 残基-残基接触（线性投影）。左侧：对保持折叠水平示例d1n3ya_的Top-L预测，具有vWA样折叠。真正阳性以蓝色显示，假阳性以红色显示，叠加在灰色的真实接触图上。ESM-1b 变换器的投影位于对角线下方，CCMpred的预测位于对角线上方。右侧：精确度分布（Top-L远程），在五个测试分区内比较ESM-1b投影与CCMpred，结构保持在折叠水平。可视化域标记为x。

由 CCMpred 预测的接触情况在所有结构概括水平上。随着结构划分的层次变得更加遥远，二级结构几乎没有退化，家族级别的表现与折叠级别相似。对于长程接触，尽管与家族级别相比，折叠级别的概括性较低，但最好的模型仍然捕获到的结构比无监督基准更多。使用多样性更高的序列（UR50 数据集）进行训练，提高了对二级结构和长程接触的学习，对长程接触的影响更为明显。

图4可视化了两个属于保留折叠的域的二类二级结构投影。在预训练之前，投影产生了不连贯的二级结构预测。在预训练之后，投影恢复了连贯的预测，大多数错误发生在二级结构区域的边界。图5比较了一幅投影接触图与CCMpredict的预测。变换器投影恢复了复杂的接触模式，包括长距离接触。进一步的投影接触可视化针对八个随机选择的测试蛋白质如图S7所示。

深度神经网络我们训练深度神经网络，从表示中预测二级结构和接触。我们为这两项任务选择最先进的神经架构。这些下游模型使用监督损失训练，以从预训练表示中预测二级结构或接触图。下游模型的架构在不同表示和基准测试的实验中保持固定，以便进行比较。

Model	Pre- Training	CB513	CASP13
HMM Profile		71.2 ± 0.1	72.3 ± 0.9
LSTM (S)	UR50/S	60.4 ± 0.1	63.2 ± 0.6
LSTM (L)	UR50/S	62.4 ± 0.2	64.1 ± 0.7
Transf-6	UR50/S	62.0 ± 0.2	64.2 ± 1.2
Transf-12	UR50/S	65.4 ± 0.1	67.2 ± 0.3
Transf-34	None	56.8 ± 0.3	60.0 ± 0.5
Transf-34	UR100	64.3 ± 0.2	66.5 ± 0.3
Transf-34	UR50/S	69.1 ± 0.2	70.7 ± 0.8
Transf-34	UR50/D	69.2 ± 0.1	70.9 ± 0.5
ESM-1B	UR50/S	71.6 ± 0.1	72.5 ± 0.2

Table 4. Eight-class secondary structure prediction accuracy on the CB513 and CASP13 test sets. A fixed neural architecture is trained to predict the secondary structure label from the language model representation of the input sequence. The Transformer has higher performance than the comparable LSTM baselines. Pre-training with the high-diversity UR50 datasets increases accuracy significantly. Features from ESM-1b Transformer are competitive with HMM profiles for supervised secondary structure prediction.

To predict secondary structure we replace the linear layer with a deep neural network, using the model architecture introduced by the Netsurf method (Klausen et al., 2019). For tertiary structure, we predict the binary contact map from the hidden representation of the sequence. We use a dilated convolutional residual network similar to recent state-of-the-art methods for tertiary structure prediction (Xu, 2018; Jones & Kandathil, 2018; Senior et al., 2018).

Table 4 compares the representations for secondary structure prediction. We evaluate models on the CB513 test set (Cuff & Barton, 1999) and the CASP13 domains (Kryshtafovych et al., 2019). For comparison we also re-implement the Netsurf method. The models are trained on the Netsurf training dataset which applies a 25% sequence identity hold-out with CB513, and a temporal hold-out with CASP13. The Transformer features are compared before and after unsupervised pre-training to features from the LSTM baselines. They are also compared to the HMM profiles used by Netsurf. The best Transformer features (71.6%) match the performance of the HMM profiles (71.2%), and exceed the published performance of RaptorX (70.6%) on the same benchmark (Klausen et al., 2019), implying that protein language models can produce features that are directly competitive with sequence profiles for secondary structure prediction.

Table 5 shows performance of the various representations for predicting Top-L long-range contacts across a panel of benchmarks using the RaptorX train set (Wang et al., 2017). For comparison we train the same architecture us-

Model	Pre- training	Test	11	12	13
LSTM (S)	UR50/S	24.1	23.6	19.9	15.3
LSTM (L)	UR50/S	27.8	26.4	24.0	16.4
Transf-6	UR50/S	30.2	29.9	25.3	19.8
Transf-12	UR50/S	37.7	33.6	27.8	20.7
Transf-34	(None)	16.3	17.7	14.8	13.3
Transf-34	UR100	32.7	28.9	24.3	19.1
Transf-34	UR50/S	50.2	42.8	34.7	30.1
Transf-34	UR50/D	50.0	43.0	33.6	28.0
ESM-1b	UR50/S	56.9	47.4	42.7	35.9

Table 5. Top-L long-range contact precision. A deep dilated convolutional residual network is trained to predict contacts using the representations from the pre-trained language model. The pre-trained Transformer representations outperform the LSTM representations in all cases. Pre-training on the high-diversity UR50 datasets boosts precision of representations over pre-training on UR100. High-capacity Transformers (34 layer) outperform lower capacity models (6/12 layer).

ing features from RaptorX (Wang et al., 2017; Xu, 2018). The Test (Wang et al., 2017) and CASP11 (Moult et al., 2016) test sets evaluate with sequence identity hold-out at 25%; the CASP12 (Moult et al., 2018) test set implements a temporal hold-out with the structural training data; and the CASP13 (Kryshtafovych et al., 2019) experiment implements a full temporal hold-out of both the pre-training and training data. For contact prediction, the best features from representation learning do not achieve comparable performance to the state-of-the-art RaptorX features (50.2 vs 59.4 respectively on the RaptorX test set).

In the secondary structure benchmarks Transformer representations produce higher accuracy than the LSTM baselines with comparable numbers of parameters. For contact prediction Transformer representations yield higher precision than LSTMs, with even the smallest Transformer representations exceeding LSTMs with more parameters. Diversity in the pre-training data also has a strong effect, with the high-diversity datasets providing significant improvements over the low-diversity dataset. Relative performance of the representations is consistent across all four of the contact benchmarks using different hold-out methodology.

Relationship between language modeling and structure learning To investigate the relationship between the language modeling objective and information about structure in the model, linear projections for secondary structure and contacts are fit using the representations from Transformer models taken from checkpoints across their pre-training trajectories. We use the Transformers trained on UR50/S. We

模型	前- 训练	CB513	CASP13
隐马尔可夫模型配置文件		71.2 ± 0.1	72.3 ± 0.9
长短期记忆网络 (S)	UR50/S	60.4 ± 0.1	63.2 ± 0.6
LSTM (L)	UR50/S	62.4 ± 0.2	64.1 ± 0.7
Transf-6	UR50/S	62.0 ± 0.2	64.2 ± 1.2
Transf-12	UR50/S	65.4 ± 0.1	67.2 ± 0.3
Transf-34	无	56.8 ± 0.3	60.0 ± 0.5
Transf-34	UR100	64.3 ± 0.2	66.5 ± 0.3
Transf-34	UR50/S	69.1 ± 0.2	70.7 ± 0.8
Transf-34	UR50/D	69.2 ± 0.1	70.9 ± 0.5
ESM-1B	UR50/S	71.6 ± 0.1	72.5 ± 0.2

表4。在CB513和CASP13测试集上的八类二级结构预测准确性。一个固定的神经架构被训练从输入序列的语言模型表示中预测二级结构标签。变换器的性能优于可比的LSTM基准。使用高度多样化的UR50数据集进行预训练显著提高了准确性。来自ESM-1b变换器的特征在监督二级结构预测中与HMM轮廓具有竞争力。

为了预测二级结构, 我们用深度神经网络替换线性层, 采用Netsurf方法介绍的模型架构 (Klausen等, 2019)。对于三级结构, 我们从序列的隐藏表示中预测二元接触图。我们使用类似于最近最先进的三级结构预测方法的扩张卷积残差网络 (Xu, 2018; Jones & Kandathil, 2018; Senior等, 2018)。表4比较了二级结构预测的表示。我们在CB513测试集上评估模型 (Cuff与Barton, 1999) 和CASP13领域 (Kryshtafovych等, 2019)。为进行比较, 我们还重新实现了Netsurf方法。这些模型在Netsurf训练数据集上训练, 该数据集对CB513应用了25%的序列同一性保持, 并对CASP13进行了时间保持。变换器特征在无监督预训练之前和之后与LSTM基线的特征进行了比较。它们还与所使用的HMM轮廓进行了比较

由Netsurf提供。最佳变换器特征 (71.6%) 与HMM轮廓的性能 (71.2%) 相匹配, 并且超过了在同一基准上RaptorX公布的性能 (70.6%) (Klausen等, 2019), 这意味着蛋白质语言模型可以产生与序列轮廓在二级结构预测中直接竞争的特征。表5显示了各种表示在预测Top-L远程接触方面的性能。

使用RaptorX训练集的基准测试 (王等, 2017)。为了比较, 我们使用相同的架构进行训练-

模型	训练	Test	11	12	13
长短期记忆网络 (S)	UR50/S	24.1	23.6	19.9	15.3
长短期记忆网络 (L)	UR50/S	27.8	26.4	24.0	16.4
Transf-6	UR50/S	30.2	29.9	25.3	19.8
Transf-12	UR50/S	37.7	33.6	27.8	20.7
Transf-34	(无)	16.3	17.7	14.8	13.3
Transf-34	UR100	32.7	28.9	24.3	19.1
Transf-34	UR50/S	50.2	42.8	34.7	30.1
Transf-34	UR50/D	50.0	43.0	33.6	28.0
ESM-1b	UR50/S	56.9	47.4	42.7	35.9

表5。Top-L 长距离接触精度。一个深度扩张卷积残差网络被训练用来预测接触, 使用来自预训练语言模型的表示。预训练的变换器表示在所有情况下都超越了长短期记忆网络 (LSTM) 表示。对高多样性的 UR50 进行预训练

数据集在预训练 UR100 的基础上提升了表示的精度。高容量变换器 (34 层) 超越了低容量模型 (6/12 层)。

使用RaptorX的特征 (王等, 2017; 徐, 2018)。测试集 (王等, 2017) 和CASP11 (Moult等, 2016) 测试集在序列同一性的25%上进行评估; CASP12 (Moult等, 2018) 测试集实施了使用结构训练数据的时间保留; 以及

CASP13 (Kryshtafovych 等人., 2019) 实验实施了对预训练和训练数据的完全时间保留。对于接触预测, 最佳特征来自表示学习未能达到与最先进的 RaptorX 特征相当的性能 (50.2 对 59.4

在RaptorX测试集上分别进行评估。在二级结构基准测试中, 变换器表示的准确率高于基于长短期记忆网络的基准, 并且参数数量相当。在接触预测方面, 变换器表示的精度高于长短期记忆网络, 即便是最小的变换器表示也超过了参数更多的长短期记忆网络。预训练数据的多样性也产生了显著影响, 高多样性的数据集相较于低多样性数据集提供了显著改善。在使用不同的留出方法的所有四个接触基准中, 表示的相对性能是一致的。

语言模型与结构学习之间的关系为了研究语言建模目标与模型中结构信息之间的关系, 针对二级结构的线性投影和

接触通过从变换器模型的检查点所获得的表示进行拟合, 这些检查点来自于它们的预训练轨迹。我们使用训练在UR50/S上的变换器。我们

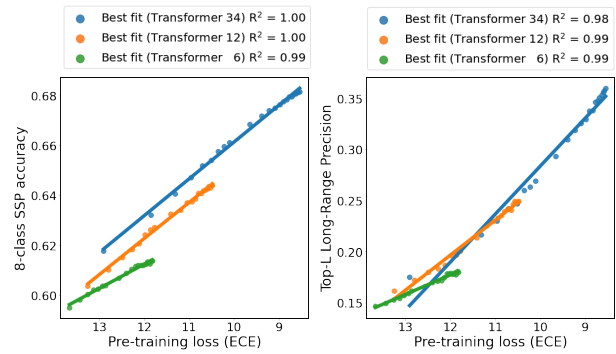


Figure 6. Relationship between the language modeling objective and structure learning. Eight-class secondary structure prediction accuracy (left) and contact prediction Top-L long-range precision (right) both as a function of pre-training ECE. Performance is evaluated on held-out folds. Linear projections are fit using model checkpoints over the course of pre-training on UR50/S. The linear relationship for each model indicates that for a given model and pre-training dataset, the language modeling ECE is a good proxy for the structural content of the representations. Improvement of the model’s ECE leads to an increase in information about structure. This establishes a link between the language modeling objective and unsupervised structure learning.

fit the projections and evaluate with the train and test split implemented by the first partition of the fold level structural hold-out dataset. For each model, Figure 6 shows a linear relationship between the language modeling objective and information about structure, which is maintained over the course of pre-training. The linear fit is close to ideal for both secondary structure and contacts. A similar experiment is also performed for secondary structure with a deep neural network instead of linear projection, using the Netsurf training sequences and CB513 test set. A linear relationship between secondary structure accuracy and language modeling ECE is also observed for the deep prediction head (Figure S4). Thus, for a given model and pre-training dataset, language modeling fidelity measured by ECE is a good proxy for the structural content of the representations. Since performance on the language modeling objective improves with model capacity, this suggests further scale may improve results on structure prediction tasks.

Single versus multi-family pre-training We compare training across evolutionary statistics to training on single protein families. We pre-train separate 12-layer Transformer models on the Pfam multiple sequence alignments of the three most common domains in nature longer than 100 amino acids, the ATP-binding domain of the ABC transporters, the protein kinase domain, and the response regulator receiver domain. We test the ability of models trained on one protein family to generalize secondary structure in-

Features	CB513	CASP13
RaptorX	70.6	
Netsurf	72.1	74
(a) Netsurf (reimpl.)	71.2 ± 0.1	72.3 ± 0.9
(b) +direct	72.1 ± 0.1	72.2 ± 0.5
(c) +avg	73.7 ± 0.2	75.1 ± 0.4

Table 6. Feature combination (secondary structure prediction). Eight-class accuracy. The language model improves state-of-the-art features for secondary structure prediction. Features from a reimplementation of Netsurf (Klausen et al., 2019) are combined with 34-layer Transformer (UR50/S) embeddings using a two layer BiLSTM architecture. (a) Performance of Netsurf features alone. (b) *Direct* adds the Transformer representation of the input sequence. (c) *Avg* adds the average of Transformer features for each position in the MSA of the input sequence. Results exceed those for state-of-the-art methods RaptorX (70.6%) and Netsurf (72.1%) on the CB513 test set, and for Netsurf (74.0%) on the CASP13 evaluation set used here.

formation within-family and out-of-family by evaluating on sequences with ground truth labels from the family the model was trained on or from the alternate families. The models are evaluated using linear projections. In all cases, the model trained on within-family sequences has higher accuracy than models trained on out-of-family sequences (Table S2), indicating poor generalization when training on single MSA families. More significantly, the model trained across the full UniParc sequence diversity has a higher accuracy than the single-family model accuracies, even on the same-family evaluation dataset. This suggests that the representations learned from the full dataset are generalizing information about secondary structure learned outside the sequence family.

6. Feature combination

Features discovered by unsupervised protein language modeling can be combined with state-of-the-art features to improve them further. Current state-of-the-art methods use information derived from MSAs. We combine this information with features from the Transformer model.

We explore three approaches for incorporating information from representation learning. For each input sequence x : (i) *direct* uses the final hidden representation from the Transformer directly; (ii) *avg* takes the average of the final hidden representation at each position across the sequences from the MSA of x ; (iii) *cov* produces features for each pair of positions, using the uncentered covariance across sequences from the MSA of x , after dimensionality reduction of the final hidden representations by PCA. Note that (i) and (ii) produce features for each position in x , while (iii) produces

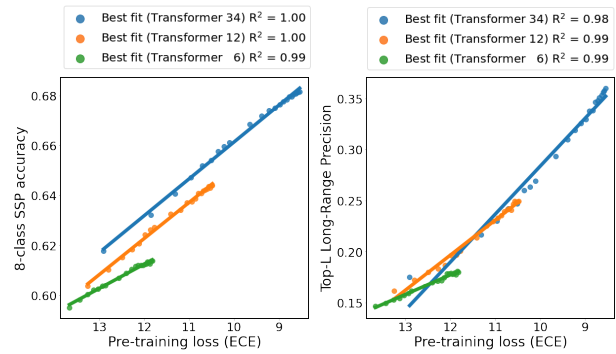


图6. 语言建模目标与结构学习之间的关系。八类二级结构预测准确性（左）和接触预测Top-L长距离精度（右）均作为预训练ECE的函数进行表示。性能在保留的折叠上进行评估。线性投影是使用UR50/S预训练过程中的模型检查点进行拟合的。每个模型的线性关系表明对于特定模型和

预训练数据集，语言建模的ECE是表示的结构内容的良好代理。模型的ECE改进会导致关于结构的信息增加。这建立了语言建模目标与无监督结构学习之间的联系。

拟合投影并使用折叠级结构保留数据集的第一个分区实现的训练和测试拆分进行评估。对于每个模型，图6显示了语言模型目标与结构信息之间的线性关系，该关系得以维持

在预训练过程中。对于二级结构和接触，线性拟合接近理想。类似的实验也使用深度神经网络而非线性投影进行二级结构的预测，使用Netsurf训练序列和CB513测试集。还观察到二级结构准确性与语言建模ECE之间的线性关系（图S4）。因此，对于特定模型和

预训练数据集，语言建模的保真度被测量通过 ECE 是表示的结构内容的良好代理。由于语言模型目标上的表现随着模型容量的提高而改善，这表明进一步的扩大可能改善结构预测任务的结果。

单家族与多家族预训练我们比较了跨进化统计的训练与单一蛋白质家族的训练。我们在自然界中最常见的三种结构域的Pfam多序列比对上，分别预训练了12层的Transformer模型，这些结构域长度超过100个氨基酸，包括ABC转运蛋白的ATP结合域、蛋白激酶结构域，以及反应调节器接收器结构域。我们测试了训练模型的能力

关于一个蛋白质家族以推广二级结构的-

特征	CB513	CASP13
RaptorX	70.6	
Netsurf	72.1	74
(a) Netsurf (重实现)	71.2 ± 0.1	72.3 ± 0.9
(b) +直接	72.1 ± 0.1	72.2 ± 0.5
(c) +平均	73.7 ± 0.2	75.1 ± 0.4

表 6.特征组合（二级结构预测）。八类准确率。语言模型改善了二级结构预测的最先进特征。从Netsurf的重新实现中获得的特征（Klausen等，2019）被组合

使用34层Transformer (UR50/S)嵌入和双层BiLSTM架构。(a) 仅 Netsurf 特征的性能。(b) 直接添加输入序列的Transformer表示。(c) 平均为输入序列的MSA中每个位置添加Transformer特征的平均值。结果超过了最先进的方法 RaptorX (70.6%)和Netsurf (72.1%)在CB513测试集上的表现，以及Netsurf (74.0%)在CASP13上的表现。

此处使用的评估集。

通过在使用来自模型训练的家族或来自其他家族的真实值标签的序列上进行评估，形成家族内和家族外的对比。模型使用线性投影进行评估。在所有情况下，基于家族内序列训练的模型具有比基于家族外序列训练的模型更高的准确性（表 S2），指示出在单一MSA家族上训练时的泛化能力较差。更重要的是，训练模型

在整个UniParc序列多样性上，准确性高于单一家族模型的准确性，即使在同一家族的评估数据集中。这表明，从完整数据集中学习到的表示包含了对在序列家族之外学习到的二级结构的信息。

6. 特征组合

通过无监督蛋白质语言建模发现的特征可以与最先进的特征结合，以进一步改善它们。目前最先进的方法使用来自MSA的信息。我们将这些信息与Transformer模型的特征结合。我们探讨三种方法将表示学习的信息纳入。对于每个输入序列 x : (i)*direct* 直接使用Transformer的最终隐藏表示；(ii) *avg* 计算来自序列的每个位置的最终隐藏表示的平均值

MSA of x ; (iii) 协方差为每对位置生成特征，使用来自于 x 的 MSA 中的未中心化协方差，在主成分分析（PCA）对最终隐藏表示进行降维之后。请注意，(i) 和 (ii) 为 x 中的每个位置生成特征，而 (iii) 生成

	Test	CASP11	CASP12	CASP13
# domains	500	105	55	34
(a) RaptorX	59.4 ± .2	53.8 ± .3	51.1 ± .2	43.4 ± .4
(b) +direct	61.7 ± .4	55.0 ± .1	51.5 ± .5	43.7 ± .4
(c) +avg	62.9 ± .4	56.6 ± .4	52.4 ± .5	44.8 ± .8
(d) +cov	63.3 ± .2	56.8 ± .2	53.0 ± .3	45.2 ± .5

Table 7. Feature combination (contact prediction). Top-L long-range contact precision. The language model improves state-of-the-art features for contact prediction. A deep ResNet with fixed architecture is trained on each feature set to predict binary contacts. (a) performance of state-of-the-art RaptorX (Xu, 2018) features including PSSM, predicted secondary structure, predicted accessibility, pairwise APC-corrected Potts model couplings and mutual information, and a pairwise contact potential. (b) Adds Transformer representation of the input sequence to the feature set. (c) Adds the average Transformer representation at each position of the MSA. (d) Adds the uncentered covariance over the MSA of a low-dimensional projection of the Transformer features. Features are from the 34-layer Transformer pre-trained on UR50/S.

features for each pair of positions.

Secondary structure Current state-of-the-art methods for secondary structure prediction have high accuracies for the eight-class prediction problem. We investigate whether performance can be improved by combining Transformer features with sequence profiles. Table 6 shows that combining the representations with profiles further boosts accuracy, resulting in state-of-the-art performance on secondary structure prediction.

We establish a baseline of performance by reimplementing the Klausen et al. (2019) method using the same features, resulting in an accuracy of 71.2% (vs. published performance of 72.1%) on the CB513 test set. Then we add the the Transformer features using the *direct* and *avg* combination methods; these achieve 0.9% and 2.5% absolute improvement in accuracy respectively. This suggests that the Transformer features contain information not present in the MSA-derived features.

Residue-residue contacts Deep neural networks have enabled recent breakthroughs in the prediction of protein contacts and tertiary structure (Xu, 2018; Senior et al., 2018). State-of-the-art neural networks for tertiary structure and contact prediction use deep residual architectures with two-dimensional convolutions over pairwise feature maps to output a contact prediction or distance potential for each pair of residues (Wang et al., 2017; Xu, 2018; Senior et al., 2018).

A variety of input features, training datasets, and supervision signals are used in state-of-the-art methods. To make a controlled comparison, we fix a standard architecture, training dataset, multiple sequence alignments, and set of base input features for all experiments, to which we add pre-trained features from the Transformer model. For the base features we use the RaptorX feature set which includes PSSM, 3-state secondary structure prediction, one-hot embedding of

sequence, APC corrected Potts model couplings, mutual information, pairwise contact potential, and predicted accessibility. RaptorX was the winning method for contact prediction in the CASP12 and CASP13 competitions (Xu, 2018). The training and evaluation sets are the same as used in the previous section.

Table 7 indicates that addition of Transformer features from the 34-layer model trained on UR50/S consistently produces an improvement across the test sets. The table shows Top-L long-range precisions reporting mean and standard deviation over 5 different model seeds. *Direct* gives a modest improvement on some test sets. *Avg* improves over *direct*, and *cov* provides further gains. For example, *cov* produces an absolute improvement of 3.9% on the RaptorX test set, and 1.8% improvement on the CASP13 test set evaluated with temporal hold-outs on both fine-tuning and pre-training data. Additional results and metrics for contact prediction are reported in Table S3.

7. Prediction of mutational effects

The mutational fitness landscape provides deep insight into biology. Coupling next generation sequencing with a mutagenesis screen allows parallel readout of tens of thousands of variants of a single protein (Fowler & Fields, 2014). The detail and coverage of these experiments provides a view into the mutational fitness landscape of individual proteins, giving quantitative relationships between sequence and protein function. We adapt the Transformer protein language model to predict the quantitative effect of mutations.

First we investigate intra-protein variant effect prediction, where a limited sampling of mutations is used to predict the effect of unobserved mutations. This setting has utility in protein engineering applications (Yang et al., 2019). We evaluate the representations on two deep mutational scanning datasets used by recent state-of-the-art methods for variant effect prediction, Envision (Gray et al., 2018)

	测试	CASP11	CASP12	CASP13
# 领域	500	105	55	34
(a) RaptorX	59.4 ± .2	53.8 ±.3	51.1 ± .2	43.4 ±.4
(b) +直接	61.7 ± .4	55.0 ±.1	51.5 ± .5	43.7 ±.4
(c) +平均	62.9 ± .4	56.6 ±.4	52.4 ± .5	44.8 ±.8
(d) +cov	63.3 ± .2	56.8 ±.2	53.0 ± .3	45.2 ±.5

表7。特征组合（接触预测）。Top-L 长距离接触精度。语言模型改善了接触预测的最先进特征。一个具有固定结构的深度 ResNet 在每个特征集上进行训练，以预测二元接触。（a）最先进的 RaptorX 的表现（徐, 2018）特征包括 PSSM、预测的二级结构、预测的可接近性、成对的 APC 校正 Potts 模型耦合和互信息，以及成对接触势。（b）将输入序列的 Transformer 表示添加到特征集。（c）在 MSA 的每个位置添加平均 Transformer 表示。（d）添加对 Transformer 特征低维投影的未中心化协方差。特征来自于在 UR50/S 上预训练的 34层 Transformer。

每对位置的特征。**二级结构** 当前最先进的二级结构预测方法在八类预测问题上具有很高的准确性。我们研究了通过结合变换器特征和序列轮廓是否可以提高性能。表6 显示，结合具有轮廓的表征进一步提高了准确性，从而在二级结构预测上实现了最先进的性能。我们通过使用相同特征重新实现Klausen等(2019) 方法建立了性能基线，在CB513测试集上得到71.2%的准确性（与发表的72.1%性能相比）。然后我们添加

变换器特征使用直接 和 平均 组合方法；这些方式分别实现0.9% 和 2.5% 的准确性绝对提升。这表明变换器特征包含MSA派生特征中不存在的信息。**残基-残基接触** 深度神经网络最近在蛋白质接触和三级结构的预测中取得了突破性进展（徐, 2018; Senior等, 2018）。最先进的三级结构神经网络和

序列、经过APC修正的Potts模型耦合、互信息信息、成对接触势和预测可及性。RaptorX 是 CASP12 和 CASP13 竞赛中接触预测的获胜方法（徐, 2018）。训练和评估集与使用的相同

在上一节中。表7说明从添加变换器特征

在UR50/S上训练的34层模型在测试集上 consistently 产生了改善。表格显示了Top-L 长期精度报告在5个不同模型种子上的均值和标准差。直接在某些测试集上有所改善。平均在直接基础上进一步改善，而*cov* 提供了更多的增益。例如，*cov* 在 RaptorX测试集上产生了3.9%的绝对改善，在评估的 CASP13测试集上改善了1.8%。

在微调和预训练数据上进行时间保留的情况下。接触预测的附加结果和指标报告在表S3。

7. 突变效应预测

突变适应性景观提供了对生物学的深入洞察。将下一代测序与突变筛选相结合，可以并行读取单个蛋白质的数万个变体（Fowler & Fields,2014）。这些实验的细节和覆盖范围提供了对个别蛋白质的突变适应性景观的视图，给出了序列和蛋白质功能之间的定量关系。我们改编了变换器蛋白语言模型，以预测突变的定量效应。首先，我们研究蛋白质内变体效应预测，其中使用有限的突变采样来预测未观察到的突变的效应。这种设置在蛋白质工程应用中具有实用性（杨等, 2019）。我们在两个深度突变

来自变换器模型的特征。对于基本特征，我们使用 RaptorX特征集，其中包括PSSM、三态二级结构预测和单热编码的

最近最先进的变体效应预测方法使用的扫描数据集，Envision （Gray等, 2018）

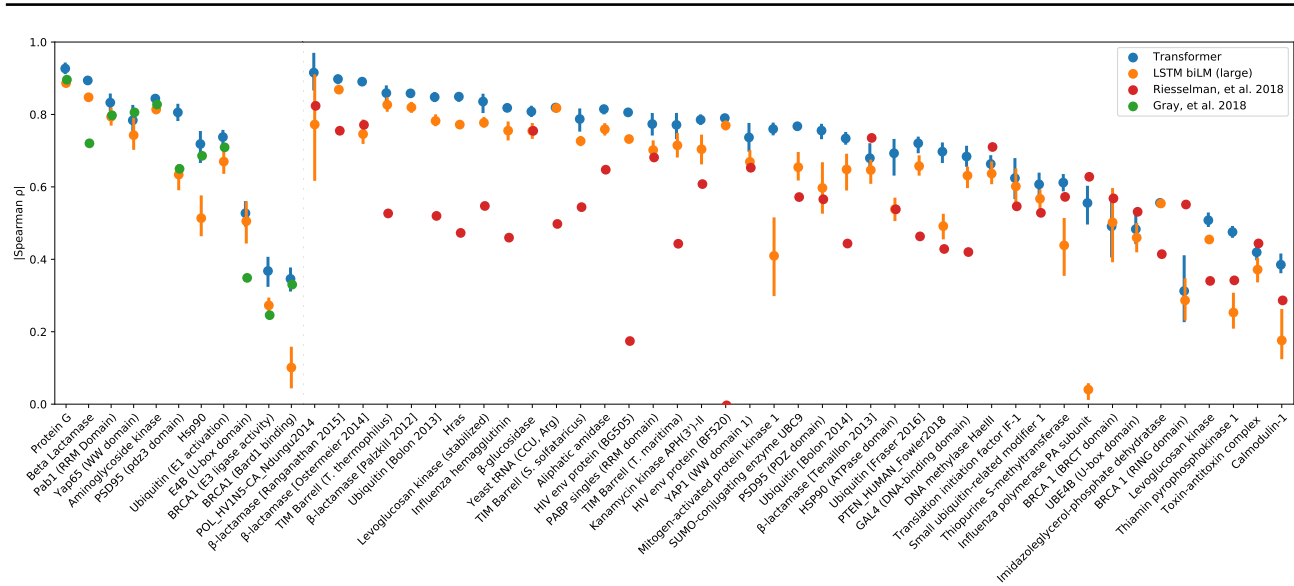


Figure 7. Representation learning enables state-of-the-art supervised prediction of the quantitative effect of mutations. Left panel: Envision dataset (Gray et al., 2018); right panel: DeepSequence dataset (Riesselman et al., 2018). Transformer representations (34-layer, UR50/S) are compared to the LSTM bidirectional language model (large model, UR50/S). The result of five-fold cross validation is reported for each protein. For each partition, supervised fine-tuning is performed on 80% of the mutational data for the protein, and results are evaluated on the remaining 20%. Transformer representations outperform baseline LSTM representations on both datasets. State-of-the-art methods are also shown for each dataset. Gray et al. (2018) is a supervised method using structural, evolutionary, and biochemical features, trained with the same protocol as used for the Transformer. Riesselman et al. (2018) is an unsupervised method trained on the MSA of each protein.

and DeepSequence (Riesselman et al., 2018). Collectively the data includes over 700,000 variant effect measurements from over 100 large-scale experimental mutagenesis datasets.

Fine-tuning the Transformer yields a mutational effect predictor that is comparable to the results of Envision. Envision (Gray et al., 2018) relies on protein structural and evolutionary features to generalize. We assess whether the Transformer can achieve similar generalization results, without direct access to structural features. The same methodology for partitioning data for training and evaluation is used as in Gray et al. (2018) to allow a comparison of the results. We use the 34-layer Transformer trained on UR50/S. Figure 7 shows the fine-tuned Transformer exceeds the performance of Envision on 10 of the 12 proteins. For each protein a fraction $p = 0.8$ of the data is used for training and the remaining data is used for testing. We report mean and standard deviations for five-fold cross-validation in Table S5. Results varying the fraction of data that is used for training are reported in Figure S5.

We also evaluate using the same five-fold cross validation methodology on the deep mutational scanning experiments assembled for DeepSequence (Riesselman et al., 2018). The fine-tuned Transformer model outperforms the fine-tuned LSTM baselines. While not directly comparable, we also in-

clude the performance of the original DeepSequence method which is unsupervised, and represents state-of-the-art for this dataset.

Generalization to a new fitness landscape We analyze the Transformer’s ability to generalize to the fitness landscape of a new protein. Following the protocol introduced in Envision, we use a leave-one-out analysis: to evaluate performance on a given protein, we train on data from the remaining $n - 1$ proteins and test on the held-out protein. Figure S6 shows that the Transformer’s predictions from raw sequences perform better than Envision on 5 of the 9 tasks.

8. Related Work

Contemporaneously with the preprint of this work, Rives et al. (2019), related preprints Alley et al. (2019) and Heinzinger et al. (2019), also proposed protein language modeling, albeit at a smaller scale. These works, along with Rao et al. (2019), evaluated on a variety of downstream tasks. Rives et al. (2019) first proposed protein language modeling with Transformers. Alley et al. (2019) and Heinzinger et al. (2019) train LSTMs on UniRef50. Rao et al. (2019) trained a 12-layer Transformer model (38M parameters) on Pfam (Bateman et al., 2013). The baselines in this paper are

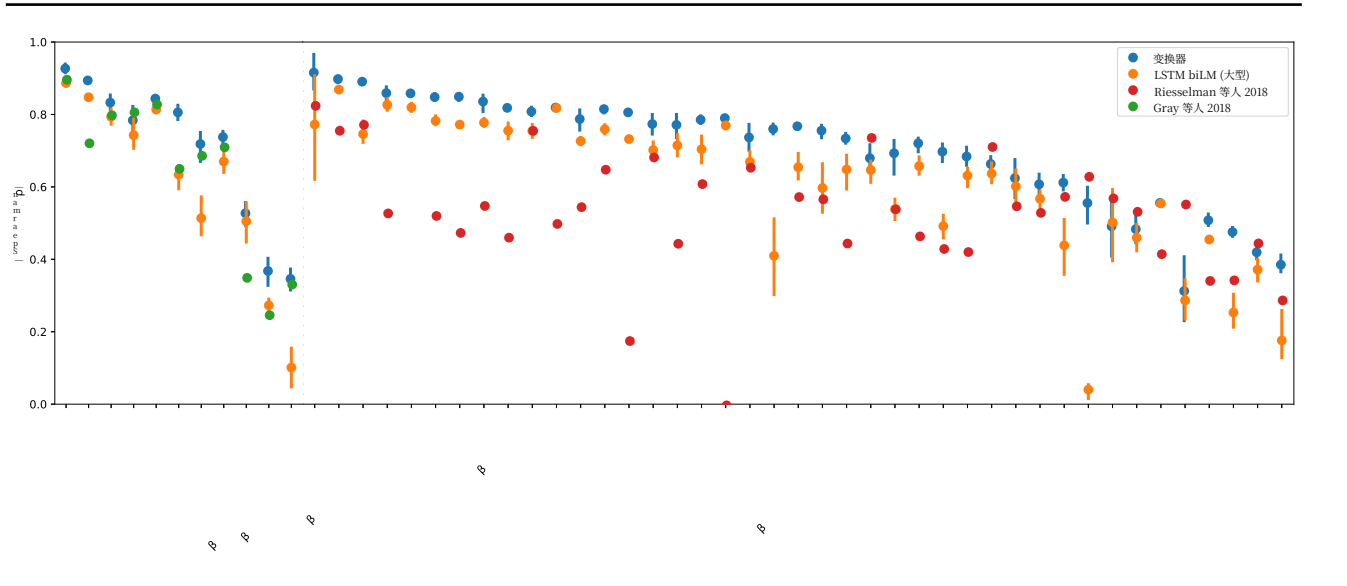


图7. 表示学习使得最先进的监督预测蛋白质突变的定量效应成为可能。左侧面板：Envision 数据集 (Gray 等人, 2018)；右侧面板：DeepSequence 数据集 (Riesselman 等人, 2018)。对比了变换器表示 (34层, UR50/S) 与 LSTM 双向语言模型 (大型模型, UR50/S)。每个蛋白质的五折交叉验证的结果已报告。对于每个划分，监督微调在蛋白质的 80% 突变数据上进行，结果在剩余的20%上进行评估。变换器表示在两个数据集上超越了基线LSTM表示。每个数据集的最先进方法也显示出来。Gray 等人(2018) 是一种监督方法，使用与变换器相同协议训练的结构、进化和生化特征。Riesselman 等人(2018) 是一种在每个蛋白质的MSA上训练的无监督方法。

和 DeepSequence (里瑟尔曼等,2018)。数据总共包括来自超过100 个大型实验突变数据集的超过 700,000 个变异效应测量值。

微调变换器产生的突变效应预测器的结果可与 Envision的结果相比。En- vision (Gray等, 2018) 依赖于蛋白质的结构和进化特征进行推广。我们评估变换器是否能够在没有直接访问结构特征的情况下实现类似的推广结果。用于训练和评估的数据分割方法与 Gray等(2018) 中使用的方法相同，以便对结果进行比较。我们使用在UR50/S上训练的34层变换器。

图7 显示微调后的变换器在 10 个 12 蛋白质的性能上超过了 Envision。对于每个蛋白质，使用数据的一部分 $p = 0.8$ 用于训练，其余数据用于测试。我们报告五折交叉验证的均值和标准差在 表 S5。结果中变更用于训练的数据比例如 图 S5 所示。

我们还使用相同的五折交叉验证方法对为 DeepSequence组装的深度突变扫描实验进行评估 (里瑟尔曼等, 2018)。经过精细调整的 Transformer模型优于经过精细调整的LSTM基线。尽管不可直接比较，我们还在-

包括了原始DeepSequence方法的性能，该方法是无监督的，并且在该数据集中代表了最先进技术。

对新适应性景观的推广我们分析了变换器在新蛋白质的适应性景观中的推广能力。遵循Envision中提出的协议，我们使用留一法分析：为了评估在给定蛋白质上的表现，我们在其余的 $n - 1$ 蛋白质的数据上进行训练，并在保留的蛋白质上进行测试。图 S6显示变换器从原始序列的预测在5 个9个任务中表现优于 Envision。

8. 相关工作与本工作的预印本同时， Rives 等人 (2019)，相关的预印本 Alley 等人(2019) 和

Heinzinger 等人(2019)，也提出了蛋白质语言建模，尽管规模较小。这些工作以及 Rao 等人 (2019)，在多种下游任务上进行了评估。Rives 等人 (2019) 首先提出了使用变换器进行蛋白质语言建模。Alley 等人(2019) 和 Heinzinger 等人 (2019) 在 UniRef50 上训练长短期记忆网络。Rao 等人(2019) 在 Pfam 上训练了一个 12 层的变换器模型 (38M 参数) (贝特曼等人, 2013)。本文的基准是

Model	Pre- Training	Params	RH	SSP	Contact
UniRep ^{1†}		18M	.527	58.4	21.9
SeqVec ^{2†}		93M	.545	62.1	29.0
Tape ^{3†}		38M	.581	58.0	23.2
LSTM (S)	UR50/S	28.4M	.558	60.4	24.1
LSTM (L)	UR50/S	113.4M	.574	62.4	27.8
Transformer-6	UR50/S	42.6M	.653	62.0	30.2
Transformer-12	UR50/S	85.1M	.639	65.4	37.7
Transformer-34	UR100	669.2M	.599	64.3	32.7
Transformer-34	UR50/S	669.2M	.639	69.2	50.2
ESM-1b	UR50/S	652.4M	.532	71.6	56.9

Table 8. Comparison to related protein language models. (RH) Remote Homology at the fold level, using Hit-10 metric on SCOP. (SSP) Secondary structure Q8 accuracy on CB513. (Contact) Top-L long range contact precision on RaptorX test set from Wang et al. (2017). Results for additional test sets in Table S6. ¹Alley et al. (2019) ²Heinzinger et al. (2019) ³Rao et al. (2019). [†]The pre-training datasets for related work have differences from ours.

comparable to these models. The large Transformer models trained in this paper are considerably larger than in these related works.

We benchmark against related work in Table 8. Heinzinger et al. (2019), Alley et al. (2019), and Rao et al. (2019), evaluate models on differing downstream tasks and test sets. We retrieve the weights for the above models, evaluating them directly in our codebase against the panel of test sets used in this paper for remote homology, secondary structure prediction, and contact prediction, with the same training data and model architectures. This allows a direct comparison between the representations. Table 8 shows that high-capacity Transformers have strong performance for secondary structure and contact prediction significantly exceeding Alley et al. (2019), Heinzinger et al. (2019), and Rao et al. (2019). The small Transformer models trained as baselines also have higher performance than the methods with comparable parameter numbers.

Protein sequence embeddings have been the subject of recent investigation for protein engineering (Yang et al., 2018). Bepler & Berger (2019) pre-trained LSTMs on protein sequences, adding supervision from contacts to produce embeddings. Subsequent to our preprint, related works have built on its exploration of protein sequence modeling, exploring generative models (Riesselman et al., 2019; Madani et al., 2020), internal representations of Transformers (Vig et al., 2020), and applications of representation learning and generative modeling such as classification (Elnaggar et al., 2019; Strodthoff et al., 2020), mutational effect prediction (Luo et al., 2020), and design of sequences (Repecka et al., 2019; Hawkins-Hooker et al., 2020; Amimeur et al., 2020).

9. Discussion

One of the goals for artificial intelligence in biology could be the creation of controllable predictive and generative models that can read and generate biology in its native language. Accordingly, research will be necessary into methods that can learn intrinsic biological properties directly from protein sequences, which can be transferred to prediction and generation.

We investigated deep learning across evolution at the scale of the largest protein sequence databases, training contextual language models across 86 billion amino acids from 250 million sequences. The space of representations learned from sequences by high-capacity networks reflects biological structure at multiple levels, including that of amino acids, proteins, and evolutionary homology. Information about secondary and tertiary structure is internalized and represented within the network. Knowledge of intrinsic biological properties emerges without supervision — no learning signal other than sequences is given during pre-training.

We find that networks that have been trained across evolutionary data generalize: information can be extracted from representations by linear projections, deep neural networks, or by adapting the model using supervision. Fine-tuning produces results that match state-of-the-art on variant activity prediction. Predictions are made directly from the sequence, using features that have been automatically learned by the language model, rather than selected by domain knowledge.

We find that pre-training discovers information that is not present in current state-of-the-art features. The learned features can be combined with features used by state-of-the-art structure prediction methods to improve results. Empirically we find that features discovered by larger models

模型	预- 训练	参数	RH	SSP	联系
UniRep ¹ †		18M	.527	58.4	21.9
SeqVec ² †		93M	.545	62.1	29.0
Tape ³ †		38M	.581	58.0	23.2
LSTM (S)	UR50/S	28.4M	.558	60.4	24.1
LSTM (L)	UR50/S	113.4M	.574	62.4	27.8
变换器-6	UR50/S	42.6M	.653	62.0	30.2
Transformer-12	UR50/S	85.1M	.639	65.4	37.7
变换器-34	UR100	669.2M	.599	64.3	32.7
变换器-34	UR50/S	669.2M	.639	69.2	50.2
ESM-1b	UR50/S	652.4M	.532	71.6	56.9

表8。与相关的蛋白质语言模型的比较。(RH) 在折叠水平上的远程同源, 使用 Hit-10 指标在 SCOP 上。(SSP) 在 CB513 上的二级结构 Q8 准确性。(Contact) 在 王等(2017) 的 RaptorX 测试集上的 Top-L 长程接触精度。表 S6中的额外测试集结果。 ¹Alley et al. (2019) ²Heinzinger et al. (2019) ³Rao et al. (2019)。

[†]相关工作的预训练数据集与我们的存在差异。

与这些模型相当。本文中训练的大型变换器模型显著大于这些相关工作中的模型。我们在表8中对相关工作进行了基准测试。Heinzinger等人(2019), Alley等人(2019), 和Rao等人(2019), 在不同的下游任务和测试集上评估模型。我们获取上述模型的权重, 直接在我们的代码库中评估它们与本文中用于远程同源、二级结构预测和接触预测的一组测试集的表现, 使用相同的训练数据和模型架构。这允许对表示进行直接比较。表8显示高容量变换器在二级结构和接触预测方面具有强大的性能, 显著超过Alley等人(2019), Heinzinger等人(2019), 和

Rao等(2019)。作为基线训练的小型Transformer模型的性能也优于参数数量相当的方法。蛋白质序列嵌入是最近针对蛋白质工程的研究主题(杨等,2018)。Bepler & Berger(2019) 在蛋白质序列上对LSTM进行了预训练, 并通过接触信息添加监督以生成嵌入。在我们的预印本之后, 相关工作在其对蛋白质序列建模的探索基础上进行了扩展, 探索生成模型(里瑟尔曼等, 2019; Madani

等,2020), 变换器的内部表示(Vig 等,2020), 以及表示学习和生成建模的应用, 如分类(Elnaggar 等, 2019; Strodthoff 等,2020), 突变效应预测(Luo 等, 2020), 以及序列设计(Repecka 等, 2019; Hawkins-Hooker 等, 2020;Amimeur 等,2020)。

9. 讨论

人工智能在生物学中的目标之一可能是创建可控的预测和生成模型, 能够以其本土语言读取和生成生物学。因此, 研究将有必要探索可以直接从蛋白质序列中学习内在生物特性的方法, 这些特性可以转移至预测和

生成。我们在最大蛋白质序列数据库的规模上研究了进化中的深度学习, 在860亿个氨基酸上训练了上下文语言模型。

2.5亿序列。学习到的表征空间通过高容量网络从序列中提取的信息反映了生物结构的多个层面, 包括氨基酸、蛋白质和进化同源性。关于二级和三级结构的信息被内化并表示。

在网络内部。对内在生物特性的知识在没有监督的情况下出现——没有学习信号在预训练期间, 除了序列之外, 没有提供其他信息。我们发现, 已经在进化数据上训练的网络具有良好的泛化能力: 可以从中提取信息通过线性投影、深度神经网络, 或通过使用监督来调整模型的表示。微调产生的结果在变异活性预测上达到最先进的水平。预测是直接从序列中进行的, 使用的是语言模型自动学习的特征, 而不是通过领域知识选择的特征。我们发现预训练发现了当前最先进特征中不存在的信息。这些学习到的特征可以与最先进的结构预测方法使用的特征相结合, 以改善结果。我们的实证研究表明, 更大模型发现的特征

perform better on downstream tasks. The Transformer outperforms LSTMs with similar capacity across benchmarks. Increasing diversity of the training data results in significant improvements to the representations.

While the protein language models we study are of comparable scale to those used in the text domain, our experiments have not yet reached the limit of scale. We observed that even the highest capacity models we trained (with approximately 650-700M parameters) under-fit the sequence datasets, due to insufficient model capacity. The relationship we find between language modeling fidelity and the information about structure encoded into the representations suggests that higher capacity models will yield better representations. These findings imply potential for further model scale and data diversity incorporating sequences from metagenomics.

Combining high-capacity generative models with gene synthesis and high throughput characterization can enable generative biology. The models we have trained can be used to generate new sequences (Wang & Cho, 2019). If neural networks can transfer knowledge learned from protein sequences to design functional proteins, this could be coupled with predictive models to jointly generate and optimize sequences for desired functions. The size of current sequence data and its projected growth point toward the possibility of a general purpose generative model that can condense the totality of sequence statistics, internalizing and integrating fundamental chemical and biological concepts including structure, function, activity, localization, binding, and dynamics, to generate new sequences that have not been seen before in nature but that are biologically active.

Pre-trained models

Transformer models and baselines are available at: <https://github.com/facebookresearch/esm>

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在下游任务中表现更好。变换器在基准测试中超越了具有相似能力的长短期记忆网络。增加训练数据的多样性会显著改善表示。

我们研究的蛋白质语言模型的规模与文本领域使用的模型相当, 但我们的实验尚未达到规模的极限。我们观察到, 即使是我们训练的最高容量模型 (大约有 650-700M 参数) 也因模型容量不足而对序列数据集存在欠拟合。我们发现的语言建模保真度与表示中编码的结构信息之间的关系表明, 容量更大的模型将产生更好的表示。这些发现暗示了进一步增加模型规模和数据多样性的潜力, 包括来自宏基因组学的序列。

将高容量生成模型与基因合成和高通量表征结合起来, 可以实现生成生物学。我们训练的模型可以用来生成新的序列 (Wang & Cho, 2019)。如果神经网络能够将从蛋白质序列中学习的知识转移到设计功能性蛋白质, 这可以与预测模型结合, 联合生成和优化具有期望功能的序列。目前的序列数据规模及其预计增长指向了一个通用生成模型的可能性, 该模型可以凝练所有序列统计信息, 内化并整合基本的化学和生物概念, 包括结构、功能、活性、定位、结合和动态, 以生成在自然界中未曾见过但具有生物活性的新的序列。

预训练模型

变换器模型和基准可在以下地址获取: <https://github.com/facebookresearch/esm>

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A. Approach & Data

1. Pre-training datasets

UniParc pre-training dataset. A series of development models are trained on UniParc (The UniProt Consortium, 2007) Release 2019_01 which contains approximately 250M sequences. 1M sequences are held-out randomly for validation. These models were used in the preprint of this paper, and representations from the models are used in Figures 1, 2, and 3.

UniRef pre-training datasets. Datasets are based on UniRef (Suzek et al., 2015) dated March 28, 2018 to permit a temporal hold-out with CASP13. 10% of UniRef50 clusters are randomly selected as a held-out evaluation set, yielding 3.02 million representative sequences for evaluation. Three training datasets are used, removing all sequences belonging to clusters selected for the evaluation set: (i) UR100, 124.9M UniRef100 representative sequences; (ii) UR50/S, 27.1M UniRef50 representative sequences; (iii) UR50/D, 124.9M UniRef50 cluster members sampled evenly by cluster. To ensure a deterministic validation set, we removed sequences longer than 1024 amino acids from the validation set.

2. Downstream tasks

Remote Homology. A dataset of remote homolog pairs is derived from SCOPe (Fox et al., 2014) containing 256,806 pairs of remote homologs at the fold level and 92,944 at the superfamily level, consisting of 217 unique folds and 366 unique superfamilies. Creation of the dataset is detailed below in the section on remote homology.

Linear projections. Five-fold cross validation datasets implementing structural hold-outs at the family, superfamily, and fold level are constructed using SCOPe (Fox et al., 2014). Independently for each level of structural hold-out, the domains are split into 5 equal sets, i.e. five sets of folds, superfamilies, or families. This ensures that for each of the five partitions, structures having the same classification do not appear in both the train and test sets. For a given classification level each structure appears in a test set once, so that in the cross validation experiment each of the structures will be evaluated exactly once. Scores reported are the mean and standard deviation over each of the five test sets. Further details on construction of the dataset are given below in the section on linear projections.

Secondary structure prediction. All downstream models are trained using the Netsurf (Klausen et al., 2019) training dataset containing 10,837 examples with labels and HMM profiles. Netsurf features are replicated for CASP13 domains using MMseqs2 (Steinegger & Söding, 2017) on

the Uniclust90 (Mirdita et al., 2017) dataset released April 2017. For test sets we use (i) the standard CB513 (Cuff & Barton, 1999) test set of 513 sequences with sequence identity hold-out at 25% identity; and (ii) the 34 publicly available CASP domains, using DSSP (Kabsch & Sander, 1983) to label secondary structure, with temporal hold-out for both pre-training and downstream data.

Contact prediction. All downstream models are trained using the training and test sets of Wang et al. (2017). Comparisons with RaptorX features use features from Wang et al. (2017) and Xu (2018). The following test sets are used: (i) RaptorX Test, 500 domains (25% sequence identity hold-out); (ii) CASP11, 105 domains (25% sequence identity hold-out); (iii) CASP12, 55 domains (temporal hold-out from training data but not pre-training data); (iv) CASP13, 34 publicly released domains (temporal hold-out from training data and pre-training data). The training set consists of 6,767 sequences with contact map targets, a subset of PDB created in February 2015 (Wang et al., 2017). The use of an earlier version of the PDB ensures a temporal hold-out w.r.t. both CASP12 and CASP13. Additionally, Wang et al. (2017) implemented a sequence identity hold-out for Test and CASP11 by removing proteins from the training set which share >25% sequence identity or have BLAST E-value <0.1 with the proteins in these test sets.

Mutational effect prediction. The model is fine-tuned on deep mutational scanning datasets compiled by Gray et al. (2018) and Riesselman et al. (2018).

3. The Transformer

We use a deep Transformer encoder model (Vaswani et al., 2017b; Devlin et al., 2018b), processing input as character sequences of amino acids. In contrast to recurrent and convolutional neural networks, the Transformer makes no assumptions on the ordering of the input and instead uses position embeddings. Particularly relevant to protein sequences is the Transformer’s ability to model long range dependencies, which are not effectively captured by RNNs or LSTMs (Khandelwal et al., 2018). A key factor affecting the performance of LSTMs on these tasks is the path lengths that must be traversed by forward activation and backward gradient signals in the network (Hochreiter et al., 2001).

It is well known that structural properties of protein sequences are reflected in long-range dependencies. Direct coupling analysis (Lapedes et al., 1999; Thomas et al., 2008; Weigt et al., 2009) which aims to detect pairwise dependencies in multiple sequence alignments uses a Markov Random Field (Potts Model) which models the complete sequence with pairwise coupling parameters. Similarly, the Transformer builds up a representation of a sequence by alternating self-attention with non-linear projections. Self-

A. 方法与数据

1. 预训练数据集 UniParc 预训练数据集。一系列开发模型在 UniParc上进行训练 (UniProt联盟, 2007) 版本 2019_01, 其中大约包含 2.5 亿个序列。100 万个序列随机保留用于验证。这些模型用于本文的预印本, 并且模型的表示用于图 1、图 2 和图 3。**UniRef 预训练数据集。**数据集基于 UniRef (Suzek等, 2015年) 定于2018年3月28日, 以便与 CASP13 进行时间上的保留。10%的 UniRef50 簇被随机选择作为保留的评估集, 生成 302 万个代表性序列用于评估。使用了三个训练数据集, 移除了属于选定评估集的簇的所有序列: (i) UR100, 1.249亿个 UniRef100 代表性序列; (ii) UR50/S, 2710万个 UniRef50 代表性序列; (iii) UR50/D, 124.9M UniRef50 簇成员按簇均匀抽样。为确保确定性的验证集, 我们移除了验证集中长度超过 1024 个氨基酸的序列。

2. 下游任务

远程同源。一个远程同源对的数据集源自 SCOPe (Fox等, 2014), 包含 256,806 对远程同源, 在折叠水平上有 92,944 对, 在超家族水平上包含 217 个独特的折叠和 366 个独特的超家族。数据集的创建细节如下在远程同源部分。

线性投影。使用 SCOPe (Fox等, 2014) 构建的五折交叉验证数据集实现了在家族、超家族和折叠水平上的结构性保留。对于每一层结构性保留, 领域被分成 5 个相等的集合, 即五组折叠、超家族或家族。这确保了在每一个五个分区中, 具有相同分类的结构不会出现在训练集和测试集中。在给定的分类水平下, 每个结构在测试集中出现一次, 因此在交叉验证实验中每个结构将被准确评估一次。报告的分数是每五个测试集的均值和标准差。有关数据集构建的进一步细节在下面的线性投影部分中给出。

二级结构预测。所有下游模型都使用Netsurf (Klausen等, 2019) 训练数据集, 该数据集包含 10,837个带标签和HMM配置文件的示例。Netsurf特征通过MMseqs2 (Steinegger & Söding, 2017) 在 CASP13领域中进行复制。

Uniclust90 (Mirdita等,2017) 数据集于2017年4月发布。对于测试集, 我们使用 (i) 标准的 CB513 (Cuff与 Barton,1999) 测试集, 该集包含 513 个序列, 序列相似度在 25% 的身份处被保留; 以及 (ii) 34个公开可用的 CASP 域, 使用 DSSP (Kabsch与 Sander,1983) 来标记二级结构, 预训练和下游数据均进行时间上的保留。

接触预测。所有下游模型都是使用王等(2017) 的训练集和测试集进行训练的。与 RaptorX 特征的比较使用了来自王等(2017) 和徐 (2018) 的特征。使用以下测试集: (i) RaptorX 测试, 500 个领域 (25% 序列相似性保留); (ii) CASP11, 105 个领域 (25% 序列相似性保留); (iii) CASP12, 55 个领域 (从训练数据中按时间保留但不是从预训练数据); (iv) CASP13, 34 个公开发表的领域 (从训练数据和预训练数据中按时间保留)。训练集由 6,767 个具有接触图目标的序列组成, 是在 2015年2月创建的 PDB 的一个子集 (王等, 2017)。使用早期版本的 PDB 确保了相对于 CASP12 和 CASP13 的时间保留。此外, 王等(2017) 为测试和 CASP11 实施了序列相似性保留, 通过从训练集中移除与这些测试集中蛋白质共享 >25% 序列相似性或具有 BLAST E 值 <0.1 的蛋白质。

突变效应预测。该模型在Gray等人(2018)和 Riesselman等人(2018)编纂的深度突变扫描数据集上进行了微调。

3. 变换器

我们使用一种深度变换器编码模型 (Vaswani等人, 2017b; Devlin等人, 2018b), 将输入处理为氨基酸的字符序列。与递归和卷积神经网络不同, 变换器对输入的排序没有假设, 而是使用位置嵌入。与蛋白质序列特别相关的是变换器建模长距离依赖的能力, 而这一点在RNN或LSTM中无法有效捕获 (Khandelwal等人, 2018)。影响LSTM在这些任务上性能的一个关键因素是网络中必须通过前向激活和反向梯度信号遍历的路径长度 (Hochreiter等人, 2001)。

众所周知, 蛋白质序列的结构特性反映在长距离依赖中。直接耦合分析 (Lapedes等人, 1999; Thomas等人, 2008; Weigt等人, 2009) 旨在检测多个序列比对中的成对依赖, 采用马尔可夫随机场 (Potts模型), 该模型用成对耦合参数来模拟完整序列。类似地, 变换器通过交替使用自注意力和非线性投影来构建序列的表示。自-

attention structures computation so that each position is represented by a weighted sum of the other positions in the sequence. The attention weights are computed dynamically and allow each position to choose what information from the rest of the sequence to integrate at every computation step.

Developed to model large contexts and long range dependencies in language data, self-attention architectures currently give state-of-the-art performance on various natural language tasks, mostly due to the Transformer’s scalability in parameters and the amount of context it can integrate (Devlin et al., 2018b). The tasks include token-level tasks like part-of-speech tagging, sentence-level tasks such as textual entailment, and paragraph-level tasks like question-answering.

Scaled dot-product attention. Self-attention takes a sequence of vectors $(h_1, \dots, h_n)$ and produces a sequence of vectors $(h'_1, \dots, h'_n)$ by computing interactions between all elements in the sequence. The Transformer model uses scaled dot-product attention (Vaswani et al., 2017b):

$$A(h) = \text{softmax}(1/\sqrt{d} \, Q(h)K(h)^T)V(h) \quad (2)$$

Here the query Q , key K , and value V , are projections of the input sequence to $n \times d$ matrices where n is the length of the sequence and d is the inner dimension of the matrix outer product between Q and K . This outer product parameterizes an $n \times n$ map of attention logits, which are rescaled, and passed through the softmax function row-wise, thereby representing each position of the sequence in the output as a convex combination of the sequence of values V . One step of self-attention directly models possible pairwise interactions between all positions in the sequence simultaneously. Note the contrast to recurrent and convolutional models which can only represent long-range context through many steps, and the parallel inductive bias with the explicit pairwise parameterization of Markov Random Fields in widespread use for modeling protein MSAs.

Multi-headed self-attention concatenates the output of t independent attention heads:

$$A_{\text{MH}}(x) = A_1(x) \dots A_t(x) \quad (3)$$

Use of multiple heads enables representation of different inter-position interaction patterns.

Architecture The Transformer models (Vaswani et al., 2017b) in this work take a sequence of tokens $(x_1, \dots, x_n)$ and output a sequence of log probabilities $(y_1, \dots, y_n)$ which are optimized using the masked language modeling objective. The computation proceeds through a series of residual blocks producing hidden states, each a sequence of vectors $(h_1, \dots, h_n)$ with embedding dimension d .

The Transformer model architecture consists of a series of encoder blocks interleaving two functions: a multi-headed self-attention computing position-position interactions across the sequence, and a feed-forward network applied independently at each position.

The attention unit:

$$U_{\text{AT}}(h) = P(A_{\text{MH}}(n(h))) \quad (4)$$

Applies one step of multi-headed scaled dot-product attention to the normalized input, denoted by $n(x)$, projecting the result into the residual path.

The feed-forward network (with the output state of P_1 defining the “MLP dimension”):

$$U_{\text{FF}}(h) = P_2(g(P_1(n(h)))) \quad (5)$$

Passes the normalized input through a position-independent multi-layered perceptron (MLP) with activation function $g(x)$.

The full Transformer block:

$$\begin{aligned} B(h) : \\ h &\leftarrow h + U_{\text{AT}}(h) \\ h &\leftarrow h + U_{\text{FF}}(h) \end{aligned}$$

Successively applies the self-attention unit, and the feed-forward network on a residual path.

The Transformer model:

$$\begin{aligned} \text{Transformer}(x) : \\ h &\leftarrow E(x) + H(x) \\ h &\leftarrow B_k(h) \text{ for } k \in 1 \dots K \\ y &\leftarrow W(h) \end{aligned}$$

Consists of an embedding step with token $E(x)$ and positional $H(x)$ embeddings, followed by K layers of Transformer blocks, before a projection W to log probabilities. The raw input sequence is represented as a sequence of 1-hot vectors of dimension 25, which is passed through $E(x)$ the learned embedding layer before being presented to the first Transformer layer.

The models trained in the paper use pre-activation blocks (He et al., 2016), where the layer normalization (Ba et al., 2016) is applied prior to the activation as in Radford et al. (2019b), enabling stable training of deep Transformer networks. No dropout is used. All projections include biases, except for the token and positional embeddings. We use learned token embeddings, and harmonic positional embeddings as in (Vaswani et al., 2017b). The feed-forward network uses the Gaussian error linear unit (Hendrycks & Gimpel, 2016) activation function. We initialize all layers from a zero centered normal distribution with standard

注意力结构计算使得序列中的每个位置都通过其他位置的加权总和来表示。注意力权重动态计算，允许每个位置选择在每个计算步骤中要整合的来自序列其余部分的信息。

为了建模语言数据中的大型上下文和长程依赖，自注意力架构目前在各种自然语言任务上表现出最先进的性能，这主要得益于Transformer在参数和整合上下文的能力上的可扩展性（Devlin等，2018b）。这些任务包括词级任务，如词性标注，句子级任务，如文本蕴含，以及段落级任务，如问答。

缩放点积注意力。自注意力接收一系列向量 $(h_1, \dots, h_n)$ 并生成一系列向量 $(h'_1, \dots, h'_n)$ ，通过计算序列中所有元素之间的交互。Transformer模型使用缩放点积注意力（Vaswani等，2017b）：

$$A(h) = \text{softmax}(1/\sqrt{d} \, Q(h)K(h)^T)V(h) \quad (2)$$

这里查询 Q 、键 K 和值 V 是输入序列投影到 $n \times d$ 矩阵的结果，其中 n 是序列的长度， d 是 Q 和 K 之间矩阵外积的内维度。这个外积参数化了一个 $n \times n$ 注意力 logits 的映射，这些 logits 被重新缩放，并逐行通过 softmax 函数处理，从而在输出中将序列的每个位置表示为值序列 V 的凸组合。自注意力的一步直接建模了序列中所有位置之间的可能成对交互。请注意与递归和卷积模型的对比，后者只能通过多步表示长程上下文，并且与马尔可夫随机场的显式成对参数化的归纳偏倚的平行性相似，后者在蛋白质MSA建模中得到了广泛应用。

多头自注意力将独立注意力头的输出连接起来 t :

$$A_{\text{MH}}(x) = A_1(x) \dots A_t(x) \quad (3)$$

使用多个头可以表示不同的跨位置交互模式。

架构Transformer模型（Vaswani等人，2017b）在本研究中接收一系列标记 $(x_1, \dots, x_n)$ 并输出一系列对数概率 $(y_1, \dots, y_n)$ ，这些概率是使用掩蔽语言建模目标进行优化的。计算过程通过一系列残差块进行，产生隐藏状态，每个状态是一个向量序列 $(h_1, \dots, h_n)$ ，其嵌入维度为 d_o 。

Transformer模型架构由一系列编码器块组成，交替进行两个功能：一个多头自注意力机制计算序列中的位置-位置交互，以及一个在每个位置独立应用的前馈网络。

注意力单元:

$$U_{\text{AT}}(h) = P(A_{\text{MH}}(n(h))) \quad (4)$$

对归一化输入应用一步多头缩放点积注意力，记作 $n(x)$ ，并将结果投影到残差路径中。

前馈网络（输出状态 P_1 定义了“多层感知机维度”）:

$$U_{\text{FF}}(h) = P_2(g(P_1(n(h)))) \quad (5)$$

将归一化输入通过一个位置无关的多层感知器（MLP），使用激活函数 $g(x)$ 。

完整的变换器块:

$$\begin{aligned} B(h) : \\ h &\leftarrow h + U_{\text{AT}}(h) \\ h &\leftarrow h + U_{\text{FF}}(h) \end{aligned}$$

依次在残差路径上应用自注意力单元和前馈网络。

变换器模型:

$$\begin{aligned} \text{Transformer}(x) : \\ h &\leftarrow E(x) + H(x) \\ h &\leftarrow B_k(h) \text{ for } k \in 1 \dots K \\ y &\leftarrow W(h) \end{aligned}$$

由带有token $E(x)$ 的嵌入步骤和位置 $H(x)$ 嵌入构成，之后是 K 层变换器块，最后进行投影 W 以获得对数概率。原始输入序列表示为维度为25的1-hot向量序列，该序列通过学习到的嵌入层 $E(x)$ 传递，然后呈现给第一个变换器层。

本文中训练的模型使用激活前块（He等人，2016年），在激活之前应用层归一化（Ba等人，2016年），正如Radford等人(2019b)中所述，从而实现深层变换器网络的稳定训练。不使用dropout。所有投影都包括偏置，除了标记和位置嵌入。我们使用学习的标记嵌入和谐波位置嵌入，如（Vaswani等人，2017b）所示。前馈网络使用高斯误差线性单元（Hendrycks&Gimpel，2016年）激活函数。我们从零中心正态分布初始化所有层，标准

deviation 0.02, and re-scale the initialization of the projections into the residual path by $1/\sqrt{L}$ where L is the number of residual layers. All biases are initialized to zero. The query, key, and value projections are to d dimensions, and the hidden dimension of the feed-forward network is $4d$.

4. Pre-trained Transformer Models

UniParc development models We experimented with Transformer models of various depths, including a 36-layer Transformer with 708.6 million parameters, and a 12-layer model with 85.1M parameters trained. Development models were trained on UniParc. Details are in [Table A.1](#).

UniRef models We train 34-layer models with 669.2M parameters across different datasets and fractions of training data. Additionally we train 6 and 12-layer models. These models are detailed in [Table A.2](#).

ESM-1b The ESM-1b hyperparameter sweep and model is described in detail in [Appendix B](#). In brief, ESM-1b is the result of an extensive hyperparameter sweep that was performed on smaller 12-layer models. ESM-1b is the result of scaling up that model to 33 layers. Compared to the Uniref models, the main changes in ESM-1b are: higher learning rate; dropout after word embedding; learned positional embeddings; final layer norm before the output; and tied input/output word embedding.

Pre-training task The masked language modeling pre-training task follows [Devlin et al. \(2018b\)](#). Specifically, we select as supervision 15% of tokens randomly sampled from the sequence. For those 15% of tokens, we change the input token to a special “masking” token with 80% probability, a randomly-chosen alternate amino acid token with 10% probability, and the original input token (i.e. no change) with 10% probability. We take the loss to be the whole batch average cross entropy loss between the model’s predictions and the true token for these 15% of amino acid tokens. In contrast to [Devlin et al. \(2018b\)](#), we do not use any additional auxiliary prediction losses. The ESM-1b models, as well as the UniParc development models used in visualizations and in the supplemental results are trained with the masking procedure above. The UniRef models used across the experiments of the main text are trained similarly, except that for the 15% of tokens selected as prediction targets, all are replaced by the mask token.

Pre-training details Our model was pre-trained using a context size of 1024 tokens. As most Uniparc sequences (96.7%) contain fewer than 1024 amino acids, the Transformer is able to model the entire context in a single model pass. For those sequences that are longer than 1024 tokens, we sampled a random crop of 1024 tokens during

each training epoch. The model was optimized using Adam ($\beta_1 = 0.9, \beta_2 = 0.999$) with learning rate 10^{-4} . We trained with 131,072 tokens per batch (128 gpus x 1024 tokens). The models follow a warm-up period of 16000 updates, during which the learning rate increases linearly. Afterwards, the learning rate follows an inverse square root decay schedule. All models were trained using the fairseq toolkit ([Ott et al., 2019](#)) on 128 NVIDIA V100 GPUs.

5. Evaluating the models for downstream tasks

After pre-training the model with unsupervised learning, we can adapt the parameters to supervised tasks. By passing the input sequence $(x_1, \dots, x_n)$ through our pre-trained model, we obtain a final vector representation of the input sequence $(h_1, \dots, h_n)$. During pre-training, this representation is projected to log probabilities $(y_1, \dots, y_n)$. Recall that a softmax over y_i represents the model’s posterior for the amino acid at position i . These final representations $(h_1, \dots, h_n)$ are used directly, or fine-tuned in a task-dependent way by adding additional layers to the model and allowing the gradients to backpropagate through the weights of the pre-trained model to adapt them to the new task. Hidden representations from intermediate layers rather than the final layer can also be used.

6. Language Modeling Baselines

In addition to comparing to past work, we also implemented deep learning baselines for our experiments.

Frequency (n-gram) models To establish a meaningful performance baseline on the sequence modeling task ([Section 3](#)), we construct n-gram frequency-based models for context sizes $1 \leq n \leq 10^4$, applying optimal Laplace smoothing for each context size. The Laplace smoothing hyperparameter in each case was tuned on the validation set. ECE is reported for the best left-conditioning n-gram model.

Bidirectional LSTM language models We trained state-of-the-art LSTM ([Hochreiter & Schmidhuber, 1997](#)) language models on the UR50 dataset. We use the ELMo model of [Peters et al. \(2018\)](#) which concatenates two independent autoregressive language models with left-to-right and right-to-left factorization. Unlike standard LSTM language models, the ELMo model receives context in both directions and is therefore comparable to the Transformers we train that also use the whole context of the sequence. We train two models: (i) the small model has approximately 28.4M parameters across 3 layers, with an embedding dimension of 512 and a hidden dimension of 1024; (ii) the large model has approximately 113.4M parameters across 3 layers, with an embedding dimension of 512 and a hidden dimension of 4096. The models are trained with a nomi-

偏差 0.02, 并通过 $1/\sqrt{L}$ 将投影的初始化重新缩放到残差路径中, 其中 L 是残差层的数量。所有偏差初始化为零。查询、键和值的投影为 d 维度, 前馈网络的隐藏维度为 $4d$ 。

4. 预训练的 Transformer 模型 UniParc 开发模型

我们实验了各种深度的 Transformer 模型, 包括一个有 7.086 亿参数的 36 层 Transformer 和一个有 8510 万参数的 12 层模型。开发模型是在 UniParc 上训练的。详细信息见[表 A.1](#)。**UniRef 模型** 我们训练了 34 层模型, 具有 6.692 亿参数, 跨不同的数据集和训练数据的分数。此外, 我们还训练了 6 层和 12 层模型。这些模型的详细信息见[表 A.2](#)。**ESM-1b** ESM-1b 超参数搜索和模型的详细描述见[附录 B](#)。简而言之, ESM-1b 是在较小的 12 层模型上进行广泛超参数搜索的结果。ESM-1b 是将该模型扩展到 33 层的成果。与 UniRef 模型相比, ESM-1b 的主要变化有: 更高的学习率; 词嵌入后的 dropout; 学习的位置信息嵌入; 输出前的最终层归一化; 以及绑定的输入/输出 词嵌入。**预训练任务** 掩码语言建模预训练任务遵循[Devlin 等人 \(2018 年\)](#)。具体而言, 我们选择随机采样自序列的 15% 令牌作为监督。对于这 15% 的令牌, 我们以 80 % 的概率将输入令牌改为特殊的“掩码”令牌, 以 10% 的概率改为随机选择的替代氨基酸令牌, 并以 10% 的概率保留原始输入令牌 (即不改变)。我们将损失定义为模型预测与这 15% 氨基酸令牌真实值之间的整个批次平均交叉熵损失。与[Devlin 等人 \(2018 年\)](#)不同, 我们不使用任何额外的辅助预测损失。ESM-1b 模型, 以及在可视化和补充结果中使用的 UniParc 开发模型, 都是采用上述掩码程序进行训练的。在主文本的实验中使用的 UniRef 模型类似地进行训练, 只是对于被选为预测目标的 15% 令牌, 所有均被替换为掩码令牌。**预训练细节** 我们的模型是在 1024 令牌的上下文大小下进行预训练的。由于大多数 UniParc 序列 (96.7%) 包含少于 1024 个氨基酸, Transformer 可以在一次模型传递中建模整个上下文。对于那些超过 1024 令牌的序列, 我们在训练过程中随机采样了 1024 令牌的裁剪。

每个训练周期。模型使用 Adam($\beta_1 = 0.9, \beta_2 = 0.999$) 的学习率为 10^{-4} 。我们以每批次 131,072 个标记进行训练 (128 个图形处理单元 x 1024 个标记)。模型经历 16000 次更新的预热期, 在此期间学习率线性增加。之后, 学习率遵循反平方根衰减调度。所有模型都是使用 fairseq 工具包 ([Ott 等人, 2019](#)) 在 128 个 NVIDIA V100 图形处理单元上进行训练。

5. 评估下游任务的模型 在使用无监督学习对模型进行预训练之后, 我们可以将参数调整为监督任务。通过将输入序列 $(x_1, \dots, x_n)$ 传递给我们的预训练模型, 我们获得了输入序列的最终向量表示 $(h_1, \dots, h_n)$ 。在预训练期间, 这个表示被投影为对数概率 $(y_1, \dots, y_n)$ 。请注意, 对 y_i 的 softmax 表示模型在位置 i 的氨基酸后验。这些最终表示 $(h_1, \dots, h_n)$ 可以直接使用, 或者通过向模型添加额外层以任务相关的方式进行微调, 并允许梯度在预训练模型的权重中反向传播, 以将其适应新任务。也可以使用来自中间层的隐藏表示, 而不是最终层。

6. 语言建模基线除了与过去的工作进行比较外, 我们还为我们的实验实施了深度学习基线。**频率 (n-gram) 模型** 为在序列建模任务上建立一个有意义的性能基线 ([第 3 节](#)), 我们为上下文大小构建了基于 n-gram 的频率模型 $1 \leq n \leq 10^4$, 为每个上下文大小应用最佳的拉普拉斯平滑。每种情况下, 拉普拉斯平滑超参数都是在验证集上调整的。最佳左条件 n-gram 模型的 ECE 被报告。**双向长短期记忆网络语言模型** 我们在 UR50 数据集上训练了最先进的长短期记忆网络 ([Hochreiter & Schmidhuber, 1997](#)) 语言模型。我们使用了 [Peters 等\(2018\)](#) 的 ELMo 模型, 该模型将两个独立的自回归语言模型与从左到右和从右到左的分解连接在一起。与标准的长短期记忆网络语言模型不同, ELMo 模型可以接收来自两个方向的上下文, 因此可以与我们训练的变换器相比较, 这些变换器同样使用序列的整个上下文。我们训练了两个模型: (i) 小型模型大约有 2840 万个参数, 跨越 3 层, 嵌入维度为 512, 隐层维度为 1024; (ii) 大型模型大约有 1.134 亿个参数, 跨越 3 层, 嵌入维度为 512, 隐层维度为 4096。模型的训练为一个名义-

# Layers	# Heads	Embedding Dim	MLP Dim	# Params	Steps
12	12	768	3072	85.1M	1.6M
24	12	768	3072	170.2M	300k
36	20	1280	5120	708.6M	300k

Table A.1. Hyperparameters for development Transformer models trained on UniParc. Embedding dim is the dimension of the hidden states at the output of each transformer block. MLP Dim refers to the width of hidden layer P_1 in the Transformer’s MLPs.

# Layers	# Heads	Embedding Dim	MLP Dim	# Params	Data	Steps
6	12	768	3072	42.6M	UR50/S	840K
12	12	768	3072	85.1M	UR50/S	840K
34	20	1280	5120	669.2M	UR100	275K
34	20	1280	5120	669.2M	UR50/S	840K
34	20	1280	5120	669.2M	UR50/D	906K

Table A.2. Hyperparameters for UniRef Transformer models. Note: UR100 model stopped making significant progress on valid loss and was stopped at 275K updates.

nal batch size of 32,768, with truncated backpropagation to 100 tokens, dropout of 0.1, learning rate of 8e-4, using the Adam optimizer with betas of (0.9, 0.999), clip norm 0.1 and warmup of 1500 updates using an inverse square root learning rate schedule. We searched across a range of learning rates and found 8e-4 to be optimal.

7. Metric structure experiments

Dataset An orthologous group dataset was constructed from eggNOG 5.0 (Huerta-Cepas et al., 2018) by selecting 25 COG orthologous groups toward maximizing the size of the intersected set of species within each orthologous group. Through a greedy algorithm, we selected 25 COG groups with an intersecting set of 2,609 species. We shrank the dataset above by selecting only one species from each of 24 phyla in order to ensure species-level diversity.

8. Remote Homology

Dataset We used the database of SCOP 2.07e filtered to 40% sequence similarity, provided by the ASTRAL compendium (Fox et al., 2014). Following standard practices (Söding & Remmert, 2011), we exclude folds that are known to be related, specifically Rossmann-like folds (c.2-c.5, c.27 and 28, c.30 and 31) and four- to eight-bladed β -propellers (b.66-b.70). This yields 256,806 pairs of remote homologs at the fold level and 92,944 at the superfamily level, consisting of 217 unique folds and 366 unique superfamilies. We then perform an 80-20 split, and tune our hyperparameters on the “training set” and report results on the held out 20% of the data.

Metrics Given a protein sequence x , with final hidden representation $(h_1, \dots, h_n)$, we define the embedding of the sequence to be a vector e which is the average of the hidden representations across the positions in the sequence:

$$e = \frac{1}{n} \sum_{i=1}^n h_i$$

We can compare the similarity of two protein sequences, x and x' having embeddings e and e' using a metric in the embedding space.

We evaluate the L2 distance $\|e - e'\|_2$ and the cosine distance $e \cdot e' / \|e\| \|e'\|$. Additionally we evaluated the L2 distance after projecting the e vectors to the unit sphere.

Evaluation To evaluate HHblits (Remmert et al., 2011), first we construct HMM profiles for each sequence using default parameters for ‘hhblits’, except we use 3 iterations. Then, we do an all-to-all alignment using ‘hhalgn’ with default parameters, and use the resulting E-value as a measure of similarity. Given a query sequence, a sequence is more similar with a smaller E-value.

The two metrics reported are Hit-10 as introduced in Ma et al. (2014) and AUC. For both metrics, for each sequence, we treat it as a query and we rank each other sequence according to the distance metric used. Following Ma et al. (2014), for evaluation at the fold level, any domain with the same fold is a positive; any domain with a different fold is a negative; and domains belonging to the same superfamily are excluded. For evaluation at the superfamily level, any domain with the same superfamily is a positive; any domain with a different superfamily is a negative; and domains belonging to the same family are excluded. This ensures

# 层	# 头	嵌入维度	多层感知机维度	# 参数	步骤
12	12	768	3072	8510万	160万
24	12	768	3072	1702万	30万
36	20	1280	5120	7086万	30万

表 A.1. 在 UniParc 上训练的开发 Transformer 模型的超参数。嵌入维度是每个变换器块输出的隐藏状态的维度。MLP维度指的是Transformer 的多层感知器中隐藏层的宽度 P_1 。

# 层数	# 头数	嵌入维度	多层感知机维度	# 参数数	数据	步骤
6	12	768	3072	4260万	UR50/S	84万
12	12	768	3072	8510万	UR50/S	840K
34	20	1280	5120	669.2M	UR100	275K
34	20	1280	5120	669.2M	UR50/S	840K
34	20	1280	5120	669.2M	UR50/D	906K

表 A.2. UniRef Transformer模型的超参数。注意：UR100模型在验证损失上停止了显著进展，并在275K更新时停止。

最终批量大小为32,768, 使用截断反向传播到100个标记, dropout为0.1, 学习率为8e-4, 使用带有 (0.9, 0.999)的Adam优化器, clip norm 使用逆平方根学习率计划进行 1500 次更新的 0.1 和热身。我们在一系列学习率中进行了搜索, 发现 8e-4 是最佳选择。

7. 度量结构实验

数据集构建了一个同源基因组数据集 来自 eggNOG 5.0 (Huerta-Cepas 等, 2018) 通过选择 25 个 COG 同源基因组, 旨在最大化每个同源基因组内涉及物种的交集集合的大小。通过贪心算法, 我们选择了 25 个 COG 组, 交集集合中包含 2,609 个物种。我们通过从每个 24

个门中选择一个物种, 从而缩小了上述数据集, 以确保物种层次的多样性。

8. 远程同源

数据集我们使用了由 SCOP 2.07e 筛选的数据库以 40% 的序列相似性提供, 由 ASTRAL 提供文献汇编 (Fox等, 2014) . 根据标准做法 (Söding & Remmert, 2011) , 我们排除了已知相关的折叠, 特别是Rossmann样折叠 (c.2-c.5, c.27和28, c.30和31) 和四翼至八翼 β -螺旋桨 (b.66-b.70) 。这产生了256,806个远程同源序列的折叠级对和92,944个超家族级对, 包含217个独特折叠和366个独特超家族。然后我们进行80-20的拆分, 在“训练集”上调整超参数, 并在保留的20%的数据上报告结果。

指标 给定一个蛋白质序列 x , 其最终隐藏表示为 $(h_1, \dots, h_n)$, 我们定义该序列的嵌入为一个向量 e , 它是序列中各位置隐藏表示的平均值:

$$e = \frac{1}{n} \sum_{i=1}^n h_i$$

我们可以比较两个蛋白质序列的相似性, x 并且 x' 拥有嵌入 e 和 e' , 使用嵌入空间中的一个度量。我们评估 L2 距离 $\|e - e'\|_2$ 和余弦距离 e

$\cdot e' / \|e\| \|e'\|$ 此外, 我们在将向量投影到单位球面后评估了 L2 距离。 e 评估HHblits (Remmert et al., 2011)的第一步是为每个序列使用默认参数构建 HMM 配置文件, 除了我们使用3次迭代。然后, 我们使用 ‘hhalgn’ 进行全对全比对, 采用默认参数, 并将得到的 E 值作为相似性度量。给定一个查询序列, 更小的 E 值表示该序列更为相似。报告的两个度量是 Hit-10, 如Maet al.(2014)中介绍, 并且 AUC。对于这两个度量, 对于每个序列, 我们将其视为查询, 并根据所使用的距离度量对其他序列进行排名。根据Ma et al.(2014)的建议, 对于在折叠级别的评估, 任何具有相同折叠的域被视为正样本; 任何具有不同折叠的域被视为负样本; 并且属于同一超家族的域会被排除。在超家族级别的评估中, 任何具有相同超家族的域被视为正样本; 任何具有不同超家族的域被视为负样本; 并且属于同一家族的域会被排除。这确保了

we specifically measure how well our models do on finding *remote* homologs.

For Hit-10, we consider the query a success if any of the top 10 sequences is a remote homolog. We report the proportion of successes averaged across all queries. For AUC, we first compute the AUC under the ROC curve when classifying sequences by vector similarity to the query. Then, we average the AUC across all query sequences.

We found that cosine similarity results in the best Hit-10 scores, while the L2 with unnormalized vectors result in the best AUC scores, so we report this in Table 2.

Implementation We used the FAISS similarity search engine (Johnson et al., 2017).

9. Representational similarity-based alignment of sequences within MSA families

Family selection We use the Pfam database (Bateman et al., 2013). We first filtered out any families whose longest sequence is less than 32 residues or greater than 1024 residues in length. We then ranked the families by the number of sequences contained in each family and selected the 128 largest families and associated MSAs. Finally, we reduced the size of each family to 128 sequences by uniform random sampling.

Aligned pair distribution For each family, we construct an empirical distribution of aligned residue pairs by enumerating all pairs of positions and indices that are aligned within the MSA and uniformly sampling 50,000 pairs.

Unaligned pair distribution We also construct for each family a background empirical distribution of unaligned residue pairs. This background distribution needs to control for within-sequence position, since the residues of two sequences that have been aligned in an MSA are likely to occupy similar positions within their respective unaligned source sequences. Without controlling for this bias, a difference in the distributions of aligned and unaligned pairs could arise from representations encoding positional information rather than actual context. We control for this effect by sampling from the unaligned-pair distribution in proportion to the observed positional differences from the aligned-pair distribution. Specifically, the following process is repeated for each pair in the empirical aligned distribution:

1. Calculate the absolute value of the difference of each residue’s within-sequence positions in the aligned pair.
2. Select a pair of sequences at random.
3. For that pair of sequences, select a pair of residues at random whose absolute value of positional difference

equals the one calculated above.

4. Verify that the residues are unaligned in the MSA; if so, add the pair to the empirical background distribution.
5. Otherwise, return to step 2.

This procedure suffices to compute a empirical background distribution of 50,000 unaligned residue pairs.

Similarity distributions Finally, for each family and each distribution, we apply the cosine similarity operator to each pair of residues to obtain the per-family aligned and unaligned distribution of representational cosine similarities.

10. Linear projections

Dataset We construct five-fold cross validation datasets with structural hold-outs at the family, superfamily, and fold level using SCOPe 2.07 (Fox et al., 2014). We use the full version of SCOPe 2.07, clustered at 90% sequence identity, generated on January 23, 2020, and extract the domain annotations with labels. There are 19,695 domains. Then, independently for each hold-out level, we split the domains at the hold-out level into 5 equal sets, i.e. five sets of folds, superfamilies, or families. This ensures that for each partition, structures having the same classification do not appear in both the train and test sets. For a given classification level each structure appears in a test set once, so that in the cross validation experiment each of the structures will be evaluated once. Scores reported are the mean and standard deviation over each of the five test sets.

For each domain, we first obtain the sequence x whose residues align with the domain specification. To construct the secondary structure labels, we take each CIF file (pulled from PDB), and run DSSP (Joosten et al., 2010; Kabsch & Sander, 1983). If x has any residues where DSSP has not provided a secondary structure label, we mark them as missing data and do not supervise for those positions.

To construct the contact map, we obtain C_β coordinates from the structure portion of the CIF file (C_α in the case of glycine), defaulting to NaN where information is missing, and finally calculating pairwise distances and thresholding at 8Å. Similar to secondary structure, we do not supervise over NaNs.

We discard any domains where (1) DSSP fails, (2) we are unable to align the sequence to the structure, or (3) the domain is longer than 1023 residues. This leaves 15,297 domains.

For the MSA baselines, we query each sequence against the Uniclust30, 2017 database (Mirdita et al., 2017) with HH-blits (Remmert et al., 2011) using the default settings with

我们特别衡量我们的模型在查找远程 同源物时的表现。

对于 Hit-10, 如果前 10 个序列中有任意一个是远缘同源体, 我们将查询视为成功。我们报告所有查询的成功比例的平均值。对于 AUC, 我们首先在根据向量相似度对序列进行分类时计算 ROC 曲线下的 AUC。然后, 我们计算所有查询序列的 AUC 平均值。

我们发现, 余弦相似度的 Hit-10 分数最好, 而 L2 与未标准化向量的组合结果的 AUC 分数最佳, 因此我们在表 2 中报告了这一结果。

实现我们使用 FAISS 相似性搜索引擎 (约翰逊等, 2017 年)。

9. 基于表示相似度的 MSA 家族序列对齐家庭选择 我们使用 PFAM 数据库 (贝特曼等, 2013年)。我们首先过滤掉最长序列少于 32 个残基或大于 1024 个残基的家族。然后, 我们根据每个家族中包含的序列数量对家族进行排名, 并选择 128 个最大的家族及其相关的 MSA。最后, 我们通过均匀随机抽样将每个家族的大小减少到 128 个序列。**已对齐的配对分布**对于每个家族, 我们通过列举所有在 MSA 中对齐的位点和指标的配对, 并均匀抽样 50,000 个配对来构建一个已对齐残基配对的经验分布。**未对齐配对分布**我们还为每个家族构建一个未对齐残基配对的背景经验分布。这个背景分布需要控制序列内的位置, 因为在 MSA 中对齐的两个序列的残基可能在其各自的未对齐源序列中占据相似的位置。如果不控制这种偏差, 已对齐配对和未对齐配对分布的差异可能源于表示编码位置信息而非实际上下文。我们通过根据从已对齐配对分布中观察到的位置信息差异的比例, 从未对齐配对分布中抽样来控制这种效果。具体而言, 以下过程会对经验已对齐分布中的每一对重复进行:

1. 计算对齐对中每个残基的序列内位置的绝对值差异。
2. 随机选择一对序列。
3. 对于这一对序列, 随机选择一对残基, 其位置差的绝对值

等于上面计算的值。

4. 验证残基在MSA中是否未对齐; 如果是, 则将该对添加到经验背景分布中。

5. 否则, 返回第2步。此过程足以计算50,000对未对齐残基的经验背景分布。**相似性分布**最后, 对于每个家族和每个分布, 我们应用余弦相似度操作符到每对残基, 以获取每个家族的对齐和未对齐的表示余弦相似度分布。

10. 线性投影数据集 我们构建了五折交叉验证数据集, 在家族、超家族和折叠层次上使用SCOPe 2.07 (福克斯等, 2014)。我们使用SCOPe 2.07的完整版本, 聚类在90%的序列相似性上, 于2020年1月23日生成, 并提取带标签的领域注释。共有19,695个领域。随后, 我们独立地将每个保持级别的领域拆分为5个相等的集合, 即五组折叠、超家族或家族。这确保对于每个划分, 具有相同分类的结构不会同时出现在训练集和测试集中。在给定的分类级别中, 每个结构仅在测试集中出现一次, 因此在交叉验证实验中, 每个结构将被评估一次。报告的分数是五个测试集的均值和标准差。对于每个领域, 我们首先获取与领域规范对齐的序列 x 。为了构建二级结构标签, 我们提取每个CIF文件 (从PDB中获取) 并运行DSSP (Joosten等, 2010;Kabsch与Sander, 1983)。如果 x 在DSSP未提供二级结构标签的任何残基上, 我们将其标记为缺失数据, 并不对此位置进行监督。为了构建接触图, 我们从CIF文件的结构部分获取 C_β 坐标 (在甘氨酸的情况下为 C_α), 在信息缺失处默认显示为NaN, 最终计算成对距离并在8Å处阈值化。与二级结构类似, 我们不对NaNs进行监督。我们丢弃任何 (1) DSSP失败, (2) 我们无法将序列与结构对齐, 或 (3) 领域超过1023个残基的领域。这留下了15,297个领域。对于MSA基准测试, 我们将每个序列与Uniclust30, 2017数据库 (Mirdita等, 2017) 使用HH-blits (Remmert等, 2011) 的默认设置进行查询。

additional parameters (n=3, 1e-3). For secondary structure prediction, we construct HMM profiles using HHmake (default settings). For contact prediction, we apply CCMpred (Seemayer et al., 2014) implementation of pseudolikelihood maximization (Balakrishnan et al., 2011; Ekeberg et al., 2013) using default settings with (n=500) to each MSA, from which we extract both an output matrix ($\mathbb{R}^{L \times L}$), as well as a sequence profile ($\mathbb{R}^{L \times K}$) where L is the length of the sequence and K is the size of the amino acid vocab, i.e. 25.

Representations To obtain sequence representations, we provide the sequence of the domain as input to a forward pass of the Transformer model. We retrieve the activations into the final multi-head attention (after the layer normalization), using this as the matrix of sequence representations ($\mathbb{R}^{L \times d}$) where d is the hidden dimension of the model.

Secondary structure projections For secondary structure, we fit a multi-class logistic regression taking as input an individual representation h_i and as output the secondary structure label from DSSP. We observe that the logistic regression model's performance does not change with penalty settings (L1, L2, no penalty); therefore we report the result where the L2 penalty is applied during training.

Contact projections For contacts, we fit two linear projections. Given two representations h_i, h_j at positions i and j , we regress to whether residues at positions i and j are in contact:

$$P(\text{i,j contact}) = \text{sigmoid}[(Ph_i + p) \cdot (Qh_j + q) + b]$$

With learned projections P, Q , vector biases p, q , and scalar bias b . We use the AdamW optimizer (Loshchilov & Hutter, 2017) to fit the projections and bias term.

For each partition we set aside approximately 12.5% of the training set as validation. We sweep over a range of projection dimensions (32 to 512), learning rates (1e-6 to 1e-2) and weight decay values (0 to 0.9). Based on the best validation Top-L, long-range precision score, we set the projection dimension to 128, the learning rate to 1e-4, and the optimizer weight decay to 0.2. We observe that the precision score does not improve with an increased projection dimension over 128.

The above setup for contact projections only applies to the Transformer models and the sequence profile baseline. No supervision is applied to the CCMpred output.

11. Single-family data and analysis

For each of the three domains used, we extracted all domain sequences from the Pfam dataset (Bateman et al., 2013) and located the subset of PDB files containing the domain, using the latter to derive ground truth secondary structure labels (Kabsch & Sander, 1983).

Pre-training is with the masked language modeling objective, using the same hyperparameters as used to train the UniParc development models. The domain sequences were randomly partitioned into training, validation, and testing datasets. For each family, the training dataset comprises 65,536 sequences, the validation dataset comprises either 16,384 sequences (PF00005 and PF00069) or 8,192 sequences (PF00072), and the test dataset comprises the remainder.

Each Pfam family also forms an evaluation dataset for linear projection; from the sequences with corresponding crystal structures, the training dataset comprises 80 sequences and the test dataset comprises the remainder.

12. Secondary structure prediction. Deep neural networks and feature combination

We use features from the final hidden representations of the models. We removed the final embedding layer, added layer norm, and applied a top-level architecture following (Klausen et al., 2019). In particular, this top-level architecture consists of two parallel convolution layers and an identity layer, whose outputs are concatenated in the feature dimension and fed to a two layer bidirectional LSTM containing 1024 hidden units and dropout $p = 0.5$. The output is then projected to an 8-dimensional feature vector at each position and the model is trained with a categorical cross-entropy loss with the Q8 labels. The training data was obtained from the (Klausen et al., 2019). Secondary structure labels for the CASP13 test set were constructed using DSSP.

In feature combination experiments, we used the features provided by (Klausen et al., 2019) which were generated using MMseqs2 on the Uniclust90 dataset released April 2017. For CASP13 experiments, we generated these features using code provided by (Klausen et al., 2019) on CASP13 domains.

As a baseline, we reimplemented (Klausen et al., 2019) by replacing the Transformer features with the MMseqs2 features and keeping the top-level architecture. For feature combination experiments, we projected (a) the features from this baseline and (b) the features from the Transformer to the same dimension (512 units), concatenated along the feature dimension, and fed the resulting tensor to a two layer bidirectional LSTM with 512 hidden units and dropout $p = 0.3$.

额外参数 (n=3, 1e-3). 对于二级结构预测, 我们使用 HHmake (默认设置) 构建 HMM 轮廓。对于接触预测, 我们应用 CCMpred (塞梅尔等,2014) 的伪似然最大化 (巴拉克里希南等,2011; 埃克伯格等,2013) 使用默认设置和 (n=500) 应用于每个 MSA, 从中我们提取一个输出矩阵($\mathbb{R}^{L \times L}$), 以及一个序列轮廓($\mathbb{R}^{L \times K}$) 其中 L 是序列的长度, K 是氨基酸词汇的大小, 即 25。

表示为了获得序列表示, 我们将域的序列作为输入传递给Transformer模型的前向传递。我们从最终的多头注意力 (在层归一化之后) 中检索激活, 将其作为序列表示的矩阵($\mathbb{R}^{L \times d}$), 其中 d 是模型的隐藏维度。

二级结构投影对于二级结构, 我们拟合一个多类逻辑回归, 以个体表示 h_i 作为输入, 以DSSP的二级结构标签作为输出。我们观察到逻辑回归模型的性能在惩罚设置 (L1、L2、无惩罚) 下没有变化; 因此, 我们报告在训练期间应用L2惩罚时的结果。

接触投影对于接触, 我们拟合两个线性投影。给定在位置 i 和 j 的两个表示 h_i 和 h_j , 我们回归判断是否在位置 i 和 j 的残基是接触的:

$$P(\text{i,j contact}) = \text{sigmoid}[(Ph_i + p) \cdot (Qh_j + q) + b]$$

使用学习到的投影 P, Q , 向量偏置 p, q , 以及标量偏置 b 。我们使用AdamW优化器 (Loshchilov & Hutter, 2017) 来拟合投影和偏置项。

对于每个分区, 我们大约保留12.5%的训练集作为验证。我们在一系列投影维度 (32到512)、学习率 (1e-6到1e-2) 和权重衰减值 (0到0.9) 之间进行搜索。基于最佳验证Top-L和长程精度分数, 我们将投影维度设置为128, 学习率设置为1e-4, 优化器的权重衰减设置为0.2。我们观察到, 当投影维度超过128时, 精度分数并没有改善。

上述接触投影的设置仅适用于Transformer模型和序列配置基线。CCMpred输出没有应用任何监督。

11. 单家族数据和分析对于使用的三个领域, 我们从PFAM数据集 (贝特曼, 2013年) 提取了所有领域序列, 并找到了包含该领域的PDB文件子集, 使用后 者来推导真实值二级结构标签 (Kabsch与Sander, 1983年)。预训练使用掩码语言模型的目标, 使用与训练UniParc开发模型相同的超参数。领域序列被随机划分为训练、验证和测试数据集。每个家族的训练数据集由65,536个序列组成, 验证数据集由16,384个序列 (PF00005和PF00069) 或8,192个序列 (PF00072) 组成, 测试数据集则由剩余部分组成。每个PFAM家族还形成线性投影的评估数据集; 从具有相应晶体结构的序列中, 训练数据集由80个序列组成, 测试数据集由剩余部分组成。

12. 二级结构预测。深度神经网络与特征组合我们使用模型最终隐藏表示的特征。我们移除了最后的嵌入层, 添加了层归一化, 并应用了遵循 (Klausen等, 2019) 的顶层架构。特别地, 这个顶层架构包含了两个并行的卷积层和一个恒等层, 其输出在特征维度上被连接, 并输入到一个包含 1024 个隐藏单元和 dropout 的双层双向 LSTM $p = 0.5$ 。输出随后被映射到每个位置的 8 维特征向量, 并且模型使用带有 Q8 标签的分类交叉熵损失进行训练。训练数据来自于 (Klausen 等, 2019)。CASP13 测试集的二级结构标签是使用 DSSP 构建的。在特征组合实验中, 我们使用了 (Klausen等, 2019) 提供的特征, 这些特征是使用 MMseqs2 在 2017年4月发布的 Uniclust90 数据集上生成的。对于 CASP13 实验, 我们使用 (Klausen等, 2019) 提供的代码在 CASP13 域上生成了这些特征。作为基线, 我们通过将 Transformer 特征替换为 MMseqs2 特征并保持顶层架构来重新实现 (Klausen等, 2019)。对于特征组合实验, 我们将 (a) 来自这个基线的特征和 (b) 来自 Transformer 的特征投影到相同维度 (512 单元), 在特征维度上连接, 并将结果张量输入到一个具有 512 个隐藏单元和 dropout 的双层双向 LSTM $p = 0.3$ 。

To check our dataset construction, we used the pretrained weights provided by (Klausen et al., 2019) and evaluated their model directly in our evaluation pipeline. We were able to reproduce the values reported in (Klausen et al., 2019).

13. Contact prediction. Deep neural networks and feature combination

Data We use the datasets and features distributed with Wang et al. (2017) and Xu (2018). The base features are those used by RaptorX (Xu, 2018) a state-of-the-art method in CASP13, including sequence features, PSSM, 3-state secondary structure prediction, predicted accessibility, one-hot embedding of sequence, and pairwise features APC-corrected Potts model couplings, mutual information, pairwise contact potential.

We use the training, standard test set, and CASP11 test set from Wang et al. (2017). We use the CASP12 test set from Xu (2018). For the CASP13 test set we use the 34 publicly released domains.

Wang et al. (2017) established training and test sets as follows. The train (6,367 proteins), valid (400 proteins) and test (500 proteins) datasets were selected as subsets of PDB25 (each protein having <25% sequence similarity). Proteins having sequence similarity >25% or BLAST E-value <0.1 with any test or CASP11 protein were excluded from training data.

All our MSAs (used for the avg and cov combination methods) are constructed by running HHblits (Remmert et al., 2011) with 3 iterations and E-value 0.001 against Uniprot20 released on 2016-02; except for CASP12 and CASP13 where we used the four different MSAs released with and described in Xu (2018). Note that for the Transformer pre-training UniRef50 from 2018-03 was used; hence no data which was not already available prior to the start of CASP13 was present during either pre-training or contact prediction training.

Model architecture On top of the sequence and pairwise features we use a depth-32 residual network (ResNet) model to predict binary contacts. The ResNet model architecture is similar to Wang et al. (2017) and Xu (2018).

The first component of the ResNet is a learned sequence pipeline $y = f_{\theta}^S(x)$ which maps sequence features $x \in \mathbb{R}^{L \times d_1}$ to $y \in \mathbb{R}^{L \times d_2}$ with L the length of the protein. Though f_{θ}^S could be a 1D convolutional network or residual network as in Wang et al. (2017), we chose our sequence net to be a simple linear projection from input dimension d_1 to $d_2 = 128$ dimensions. The input dimension d_1 is either 46 (RaptorX only), 1280 (Transformer hidden state), or 1326 (feature combination). We varied d_2 and empirically determined 128 to be optimal.

The 128-D output y of the sequence net gets converted to pairwise matrix features z_1 with 256 feature maps, by an outer concatenation operation; i.e. at position i, j we concatenate y_i and y_j along the feature dimension, giving rise to $2 \times d_2$ feature maps. This $z_1 \in \mathbb{R}^{2d_2 \times L \times L}$ is then concatenated in the first (feature map or channel) dimension, with the pairwise features $z_2 \in \mathbb{R}^{6 \times L \times L}$ i.e. the pairwise RaptorX features described in previous subsection and/or the msa embedding covariance features ($z_3 \in \mathbb{R}^{256 \times L \times L}$) described in the next subsection. As such the concatenated $z \in \mathbb{R}^{262 \times L \times L}$ or $z \in \mathbb{R}^{518 \times L \times L}$.

The final component is the actual 2D residual network operating in z , which computes the binary contact probability $\hat{p} = g_{\theta}^R(z)$ with $\hat{p} \in \mathbb{R}^{L \times L}$ and $\hat{p}_{ij}$ the continuous predicted probability of position i and j of the protein being in contact. The ResNet has an initial 1×1 convolutional layer going to $d_3 = 128$ feature maps, followed by MaxOut over the feature maps with stride 2, reducing to 64 feature maps. After this, there are 32 residual blocks. Each residual block has on its weight path consecutively BatchNorm - ReLU - Conv 3×3 (64 feature maps) - Dropout (0.3) - ReLU - Conv 3×3 (64 feature maps). The residual blocks have consecutive dilation rates of 1,2,4. This follows Adhikari (2019). The final output is computed with a Conv 3×3 (1 output feature map) and sigmoid to produce probability of contact $\hat{p}_{ij} \in [0, 1]$. As such there are 66 convolutional layers in the main 2D ResNet.

Note that a number of shortcomings exist from our pipeline to CASP13 winners (Senior et al., 2018; Xu, 2018); most importantly we use an earlier training dataset of PDB structures compiled from PDB dated Feb 2015 by Wang et al. (2017), additionally we do not incorporate more recent developments like distance distribution prediction, sliding window on small crops allowing deeper ResNets, auxiliary losses like torsion angles, or data augmentation.

For reference, the officially released AlphaFold (Senior et al., 2018) predictions achieve a top-L/5,LR and top-L,LR precision on the same subset of CASP-13 targets of 75.2% and 52.2% respectively. The discrepancies in the pipeline explain why our best precisions using RaptorX features are about 7-9% lower (compare CASP13-AVG (a): 68.0% / 43.4%)

MSA Embedding feature combination. We construct features based on the embedding of the MSA of a protein sequence in our training data. We denote the original protein in our labeled dataset, i.e. query sequence x of length L , to have corresponding embedding $h = \text{Transformer}(x) \in \mathbb{R}^{L \times d}$, and the embedding of the i -th position to be $h_i \in \mathbb{R}^d$. Typically h is the last hidden state from the pre-trained Transformer model. The m th sequence in the MSA is x^m , with corresponding embedding h^m . $m \in [0, M[$ with M

为了检查我们的数据集构建，我们使用了由（Klausen等，2019）提供的预训练权重，并直接在我们的评估流程中评估了他们的模型。我们能够重现（Klausen等，2019）中报告的值。

13. 接触预测。深度神经网络与特征组合

数据 我们使用随附的数据集和特征王等(2017) 和 徐(2018)。基本特征是RaptorX使用的特征（徐，2018），这是CASP13中的最先进方法，包括序列特征、PSSM、三态二级结构预测、预测的可接近性、序列的一热编码和成对特征APC-修正的Potts模型耦合、互信息、成对接触势能。

我们使用来自 王等(2017) 的训练集、标准测试集和 CASP11测试集。我们使用来自 徐(2018) 的CASP12测试集。对于CASP13测试集，我们使用34个公开发布的领域。

王等(2017) 建立训练和测试集如下。训练集（6,367个蛋白质），验证集（400个蛋白质）和测试集（500个蛋白质）数据集作为PDB25的子集选择（每个蛋白质的序列相似性小于 <25%）。序列相似性大于 >25%或 BLAST E值小于 <0.1的蛋白质将被排除在训练数据之外。

我们所有的MSAs（用于平均和覆盖组方法）是通过运行HHblits（Remmert等，2011）与3次迭代和E值为0.001针对2016年2月发布的Uniprot20构建的；除了CASP12和CASP13，我们在这些情况下使用了四个不同的MSAs，这些MSAs是与Xu(2018)一起发布并描述的。请注意，对于变换器预训练，使用了2018年3月的UniRef50；因此，在CASP13开始之前没有可用的数据在预训练或接触预测期间出现过。

训练。

模型架构在序列和成对特征之上，我们使用一个深度为32的残差网络（ResNet）模型来预测二元接触。ResNet模型架构类似于Wang等(2017)和Xu (2018)。

ResNet的第一个组成部分是一个学习的序列管道 $y = f_{\theta}^S(x)$ ，它将序列特征映射 $x \in \mathbb{R}^{L \times d_1}$ 到 $y \in \mathbb{R}^{L \times d_2}$ 与 L 蛋白质的长度。尽管 f_{θ}^S 可以是1D卷积网络或如王等(2017)中的残差网络，我们选择将我们的序列网络设置为从输入维度的简单线性投影 d_1 到 $d_2 = 128$ 维度。输入维度 d_1 可以是46（仅限RaptorX），1280（变换器隐藏状态）或1326（特征组合）。我们进行了变化 d_2 并实证确定128是最佳的。

序列网络的128维输出 y 被转换为成对矩阵特征 z_1 ，具有256个特征图，通过外部连接操作；即在位置 i, j 我们沿特征维度连接 y_i 和 y_j ，产生 $2 \times d_2$ 特征图。然后，这 $z_1 \in \mathbb{R}^{2d_2 \times L \times L}$ 被在第一个（特征图或通道）维度上与成对特征连接，即前一小节中描述的成对RaptorX特征和/或下一小节中描述的msa嵌入协方差特征($z_3 \in \mathbb{R}^{256 \times L \times L}$)。因此，连接的 $z \in \mathbb{R}^{262 \times L \times L}$ 或 $z \in \mathbb{R}^{518 \times L \times L}$ 。

最后一个组件是实际的2D残差网络，在 z 中运行，它计算二元接触概率 $\hat{p} = g_{\theta}^R(z)$ ，与 $\hat{p} \in \mathbb{R}^{L \times L}$ 和 $\hat{p}_{ij}$ 蛋白质在接触时的位置 i 和 j 的连续预测概率。ResNet有一个初始的 1×1 卷积层，输出 $d_3 = 128$ 特征图，接着是在特征图上进行MaxOut，步幅为2，减少到64个特征图。之后，有32个残差块。每个残差块的权重路径依次为BatchNorm - ReLU。

- 卷积 3×3 (64 特征图) - Dropout (0.3) - ReLU - 卷积 3×3 (64 特征图)。残差块的扩张率依次为 1,2,4。这遵循Adhikari(2019)。最终输出通过一个卷积 3×3 (1 输出特征图) 和 sigmoid 计算，产生接触的概率 $\hat{p}_{ij} \in [0, 1]$ 。因此，主 2D 残差网络中有 66 个卷积层。

请注意，从我们的管道到CASP13获胜者（Senior et al., 2018; Xu, 2018）存在许多缺陷；最重要的是，我们使用的是一个早期的PDB结构训练数据集，该数据集来自于2015年2月的PDB，由Wang et al. (2017)汇编而成。此外，我们没有纳入更近期的发展，例如距离分布预测、小型裁剪上的滑动窗口以允许更深的残差网络、辅助损失如扭转角度，或数据增强。

作为参考，官方发布的 AlphaFold (Senioret al., 2018) 预测在相同的 CASP-13 目标子集上实现了 75.2% 和 52.2% 的 top-L/5,LR 和 top-L,LR 精度。管道中的差异解释了为什么我们使用 RaptorX 特征的最佳精度大约低了 7-9%（比较 CASP13-AVG (a): 68.0% / 43.4%）

MSA 嵌入特征组合。 我们基于训练数据中蛋白质序列的 MSA 嵌入构建特征。我们将我们标记数据集中原始蛋白质表示为，即查询序列 x ，长度为 L ，对应的嵌入为 $h = \text{Transformer}(x) \in \mathbb{R}^{L \times d}$ ，第 i 个位置的嵌入为 $h_i \in \mathbb{R}^d$ 。通常 h 是来自预训练 Transformer 模型的最后隐藏状态。MSA 中第 m 个序列为 x^m ，对应的嵌入为 h^m 。 $m \in [0, M[$ 与 M

the MSA depth. The embeddings are computed by embedding the original sequence x^m without inserting gaps (there is no gap character in our vocabulary), then realigning the embedding according to the alignment between x^m and query sequence x by inserting 0-vectors at position i if the x_i^m is the gap character; ie $h_{i,k}^m = 0$. We also use indicator variable $\alpha_i^m = 1$ if x_i^m is non-gap (match state), or $\alpha_i^m = 0$ if x_i^m is gap. We further compute sequence weights w^m as the commonly used debiasing heuristic to reduce the influence of the oversampling of many similar sequences. The weights are defined in the usual way with 70% sequence similarity threshold: sequence weight $w^m = 1/|\{x^{m'} | seqid(x^m, x^{m'}) > 70\% \}|$ which is the inverse of the number of sequences $x^{m'}$ that are more than 70% similar to the sequence x^m i.e. hamming distance less than 0.3L.

Now we introduce the average embedding over an MSA:

$$h_{ik}^{\text{avg}} = \frac{1}{M_{\text{eff}}(i)} \sum_{m=0}^{M-1} w^m \alpha_i^m h_{i,k}^m$$

with per-position denominator $M_{\text{eff}}(i) = \sum_{m=0}^{M-1} w^m \alpha_i^m$. This is effectively a weighted average over the sequence embeddings in the MSA. Note that if the embeddings were one-hot encodings of AA identities, we would recover the position probability matrix (except the absence of a pseudo-count).

Similarly; we introduce the (uncentered) covariance of the embeddings, with PCA-projected $\bar{h}$:

$$C_{ijkl} = \frac{1}{M_{\text{eff}}(i,j)} \sum_{m=0}^{M-1} w^m \alpha_i^m \bar{h}_{i,k}^m \alpha_j^m \bar{h}_{j,l}^m$$

With pairwise position denominator $M_{\text{eff}}(i,j) = \sum_{m=0}^{M-1} w^m \alpha_i^m \alpha_j^m$.

Note that to make above covariance embedding feasible, we first reduce the dimensionality of the embeddings by projecting onto the first 16 PCA directions: $\bar{h}_i = P h_i$ with $P \in \mathbb{R}^{16 \times d}$, giving rise to a covariance per pair of positions i, j and pair of interacting PCA components k, l of $L \times L \times 16 \times 16$. The 256 different k, l pairs of $C_{ijkl} \in \mathbb{R}^{L \times L \times 16 \times 16}$ will now become the feature maps of $z \in \mathbb{R}^{256 \times L \times L}$, such that $z_{16k+l,i,j} = C_{ijkl}$. We tried training (rather than fixed PCA) the projection of the features $h \rightarrow \bar{h}$ before covariance (learned linear projection P or training a 3-layer MLP). We also varied the formulation to center the embeddings over the MSA (normal covariance) and to rescale the feature maps with a pre-computed mean and standard deviation for each feature map corresponding to a pair of k, l . We found no gains from these variations over the current formulation. Note that centering with the average h^{avg} as in normal empirical covariance calculation,

introduces a shift that is independent per protein (because specific to the MSA), and independent per position. Therefore it is not unexpected that the uncentered covariance gives better (more consistent) features.

14. Mutational Effect

Datasets We used two datasets of variant effect measurements compiled by [Gray et al. \(2018\)](#) and [Riesselman et al. \(2018\)](#). The first dataset is a collection of 21,026 measurements from nine experimental deep mutational scans. The second dataset contains 712,218 mutations across 42 deep mutational scans.

Fine-tuning procedure To fine-tune the model to predict the effect of changing a single amino acid or combination of amino acids we regress the scaled mutational effect with:

$$y = \sum_i \log p_i(\text{mt}(i)) - \log p_i(\text{wt}(i))$$

Where $\text{mt}(i)$ is the mutated amino acid at position i , and $\text{wt}(i)$ is the wildtype amino acid. The sum runs over the indices of the mutated positions. As an evaluation metric, we report the Spearman ρ between the model's predictions and experimentally measured values.

15. Area under the ROC curve

For a binary classification task, the ROC curve plots the true positive rate against the false positive rate at various classification thresholds. The area under the ROC curve gives a measure that quantifies the model's ability to distinguish between classes. Intuitively a perfect classifier has an AUC of 1, while a uniform random classifier has an AUC of 0.5.

B. ESM-1b Hyperparameter Optimization

Experimental setup We perform a systematic analysis on Transformer models with 100M parameters. We train models on the UniRef50 dataset following the same methodology described in the rest of this work.

After identifying the best performing settings on 100M parameter models, we explore scale by training 650M parameter models. All models are trained with a target batch size of 1M tokens. To accommodate the large batch size, we use gradient accumulation and distributed data parallel. Under this setup, each epoch of a 100M parameter model completes in 1.8 hours on 32 GPUs. Each epoch of a 650M parameter model completes in 8.5 hours on 64 GPUs.

When studying architectural variants, we assess the quality of representations from each model after 10 or 12 epochs of pre-training. We observed that relative performance ranking of the models does not change after this point. Notably, this

MSA深度。嵌入是通过在不插入间隙的情况下嵌入原始序列 x^m 来计算的（我们的词汇中没有间隙字符），然后根据 x^m 和查询序列 x 之间的对齐重新对齐嵌入，如果 x_i^m 是间隙字符，则在位置 i 插入0向量；即 $h_{i,k}^m = 0$ 。我们还使用指示变量 $\alpha_i^m = 1$ 如果 x_i^m 是非间隙（匹配状态），或者 $\alpha_i^m = 0$ 如果 x_i^m 是间隙。我们进一步计算序列权重 w^m ，作为常用的去偏差启发式，以减少许多相似序列的过采样影响。权重以通常的方式定义，序列相似性阈值为70%：序列权重 $w^m = 1/|\{x^{m'} | seqid(x^m, x^{m'}) > 70\% \}|$ ，它是与序列 x^m 相似度超过70%的序列 $x^{m'}$ 的数量的倒数，即汉明距离小于0.3L。

现在我们介绍 MSAs 的平均嵌入：

$$h_{ik}^{\text{avg}} = \frac{1}{M_{\text{eff}}(i)} \sum_{m=0}^{M-1} w^m \alpha_i^m h_{i,k}^m$$

每个位置的分母是 $M_{\text{eff}}(i) = \sum_{m=0}^{M-1} w^m \alpha_i^m$ 这实际上是 MSA 中序列嵌入的加权平均。请注意，如果这些嵌入是氨基酸身份的一热编码，我们将恢复位置概率矩阵（除了缺少伪计数）。

同样地；我们介绍 $\bar{h}$ 嵌入的（未中心化）协方差，使用 PCA 投影的 h ：

$$C_{ijkl} = \frac{1}{M_{\text{eff}}(i,j)} \sum_{m=0}^{M-1} w^m \alpha_i^m \bar{h}_{i,k}^m \alpha_j^m \bar{h}_{j,l}^m$$

使用成对位置分母 $M_{\text{eff}}(i,j) = \sum_{m=0}^{M-1} w^m \alpha_i^m \alpha_j^m$.

请注意，为了使上述协方差嵌入可行，我们首先通过投影到前16个主成分分析方向来减少嵌入的维度： $h_i = P h_i$ ，与 $P \in \mathbb{R}^{16 \times d}$ 结合，产生每对位置的协方差 i, j 和一对相互作用的主成分分析组件 k, l 。这 256 对不同的 k, l 对 $C_{ijkl} \in \mathbb{R}^{L \times L \times 16 \times 16}$ 现在将成为 $z \in \mathbb{R}^{256 \times L \times L}$ 的特征图，使得 $z_{16k+l,i,j} = C_{ijkl}$ 。我们尝试训练（而不是固定主成分分析）特征的投影 $h \rightarrow \bar{h}$ ，在协方差之前（学习线性投影 P 或训练一个 3 层多层感知器）。我们还改变了公式，以使嵌入围绕 MSA 居中（正常协方差）并使用预先计算的每个特征图对应于一对 k, l 的均值和标准差来重新缩放特征图。我们发现这些变化对当前公式没有提高。请注意，像在正常经验协方差计算中一样，使用平均值居中 h^{avg} ，

引入了一个与每个蛋白质独立的偏移（因为特定于 MSA），并且在每个位置也是独立的。因此，未中心化协方差提供更好的（更一致的）特征并不令人意外。

14. 突变效应数据集 我们使用了两个由 [Gray 等 \(2018\)](#) 和 [里瑟尔曼等 \(2018\)](#) 编译的变体效应测量数据集。第一个数据集是来自九个实验深突变扫描的 21,026 个测量的集合。第二个数据集包含 712,218 个变异，分布在 42 个深突变扫描中。**微调程序** 为了微调模型以预测更改单个氨基酸或氨基酸组合的效果，我们用缩放的突变效应进行回归：

$$y = \sum_i \log p_i(\text{mt}(i)) - \log p_i(\text{wt}(i))$$

其中 $\text{mt}(i)$ 是位置 i 的突变氨基酸，而 $\text{wt}(i)$ 是野生型氨基酸。总和是对突变位置的索引求和。作为评估指标，我们报告模型预测值与实验测量值之间的斯皮尔曼 ρ 。

15. ROC 曲线下面积 对于二分类任务，ROC 曲线将真阳性率与在不同分类阈值下的假阳性率进行绘制。ROC 曲线下面积提供了一个衡量模型区分类别能力的指标。直观上，一个完美的分类器的 AUC 为 1，而一个均匀随机的分类器的 AUC 为 0.5。

B. ESM-1b 超参数优化

实验设置 我们对具有 100M 参数的 Transformer 模型进行系统分析。我们在 UniRef50 数据集上训练模型，遵循本研究其余部分所描述的方法。

在确定 100M 参数模型的最佳性能设置后，我们通过训练具有 650M 参数的模型来探索规模。所有模型的目标批量大小为 1M 标记。为了适应大型批量大小，我们使用梯度累积和分布式数据并行。在这种设置下，100M 参数模型的每个训练周期在 32 个 GPU 上完成需要 1.8 小时。650M 参数模型的每个训练周期在 64 个 GPU 上完成需要 8.5 小时。

在研究建筑变体时，我们评估每个模型在 10 或 12 个时期的预训练后表示的质量。我们观察到模型的相对性能排名在此之后并没有变化。值得注意的是，这

is still early in training; our best performing model, ESM-1b, is trained for 56 epochs.

Hyperparameter: Masking and data setup Protein language models are trained with a masked language modeling objective, wherein each input sequence is corrupted by replacing a fraction of the tokens with a special mask token. We train 100M parameter models for 10 epochs and compare their performance on the CB513 test set. We investigate four masking strategies:

- *All masks*: following supplemental section 4, 15% of the input tokens are replaced with a mask token and predicted.
- *All random (uniform)*: 15% of the input tokens are replaced with an amino acid selected uniform randomly and predicted.
- *All random (frequency)*: 15% of the input tokens are replaced with an amino acid selected according to their frequency in the dataset and predicted.
- *BERT*: 15% of the input tokens are selected and predicted. Of these, 80% are replaced with mask token; 10% with a uniform random amino acid; 10% not changed.

In all cases, we follow [Liu et al. \(2019\)](#) and dynamically mask the sequences, such that a new mask is randomly selected at each epoch. Since the input data changes across runs, language modeling perplexities cannot be fairly compared. Therefore, we evaluate the downstream performance of the models on a secondary structure benchmark. Table B.1 finds that the *BERT* masking pattern performs better than the other masking patterns and is therefore used for all model variations below.

Masking pattern	CB513
All masks	60.4
All random (uniform)	59.3
All random (frequency)	59.0
BERT	60.8

Table B.1. Comparison of masking patterns, 8-class secondary structure prediction accuracy. Models are pre-trained for 10 epochs on Uniref50. The model and all other hyperparameters remain fixed between experiments. The BERT masking pattern performs best and is used for future experiments.

Hyperparameter: Dynamic batching Models in section 4 were trained with dynamic batching, which results in a single sequence in each sample in the batch. This design choice contrasts with existing language modeling works.

For example, in NLP, [Liu et al. \(2019\)](#) and [Devlin et al. \(2018a\)](#), use a static batching approach, wherein multiple proteins are concatenated in the same batch along the sequence dimension. This approach is common in NLP, as sentences that are nearby in a corpus generally relate to the same topic. As this situation is not the case in protein language modeling, we analyze the impact of static batching schemes in protein language models, finding that they reduce model performance (Table B.2). 100M parameter models are evaluated on the secondary structure downstream task after 10 epochs.

Batching mode	CB513
Static	56.2
Static [cropping long sequences]	57.0
Dynamic	60.8

Table B.2. Comparison of batching modes, 8-class secondary structure accuracy. Batching modes are investigated on 100M parameter models and trained for 10 epochs. All models are trained with a context size of 1024 tokens. In the static batching mode, sequences longer than 1024 can span multiple batches. We crop long sequences for the other batching modes considered. The dynamic batching scheme we used performs significantly better than static batching modes.

Hyperparameters: further sweeps After fixing the data distribution to use the BERT masking scheme with dynamic batching, we next perform a hyperparameter sweep to identify the best learning rates, initializations and layer norm placement. We also propose a token dropout scheme which further improves performance on downstream tasks.

Initializations We compare our initializations to the initializations presented in [Liu et al. \(2019\)](#), finding that they perform similarly (Table B.3).

Initializations	CB513
Radford et al. (2019a)	60.9
Liu et al. (2019)	60.8

Table B.3. Comparison of initializations, 8-class secondary structure accuracy. 100M parameter models are trained for 10 epochs and evaluated on CB513.

Layer norm placement Recent works [Child et al. \(2019\)](#) and [Shoeybi et al. \(2020\)](#) have suggested that pre-activation layer norm results in more stable training for larger models. We investigate the impact of this choice in a smaller controlled setting using 100M parameter transformer models, finding that pre-activation layer norm improves performance on downstream tasks (Table B.4). To account for the lack

在训练早期; 我们表现最好的模型 ESM-1b, 训练了 56 轮。

超参数: 掩蔽和数据设置 蛋白质语言模型是通过掩蔽语言建模目标进行训练的, 其中每个输入序列通过用特殊掩码标记替换部分标记而被破坏。我们训练1亿参数模型进行10个周期, 并比较它们在CB513测试集上的性能。我们研究了四种掩蔽策略:

- 所有掩码: 遵循补充部分 4, 15%的输入标记被替换为掩码标记并进行预测。
- 全部随机 (均匀): 15%的输入标记被随机均匀选择的氨基酸替换并进行预测。
- 全部随机 (频率): 15%的输入标记被根据数据集中的频率选择的氨基酸替换并进行预测。
- *BERT*: 15%的输入标记被选择并进行预测。其中, 80%被替换为掩码标记; 10%被替换为随机均匀选择的氨基酸; 10%保持不变。

在所有情况下, 我们遵循[Liu et al.\(2019\)](#)并动态地掩膜序列, 因此在每个时代随机选择新的掩膜。由于输入数据在运行间变化, 语言模型的困惑度不能被公平比较。因此, 我们在二级结构基准上评估模型的下游性能。表B.1发现*BERT* 掩膜模式比其他掩膜模式表现更好, 因此在下面所有模型变体中使用该模式。

掩膜模式	CB513
所有掩膜	60.4
所有随机 (均匀)	59.3
所有随机 (频率)	59.0
BERT	60.8

表 B.1. 掩码模式的比较, 8类二级结构预测准确性。模型在 UniRef50 上进行了 10 个周期的预训练。模型及所有其他超参数在实验之间保持不变。BERT 掩码模式表现最佳, 并用于未来的实验。

超参数: 动态批处理第4节中的模型经过动态批处理训练, 每个样本在批处理中的结果是单一序列。这种设计选择与现有的语言建模工作形成对比。

例如, 在自然语言处理领域, [Liu等人\(2019\)](#) 和 [Devlin等人\(2018a\)](#)使用静态批处理方法, 多个蛋白质在相同批处理中的序列维度上进行串联。这种方法在自然语言处理领域很常见, 因为语料库中相邻的句子通常相关于相同的主题。由于在蛋白质语言建模中情况并非如此, 我们分析了静态批处理方案对蛋白质语言模型的影响, 发现它们降低了模型性能 (表B.2)。经过10个周期后, 1亿参数模型在二级结构下游任务上进行了评估。

批处理模式	CB513
静态	56.2
静态 [裁剪长序列]	57.0
动态	60.8

表 B.2. 批处理模式比较, 8 类二级结构准确性。批处理模式在 1 亿参数模型上进行研究, 并训练 10 个周期。所有模型均以 1024 个令牌的上下文大小进行训练。在静态批处理模式中, 长度超过 1024 的序列可以跨多个批处理。我们为其他考虑的批处理模式修剪较长的序列。我们所使用的动态批处理方案的性能明显优于静态批处理模式。

超参数: 进一步的搜索 在固定数据分布以使用 BERT 掩蔽方案和动态批处理后, 我们接下来进行超参数搜索, 以确定最佳学习率、初始化和层归一化位置。我们还提出了一种令牌丢弃方案, 进一步提高了下游任务的性能。

初始化 我们将我们的初始化与 [Liu et al.\(2019\)](#) 中提出的初始化进行比较, 发现它们的表现相似 (表B.3)。

初始化	CB513
Radford等人(2019a)	60.9
Liu等人(2019)	60.8

表 B.3. 初始值比较, 8类二级结构准确性。1亿参数模型训练 10个周期并在CB513上进行评估。

层归一化位置近期研究[查德等人\(2019\)](#) 和 [Shoeybi等人\(2020\)](#) 提出了激活前层归一化能在更大模型中实现更稳定的训练。我们一个更小的受控环境中使用 100M参数的变换器模型研究了这个选择的影响, 发现激活前层归一化在下游任务中的表现有所改善 (表 B.4)。为了解释缺乏

of a final layer norm, we additionally add a final layer norm before the linear output projection, improving performance (Table B.5).

Model	CB513
Post-activation	60.8
Pre-activation	61.1

Table B.4. Comparison of layer norm placement, 8 class secondary structure accuracy. 100M parameter models are trained for 10 epochs and evaluated on CB513.

Final layer norm	CB513
No	61.3
Yes	61.9

Table B.5. Including a final layer norm before the language modeling head performs best, 8-class secondary structure accuracy. 100M parameter models are trained for 12 epochs.

Token dropout Usually, masked language models are pre-trained with corrupted inputs and fine-tuned with complete sequences. We hypothesize that this shift in data distribution reduces performance on downstream tasks. Therefore, we propose a token dropout scheme which replaces the mask token embedding with a fixed tensor of zeros. As 10 – 15% of positions are masked, the zero tensors cause a change in the mean statistics of the word embeddings. We therefore adjust the distribution during fine-tuning by multiplying the word embeddings by a fixed constant. Formally, if a mask token is introduced during pre-training with probability $p = 0.15 \cdot 0.8$, then during fine-tuning, we multiply the embeddings by $1/p$. We find that this significantly improves performance (Table B.6).

Masking	CB513
Mask Tokens	61.9
Token Dropout	62.6

Table B.6. Token dropout performs better than mask tokens, 8-class secondary structure accuracy. 100M parameter models are trained for 12 epochs.

Learning rate Next, we investigate learning rate, finding that a peak learning rate of $4 \cdot 10^{-4}$ performs best (Table B.7). Higher learning rates result in instability, while lower learning rates result in lower performance. In all cases, learning rate is warmed up linearly for 16000 steps and then decayed following an inverse square-root schedule (Raffel et al., 2020).

Learning rate	CB513
$1 \cdot 10^{-4}$	61.9
$2 \cdot 10^{-4}$	63.7
$4 \cdot 10^{-4}$	65.2

Table B.7. A peak learning rate of $4 \cdot 10^{-4}$ performs best, 8 class secondary structure accuracy. 100M parameter models are trained for 10 epochs.

Positional embeddings	Weights	CB513
Harmonic	Untied	63.7
Harmonic	Tied	64.8
Learned	Untied	65.7
Learned	Tied	66.0

Table B.8. Comparison of tied embeddings and learned or harmonic positional embeddings, 8 class secondary structure accuracy. 100M parameter models are trained for 17 epochs.

Tying embeddings Although all experiments above are performed with tied input and output embeddings, we investigate whether learning these separately could improve performance. Our results (Table B.8) indicate that sharing the weights negatively impacts performance. Therefore, we maintain our initial setup.

Positional embeddings We also investigate the impact of learning positional embeddings compared to fixed harmonic embeddings (Vaswani et al., 2017a), finding that learned positional embeddings perform better (Table B.8).

在最终层归一化之前, 我们还增加了最终层归一化, 在线性输出投影之前, 提高了性能 (表B.5).

模型	CB513
激活后	60.8
激活前	61.1

表 B.4. 层归一化放置的比较, 8类二级结构准确性。1亿参数模型经过10个周期的训练, 并在CB513上进行评估。

最终层归一化	CB513
No	61.3
是的	61.9

表 B.5. 在语言模型头部之前包含最终层归一化表现最佳, 8类二级结构的准确率。1亿参数模型训练了12 训练轮。

标记丢弃 通常, 掩码语言模型用损坏的输入进行预训练, 并用完整序列进行微调。我们假设这种数据分布的变化会降低下游任务的性能。因此, 我们提出一种标记丢弃方案, 将掩码标记嵌入替换为固定的零张量。由于 10 – 15%的位置被掩盖, 零张量导致词嵌入的均值统计发生变化。因此, 我们在微调期间通过将词嵌入乘以一个固定常数来调整分布。正式来说, 如果掩码标记在预训练期间以概率 $p = 0.15$ 引入,

$\cdot 0.8$, 然后在微调期间, 我们将嵌入乘以 $1/p$ 。我们发现这显著提高了性能 (表B.6)。

掩码	CB513
掩码标记	61.9
标记丢弃	62.6

表 B.6. 标记丢弃的表现优于掩码标记, 8类二级结构的准确率。1亿参数模型经过12个训练轮的训练。

学习率 接下来, 我们研究学习率, 发现峰值学习率为 $4 \cdot 10^{-4}$ 表现最佳 (表B.7)。较高的学习率会导致不稳定, 而较低的学习率则会导致性能下降。在所有情况下, 学习率在16000 步中线性预热, 然后按反平方根调度降低 (Raffel 等人, 2020)。

学习率	CB513
$1 \cdot 10^{-4}$	61.9
$2 \cdot 10^{-4}$	63.7
$4 \cdot 10^{-4}$	65.2

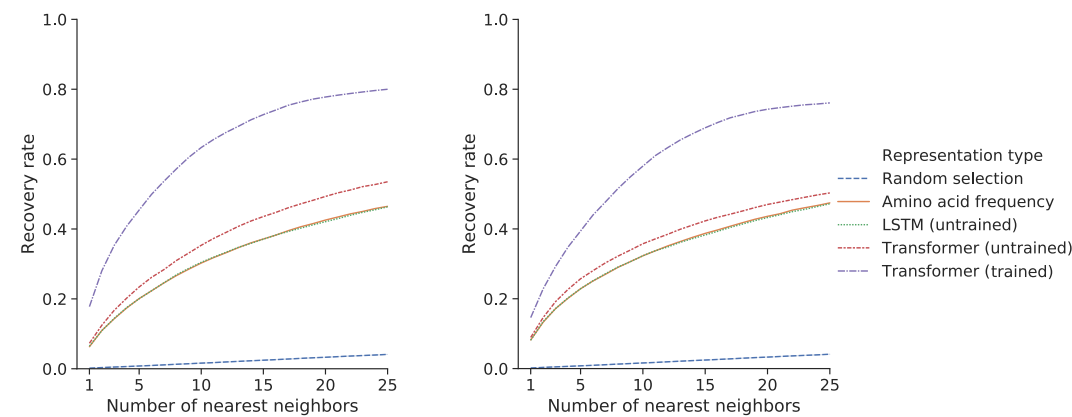
表 B.7. 10^{-4} 表现最佳, 8类二级结构准确性。1亿参数模型训练了10个周期。

位置嵌入	权重	CB513
和谐	解开	63.7
和谐	绑扎	64.8
学习到的	未绑定	65.7
已学习	已绑定	66.0

表 B.8. 对比已绑定嵌入和已学习或谐波位置嵌入的8类二级结构准确性。1亿参数模型训练了17个周期。

捆绑嵌入 尽管以上所有实验使用的是捆绑的输入和输出嵌入, 但我们调查了单独学习这些是否能够提高性能。我们的结果 (表B.8) 表明共享权重对性能产生负面影响。因此, 我们维持最初的设置。

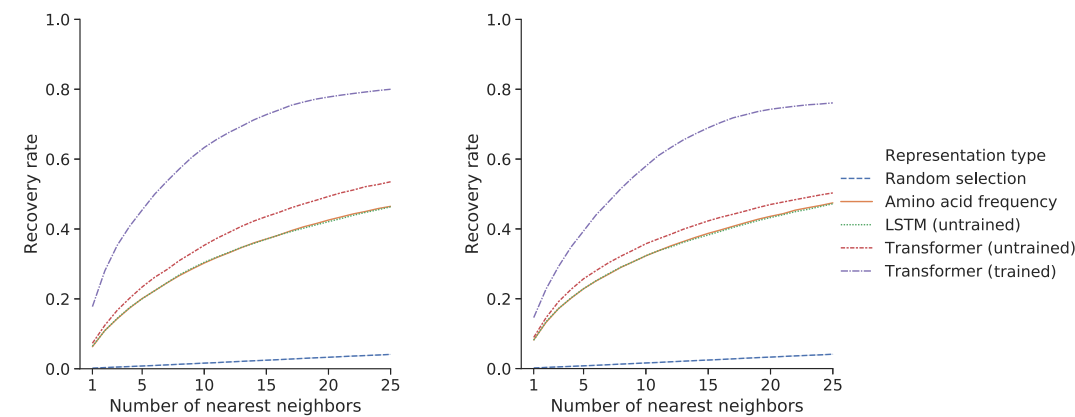
位置嵌入 我们还研究了学习位置嵌入与固定谐波嵌入的影响 (Vaswani等人., 2017a), 发现学习的位置嵌入性能更优 (表B.8)。



(a) Recovery rates under group translation

(b) Recovery rate under species translation

Figure S1. Learned sequence representations can be translated between orthologous groups and species. Depicted is the recovery rate of nearest-neighbor search under (a) orthologous group translation, and (b) species translation. In both settings, the trained Transformer representation space has a higher recovery rate. Results shown for 36-layer dev Transformer pre-trained on UniParc. To define a linear translation between protein *a* and protein *b* of the same species, we define the source and target sets as the average of protein *a* or protein *b* across all 24 diverse species. If representation space linearly encodes orthology, then adding the difference in these averages to protein *a* of some species will recover protein *b* in the same species. We use an analogous approach to translate a protein of a source species *s* to its ortholog in the target species *t*. Here, we consider the average representation of the proteins in *s* and in *t*. If representation space is organized linearly by species, then adding the difference in average representations to a protein in species *s* will recover the corresponding protein in species *t*.



(a) 组翻译下的回收率

(b) 物种翻译下的回收率

图 S1. 学习到的序列表示可以在同源组和物种之间进行翻译。图中显示了在 (a) 同源组翻译和 (b) 物种翻译下最近邻搜索的回收率。在这两种设置中, 训练的变换器表示空间具有更高的回收率。结果显示为在 UniParc 上预训练的 36 层变换器。为了定义同一物种的蛋白质 *a* 和蛋白质 *b* 之间的线性翻译, 我们将源集和目标集定义为 24 种不同物种中蛋白质 *a* 或蛋白质 *b* 的平均值。如果表示空间线性编码同源性, 那么将这些平均值的差添加到某种物种的蛋白质 *a* 将能够恢复同一物种中的蛋白质 *b*。我们使用类似的方法将源物种 *s* 的蛋白质翻译为目标物种 *t* 中的同源物。在这里, 我们考虑 *s* 和 *t* 中蛋白质的平均表示。如果表示空间按物种线性组织, 那么将平均表示的差添加到物种 *s* 中的蛋白质将恢复物种 *t* 中对应的蛋白质。

Representation type	Overall	Identical amino acid pairs	Distinct amino acid pairs
Transformer (trained)	0.841	0.870	0.792
Transformer (untrained)	0.656	0.588	0.468

Table S1. Area under the ROC curve (AUC) of per-residue representational cosine similarities in distinguishing between aligned and unaligned pairs of residues within a Pfam family. Results displayed are averaged across 128 families.

表示类型	总体	相同氨基酸对	不同氨基酸对
变换器（经过训练）	0.841	0.870	0.792
变换器（未经过训练）	0.656	0.588	0.468

表 S1.在 PFAM 家族中，通过基于每个残基的表征余弦相似度区分对齐和未对齐的残基对，ROC 曲线下面积 (AUC) 的结果。所显示的结果是128个家族的平均值。

Representation	PF00005	PF00069	PF00072
Amino acid identity	0.516	0.506	0.536
12-layer (untrained)	0.818	0.719	0.835
12-layer (PF00005)	0.864	0.725	0.842
12-layer (PF00069)	0.816	<u>0.842</u>	0.850
12-layer (PF00072)	0.789	0.688	<u>0.888</u>
12-layer (UniParc)	0.900	0.872	0.906
36-layer (UniParc)	0.902	0.884	0.902

Table S2. Three-class secondary structure prediction accuracy by linear projection. Learning across many protein families produces better representations than learning from single protein families. Transformer models are trained on three PFAM families: ATP-binding domain of the ABC transporters (PF00005), Protein kinase domain (PF00069), and Response regulator receiver domain (PF00072). The single-family models are contrasted with models trained on the full UniParc data. Comparisons are relative to the family (columnwise), since each of the families differ in difficulty. Underline indicates models trained and evaluated on the same family. Representations learned from single families perform well within the family, but do not generalize as well to sequences outside the family. Representations trained on UniParc outperform the single-family representations in all cases.

表征	PF00005	PF00069	PF00072
氨基酸同一性	0.516	0.506	0.536
12层 (未训练)	0.818	0.719	0.835
12层 (PF00005)	0.864	0.725	0.842
12层 (PF00069)	0.816	0.842	0.850
12层 (PF00072)	0.789	0.688	<u>0.888</u>
12层 (UniParc)	0.900	0.872	0.906
36层 (UniParc)	0.902	0.884	0.902

表 S2. 三类二级结构预测的准确性通过线性投影。在许多蛋白质家族之间学习产生的表示优于单一蛋白质家族的学习。Transformer 模型在三个 PFAM 家族上进行训练: ABC转运蛋白的 ATP结合域 (PF00005)、蛋白激酶结构域 (PF00069) 和响应调节接收器结构域 (PF00072)。单一家族模型与在完整的 UniParc 数据上训练的模型进行对比。比较是相对于家族 (逐列), 因为每个家族的难度不同。下划线表示在同一家族上训练和评估的模型。从单一家族学习的表示在家族内表现良好, 但对家族外的序列的泛化能力较差。在所有情况下, 基于 UniParc 训练的表示优于单一家族的表示。

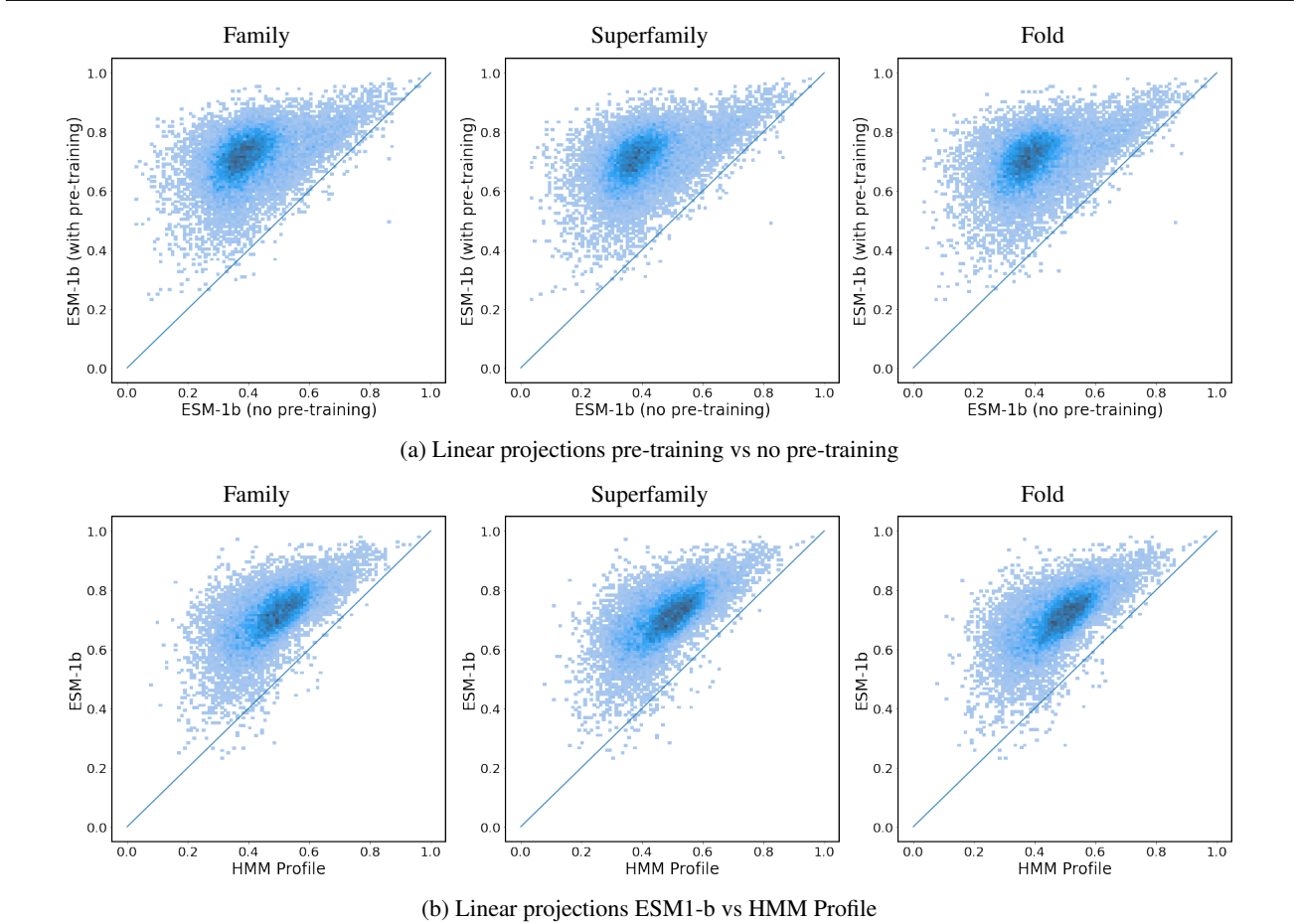


Figure S2. Secondary structure prediction 8-class accuracy distributions for linear projections. (a) Comparison with and without pretraining; (b) comparison of ESM-1b Transformer representations with HMM sequence profiles. Density is indicated by color.

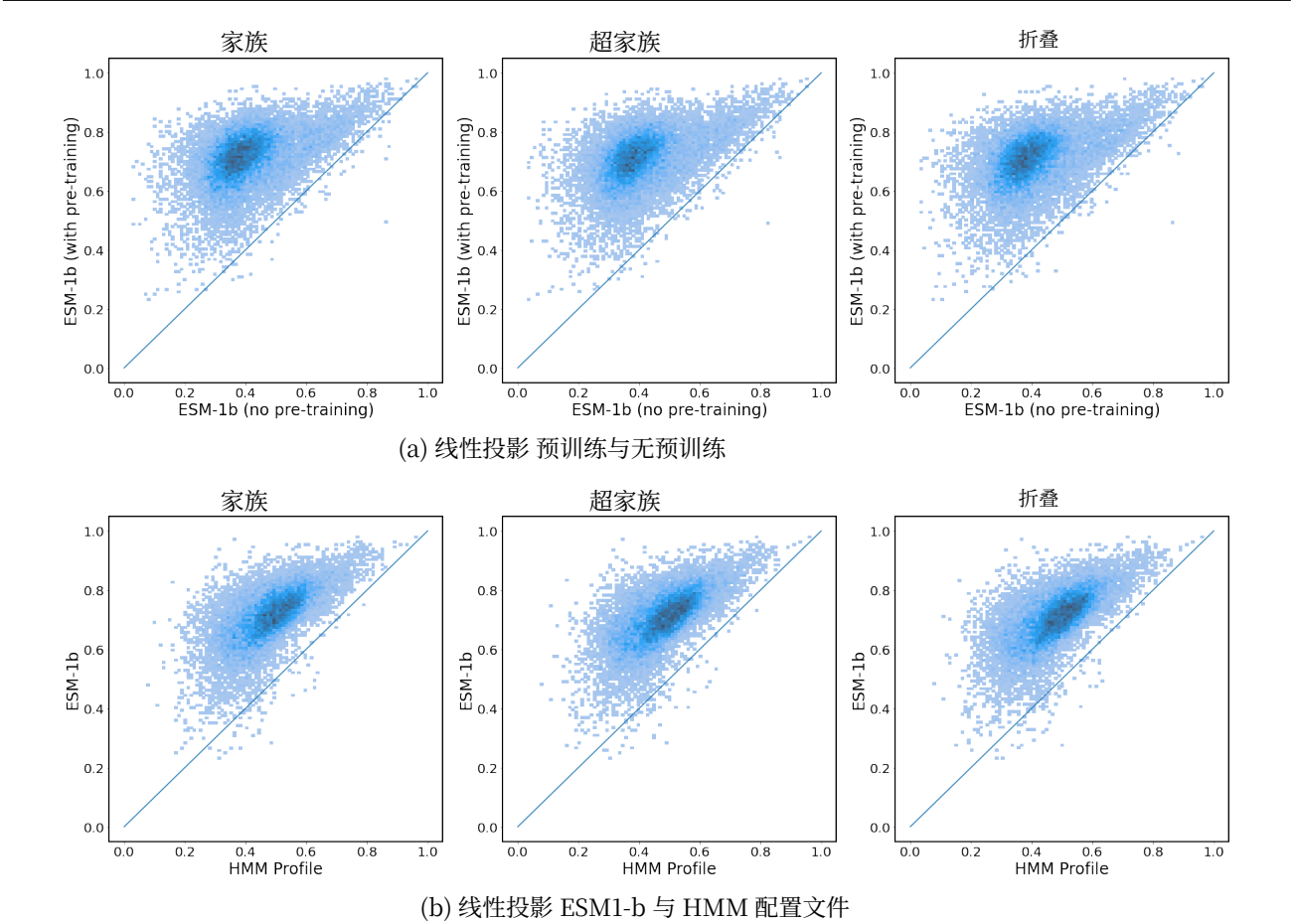


图 S2. 线性投影的八类准确率分布的二级结构预测。(a) 有预训练与无预训练的比较；(b) ESM-1b 变换器表示与 HMM 序列配置文件的比较。密度由颜色表示。

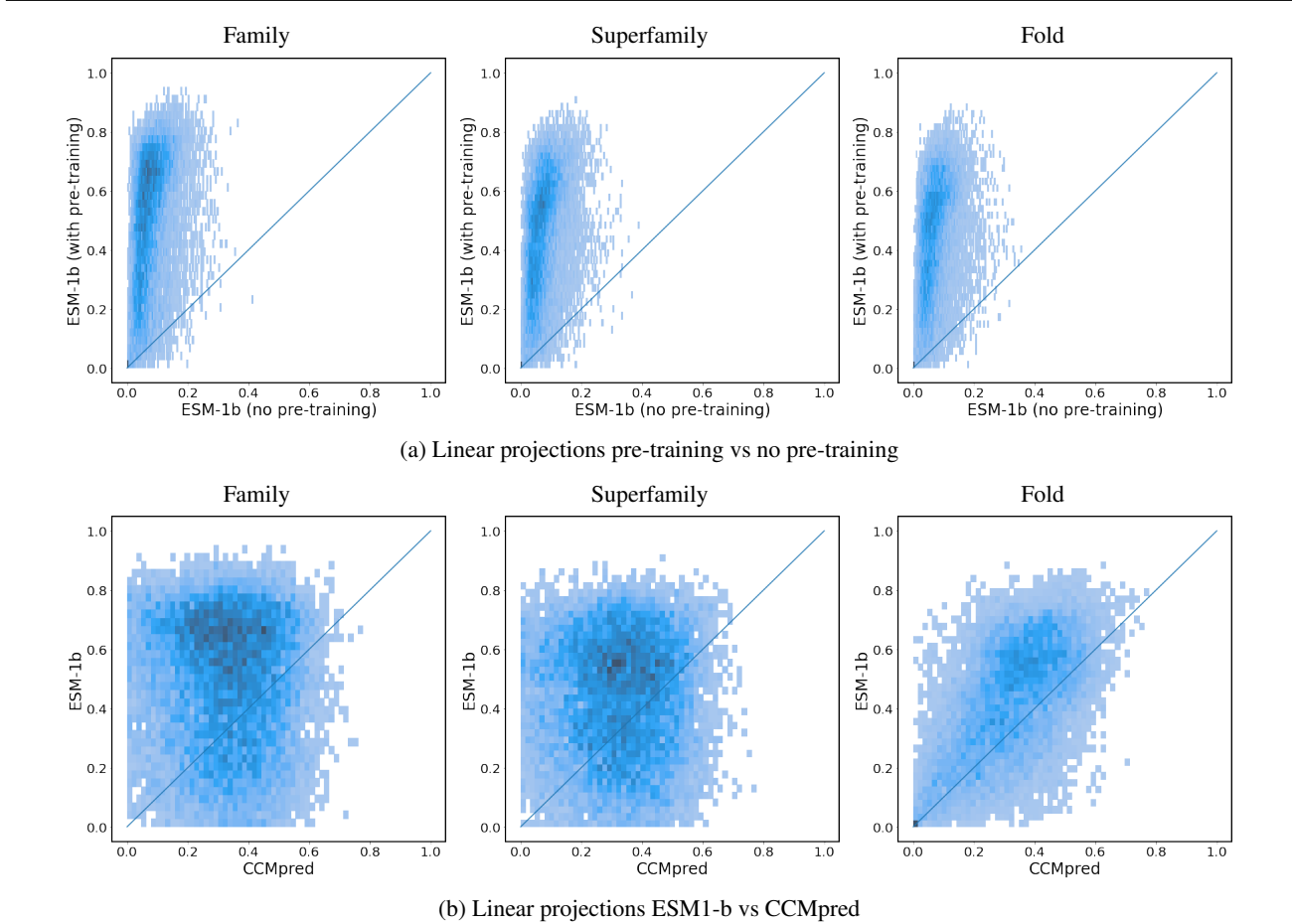


Figure S3. Contact prediction Top-L long-range precision distributions for linear projections. (a) Comparison with and without pretraining; (b) comparison of ESM-1b Transformer representations with CCMpred predictions. Density is indicated by color.

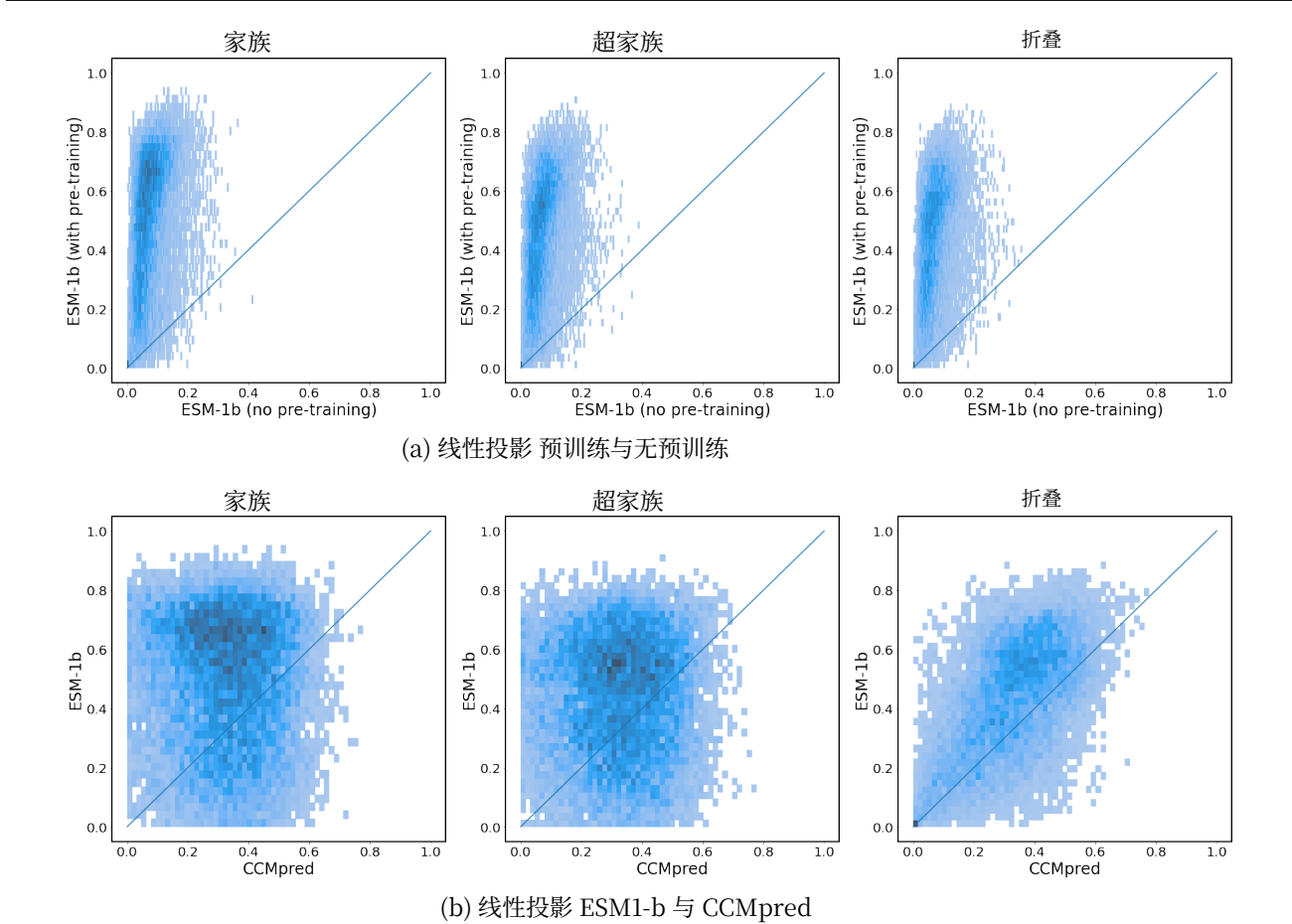


图 S3. 接触预测 Top-L 长距离精度分布的线性投影。(a) 有和没有预训练的比较；(b) ESM-1b 变换器表示与 CCMpred 预测的比较。密度由颜色表示。

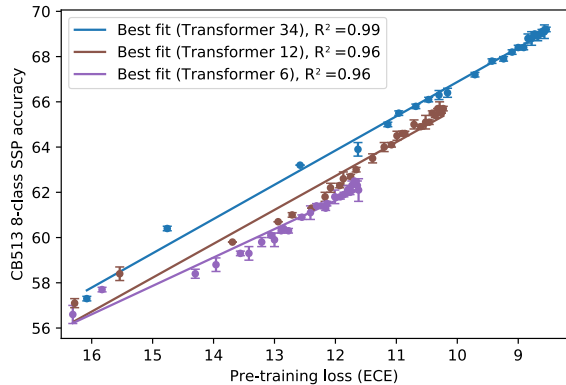


Figure S4. Eight-class secondary structure prediction accuracy as a function of pre-training ECE. A deep secondary structure predictor is trained using features from Transformer models over the course of pre-training on UR50/S. The Netsurf training sequences and CB513 test set are used. Averages across three seeds of the downstream model per pre-training checkpoint are plotted, with line of best fit for each Transformer.

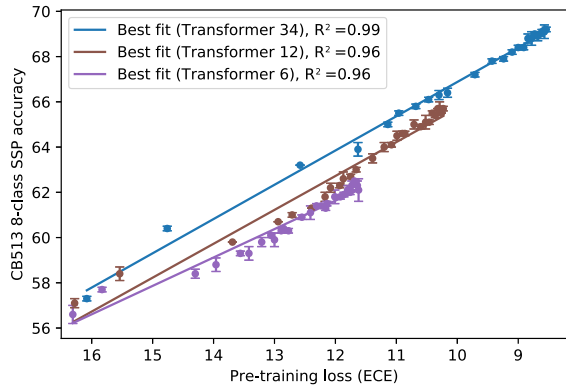


图 S4. 八类二级结构预测准确度与预训练 ECE 的关系。使用 Transformer 模型的特征, 在 UR50/S 上进行预训练的过程中训练一个深度二级结构预测器。使用 Netsurf 训练序列和 CB513 测试集。绘制了每个预训练检查点的下游模型三个种子的平均值, 并为每个 Transformer 绘制了最佳拟合线。

Metric: top-		L/5,LR	L,LR	L/5,MR	L,MR	L/5,SR	L,SR
Test	(a) RaptorX	84.3 ± .2	59.4 ± .2	74.4 ± .1	33.0 ± .0	71.6 ± .1	25.8 ± .0
	(b) +direct	86.7 ± .3	61.7 ± .4	76.5 ± .3	33.8 ± .1	73.6 ± .2	26.1 ± .1
	(c) +avg	87.7 ± .3	62.9 ± .4	77.4 ± .2	34.0 ± .1	73.7 ± .3	26.1 ± .1
	(d) +cov	87.8 ± .3	63.3 ± .2	77.6 ± .2	34.0 ± .1	73.7 ± .2	26.1 ± .0
CASP11	(a) RaptorX	77.5 ± .4	53.8 ± .3	75.0 ± .5	35.6 ± .2	72.1 ± .6	28.6 ± .2
	(b) +direct	78.3 ± .1	55.0 ± .1	76.2 ± .4	35.9 ± .2	74.0 ± .5	28.8 ± .2
	(c) +avg	80.4 ± .5	56.6 ± .4	76.5 ± .4	36.3 ± .2	73.8 ± .5	28.8 ± .1
	(d) +cov	80.3 ± .3	56.8 ± .2	76.6 ± .4	36.5 ± .2	74.0 ± .4	29.0 ± .0
CASP12-AVG	(a) RaptorX	72.7 ± .6	51.1 ± .2	68.3 ± .5	31.2 ± .3	66.5 ± .2	26.3 ± .1
	(b) +direct	74.0 ± .8	51.5 ± .5	70.7 ± .7	32.4 ± .3	68.9 ± .9	27.2 ± .2
	(c) +avg	74.4 ± 1.4	52.4 ± .5	71.7 ± .6	32.2 ± .3	70.1 ± .2	26.9 ± .2
	(d) +cov	76.6 ± .7	53.0 ± .3	70.1 ± .3	31.9 ± .3	69.1 ± .5	26.6 ± .1
CASP12-ENS	(a) RaptorX	77.1 ± .9	54.5 ± .4	70.6 ± .6	32.4 ± .4	68.6 ± .4	27.0 ± .1
	(b) +direct	77.0 ± .6	53.5 ± .6	71.9 ± .9	33.1 ± .3	69.8 ± .7	27.6 ± .2
	(c) +avg	76.7 ± 1.4	54.4 ± .7	74.1 ± .8	33.0 ± .3	71.5 ± .3	27.4 ± .2
	(d) +cov	79.7 ± .8	55.3 ± .2	72.7 ± .5	32.7 ± .3	71.0 ± .6	27.2 ± .2
CASP13-AVG	(a) RaptorX	68.0 ± .9	43.4 ± .4	71.3 ± .4	36.5 ± .3	68.8 ± 1.0	28.4 ± .0
	(b) +direct	67.4 ± .8	43.7 ± .4	69.5 ± .9	35.5 ± .4	68.1 ± .4	28.3 ± .2
	(c) +avg	68.1 ± 1.6	44.8 ± .8	73.0 ± .6	36.9 ± .1	71.2 ± .6	28.6 ± .2
	(d) +cov	70.3 ± 1.3	45.2 ± .5	73.5 ± 1.4	37.0 ± .2	70.2 ± .8	28.6 ± .3
CASP13-ENS	(a) RaptorX	72.1 ± .8	46.3 ± .5	73.6 ± .4	38.1 ± .3	71.0 ± 1.4	29.6 ± .1
	(b) +direct	68.0 ± .7	45.0 ± .6	71.4 ± 1.2	36.6 ± .4	70.2 ± .2	29.1 ± .2
	(c) +avg	70.8 ± 2.2	46.4 ± 1.1	75.4 ± .5	38.1 ± .2	73.6 ± .5	29.3 ± .2
	(d) +cov	71.9 ± 1.9	47.2 ± .4	75.2 ± 1.5	38.3 ± .4	72.3 ± .8	29.3 ± .2

Table S3. Additional metrics. Feature combination for supervised contact prediction. AVG corresponds to the average of the metrics over the different MSAs, while in ENS the probabilities are averaged (ensembled) over the different MSA predictions before computing the metrics.

指标: 顶部-		L/5, LR	L, LR	L/5, MR	L, MR	L/5, SR	L,SR
测试	(a) RaptorX	84.3 ± .2	59.4 ± .2	74.4 ± .1	33.0 ±.0	71.6 ± .1	25.8 ± .0
	(b) +直接	86.7 ± .3	61.7 ± .4	76.5 ± .3	33.8 ±.1	73.6 ± .2	26.1 ± .1
	(c) +平均值	87.7 ± .3	62.9 ± .4	77.4 ± .2	34.0 ±.1	73.7 ± .3	26.1 ± .1
	(d) +cov	87.8 ± .3	63.3 ± .2	77.6 ± .2	34.0 ±.1	73.7 ± .2	26.1 ± .0
CASP11	(a) RaptorX	77.5 ± .4	53.8 ± .3	75.0 ± .5	35.6 ±.2	72.1 ± .6	28.6 ± .2
	(b) +直接	78.3 ± .1	55.0 ± .1	76.2 ± .4	35.9 ±.2	74.0 ± .5	28.8 ± .2
	(c) +平均	80.4 ± .5	56.6 ± .4	76.5 ± .4	36.3 ±.2	73.8 ± .5	28.8 ± .1
	(d) +cov	80.3 ± .3	56.8 ± .2	76.6 ± .4	36.5 ±.2	74.0 ± .4	29.0 ± .0
CASP12-AVG	(a) RaptorX	72.7 ± .6	51.1 ± .2	68.3 ± .5	31.2 ±.3	66.5 ± .2	26.3 ± .1
	(b) +直接	74.0 ± .8	51.5 ± .5	70.7 ± .7	32.4 ±.3	68.9 ± .9	27.2 ± .2
	(c) +平均	74.4 ± 1.4	52.4 ± .5	71.7 ± .6	32.2 ±.3	70.1 ± .2	26.9 ± .2
	(d) +cov	76.6 ± .7	53.0 ± .3	70.1 ± .3	31.9 ±.3	69.1 ± .5	26.6 ± .1
CASP12-ENS	(a) RaptorX	77.1 ± .9	54.5 ± .4	70.6 ± .6	32.4 ±.4	68.6 ± .4	27.0 ± .1
	(b) +直接	77.0 ± .6	53.5 ± .6	71.9 ± .9	33.1 ±.3	69.8 ± .7	27.6 ± .2
	(c) +平均	76.7 ± 1.4	54.4 ± .7	74.1 ± .8	33.0 ±.3	71.5 ± .3	27.4 ± .2
	(d) +cov	79.7 ± .8	55.3 ± .2	72.7 ± .5	32.7 ±.3	71.0 ± .6	27.2 ± .2
CASP13-AVG	(a) RaptorX	68.0 ± .9	43.4 ± .4	71.3 ± .4	36.5 ±.3	68.8 ± 1.0	28.4 ± .0
	(b) +直接	67.4 ± .8	43.7 ± .4	69.5 ± .9	35.5 ±.4	68.1 ± .4	28.3 ± .2
	(c) +平均值	68.1 ± 1.6	44.8 ± .8	73.0 ± .6	36.9 ±.1	71.2 ± .6	28.6 ± .2
	(d) +cov	70.3 ± 1.3	45.2 ± .5	73.5 ± 1.4	37.0 ±.2	70.2 ± .8	28.6 ± .3
CASP13-ENS	(a) RaptorX	72.1 ± .8	46.3 ± .5	73.6 ± .4	38.1 ±.3	71.0 ± 1.4	29.6 ± .1
	(b) +直接	68.0 ± .7	45.0 ± .6	71.4 ± 1.2	36.6 ±.4	70.2 ± .2	29.1 ± .2
	(c) +平均值	70.8 ± 2.2	46.4 ± 1.1	75.4 ± .5	38.1 ±.2	73.6 ± .5	29.3 ± .2
	(d) +cov	71.9 ± 1.9	47.2 ± .4	75.2 ± 1.5	38.3 ±.4	72.3 ± .8	29.3 ± .2

表 S3. 附加指标。用于监督接触预测的特征组合。AVG 表示不同 MSA 上指标的平均值，而 ENS 是在计算指标之前，先对不同 MSA 预测的概率进行平均（集成）。



Figure S5. After pre-training, the Transformer can be adapted to predict mutational effects on protein function. The 34-layer Transformer model pre-trained on UR50/S is fine-tuned on mutagenesis data. Spearman ρ on each protein when supervised with smaller fractions of the data.

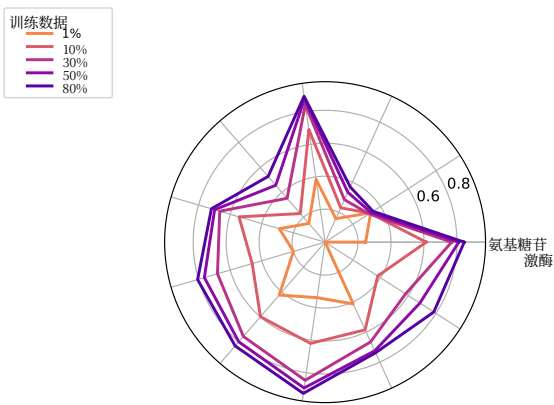


图 S5. 经过预训练后，变换器可以适应预测对蛋白质功能的突变效应。该34层变换器模型在UR50/S上进行了预训练，并在诱变数据上进行了微调。当用较小的数据比例进行监督时，斯皮尔曼 ρ 在每个蛋白质上进行评估。

Amount of training data Protein	1% data	10% data	30% data	50% data	80% data
Aminoglycoside kinase	0.25 ± 0.07	0.61 ± 0.02	0.77 ± 0.01	0.81 ± 0.01	0.84 ± 0.01
BRCA1 (Bard1 binding)	0.33 ± 0.02	0.32 ± 0.01	0.32 ± 0.03	0.33 ± 0.03	0.35 ± 0.03
BRCA1 (E3 ligase activity)	0.16 ± 0.01	0.23 ± 0.03	0.28 ± 0.07	0.33 ± 0.05	0.37 ± 0.04
Beta Lactamase	0.39 ± 0.03	0.69 ± 0.01	0.84 ± 0.01	0.88 ± 0.01	0.89 ± 0.01
E4B (U-box domain)	0.15 ± 0.03	0.23 ± 0.03	0.35 ± 0.01	0.46 ± 0.03	0.53 ± 0.04
Hsp90	0.29 ± 0.02	0.54 ± 0.01	0.67 ± 0.01	0.70 ± 0.02	0.72 ± 0.05
PSD95 (pdz3 domain)	0.20 ± 0.02	0.46 ± 0.05	0.68 ± 0.01	0.76 ± 0.01	0.81 ± 0.02
Pab1 (RRM Domain)	0.42 ± 0.07	0.60 ± 0.02	0.76 ± 0.01	0.80 ± 0.01	0.83 ± 0.02
Protein G	0.34 ± 0.15	0.62 ± 0.03	0.85 ± 0.02	0.89 ± 0.02	0.93 ± 0.01
Ubiquitin (E1 activation)	0.41 ± 0.06	0.58 ± 0.02	0.67 ± 0.02	0.72 ± 0.02	0.74 ± 0.02
Yap65 (WW domain)		0.38 ± 0.06	0.58 ± 0.06	0.68 ± 0.05	0.78 ± 0.05

Table S4. Aggregate spearman ρ measured across models and datasets. Mean and standard deviations of spearman ρ performance for the fine-tuned Transformer-34 on intraprotein tasks. Performance was assessed on five random partitions of the validation set. Model pre-trained on UR50/S.

训练数据的量	1% 的数据	10% 的数据	30% 的数据	50% 的数据	80% 的数据
蛋白质氨基糖苷激酶					
	0.25 ± 0.07	0.61 ± 0.02	0.77 ± 0.01	0.81 ± 0.01	0.84 ± 0.01
BRCA1 (Bard1结合)	0.33 ± 0.02	0.32 ± 0.01	0.32 ± 0.03	0.33 ± 0.03	0.35 ± 0.03
BRCA1 (E3连接酶活性)	0.16 ± 0.01	0.23 ± 0.03	0.28 ± 0.07	0.33 ± 0.05	0.37 ± 0.04
β-内酰胺酶	0.39 ± 0.03	0.69 ± 0.01	0.84 ± 0.01	0.88 ± 0.01	0.89 ± 0.01
E4B (U-box 结构域)	0.15 ± 0.03	0.23 ± 0.03	0.35 ± 0.01	0.46 ± 0.03	0.53 ± 0.04
Hsp90	0.29 ± 0.02	0.54 ± 0.01	0.67 ± 0.01	0.70 ± 0.02	0.72 ± 0.05
PSD95 (pdz3 领域)	0.20 ± 0.02	0.46 ± 0.05	0.68 ± 0.01	0.76 ± 0.01	0.81 ± 0.02
Pab1 (RRM 结构域)	0.42 ± 0.07	0.60 ± 0.02	0.76 ± 0.01	0.80 ± 0.01	0.83 ± 0.02
蛋白质G	0.34 ± 0.15	0.62 ± 0.03	0.85 ± 0.02	0.89 ± 0.02	0.93 ± 0.01
泛素 (E1 激活)	0.41 ± 0.06	0.58 ± 0.02	0.67 ± 0.02	0.72 ± 0.02	0.74 ± 0.02
Yap65 (WW结构域)		0.38 ± 0.06	0.58 ± 0.06	0.68 ± 0.05	0.78 ± 0.05

表 S4.聚合 Spearman ρ 在模型和数据集之间的测量。调优后的 Transformer-34 在内蛋白质任务上 Spearman ρ 性能的均值和标准差。性能是在验证集五个随机划分上评估的。模型在 UR50/S 上进行预训练。

dataset model	spearmanr		
	Envision	Envision (LOPO)	DeepSequence
Transformer	0.71 ± 0.20	0.51	0.70 ± 0.15
LSTM biLM (Large)	0.65 ± 0.22		0.62 ± 0.19
Gray, et al. 2018	0.64 ± 0.21	0.45	
Riesselman, et al. 2018			0.48 ± 0.26

Table S5. Aggregate spearman ρ measure across models and datasets. 34-layer Transformer pre-trained on UR50/S. For intraprotein models, the train/valid data was randomly partitioned five times. The mean $\pm$ standard deviation across the five runs is reported. No standard deviations are reported for LOPO experiments, as the evaluation is performed across all proteins.

数据集 模型	斯皮尔曼相关系数		
	Envision	Envision (LOPO)	DeepSequence
变换器	0.71 ± 0.20	0.51	0.70 ± 0.15
LSTM biLM (大)	0.65 ± 0.22		0.62 ± 0.19
Gray 等人 2018	0.64 ± 0.21	0.45	
Riesselman 等人 2018			0.48 ± 0.26

表 S5. 在模型和数据集上聚合的Spearman ρ 度量。34层 Transformer 在UR50/S上进行预训练。对于蛋白质内部模型，训练/验证数据随机划分五次。报告五次运行的平均值 $\pm$ 标准偏差。对于LOPO实验未报告标准偏差，因为评估在所有蛋白质上进行。

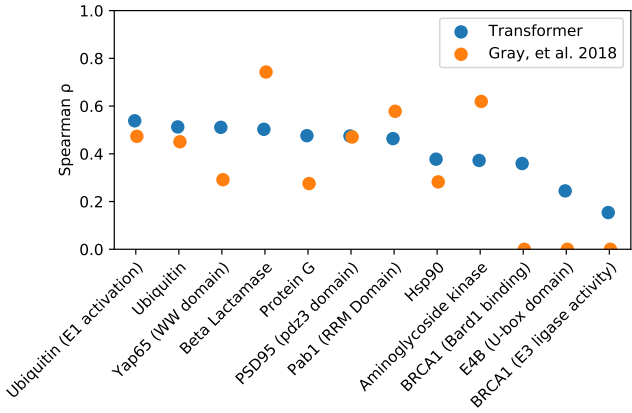


Figure S6. Leave-one-out experiment on Envision dataset (Gray et al., 2018). Pre-training improves the ability of the Transformer to generalize to the mutational fitness landscape of held-out proteins. All mutagenesis data from the protein selected for evaluation are held out, and the model is supervised with data from the remaining proteins. For each evaluation protein, a comparison is shown for the 34-layer Transformer pre-trained on UR50/S.

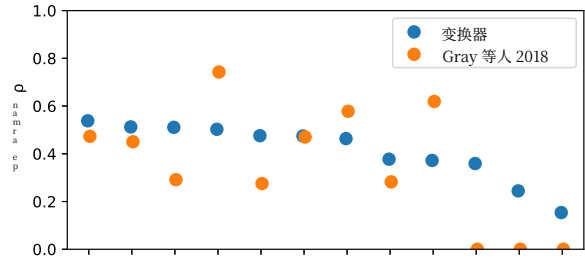


图 S6. 在 Envision 数据集上的留一交叉验证实验（Gray 等人，2018）。预训练提高了变换器在未包含蛋白质的突变适应性景观上的泛化能力。选定用于评估的蛋白质的所有突变数据均被保留，模型使用其余蛋白质的数据进行监督。对于每个评估蛋白质，展示了在 UR50/S 上预训练的 34 层变换器的比较。

Model	Pre- Training	Params	SSP		Contact			
			CB513	CASP13	Test	CASP11	CASP12	CASP13
Transformer-34	(None)	669.2M	56.8	60.0	16.3	17.7	14.8	13.3
UniRep (LSTM)		18.2M	58.4	60.1	21.9	21.4	16.8	14.3
SeqVec (LSTM)		93M	62.1	64.0	29.0	25.5	23.6	17.9
TAPE (Transformer)		38M	58.0	61.5	23.2	23.8	20.3	16.0
LSTM (S)	UR50/S	28.4M	60.4	63.2	24.1	23.6	19.9	15.3
LSTM (L)	UR50/S	113.4M	62.4	64.1	27.8	26.4	24.0	16.4
Transformer-6	UR50/S	42.6M	62.0	64.2	30.2	29.9	25.3	19.8
Transformer-12	UR50/S	85.1M	65.4	67.2	37.7	33.6	27.8	20.7
Transformer-34	UR100	669.2M	64.3	66.5	32.7	28.9	24.3	19.1
Transformer-34	UR50/S	669.2M	69.1	70.7	50.2	42.8	34.7	30.1
ESM-1b	UR50/S	652.4M	71.6	72.5	56.9	47.4	42.7	35.9

Table S6. Comparison to related methods. Top-L long-range contact precision. Predictions are directly from protein sequence, no coevolutionary features or MSAs used. Test is RaptorX test set of Wang et al. (2017). Model weights for related work are obtained and evaluated in our codebase with same downstream architecture, training, and test data.

模型	前- 训练	参数	SSP		Test	联系		
			CB513	CASP13		CASP11	CASP12	CASP13
变换器-34	(无)	669.2M	56.8	60.0	16.3	17.7	14.8	13.3
UniRep (长短期记忆网络)		18.2M	58.4	60.1	21.9	21.4	16.8	14.3
SeqVec (长短期记忆网络)		93M	62.1	64.0	29.0	25.5	23.6	17.9
TAPE (变换器)		38M	58.0	61.5	23.2	23.8	20.3	16.0
LSTM (S)	UR50/S	28.4M	60.4	63.2	24.1	23.6	19.9	15.3
LSTM (L)	UR50/S	113.4M	62.4	64.1	27.8	26.4	24.0	16.4
变换器-6	UR50/S	42.6M	62.0	64.2	30.2	29.9	25.3	19.8
变换器-12	UR50/S	85.1M	65.4	67.2	37.7	33.6	27.8	20.7
变换器-34	UR100	669.2M	64.3	66.5	32.7	28.9	24.3	19.1
变换器-34	UR50/S	669.2M	69.1	70.7	50.2	42.8	34.7	30.1
ESM-1b	UR50/S	652.4M	71.6	72.5	56.9	47.4	42.7	35.9

表 S6.与相关方法的比较。Top-L 长距离接触精度。预测直接基于蛋白质序列，没有使用共进化特征或 MSA。测试是 RaptorX 测试集王等(2017)。相关工作的模型权重在我们的代码库中获得并评估，使用相同的下游架构、训练和测试数据。

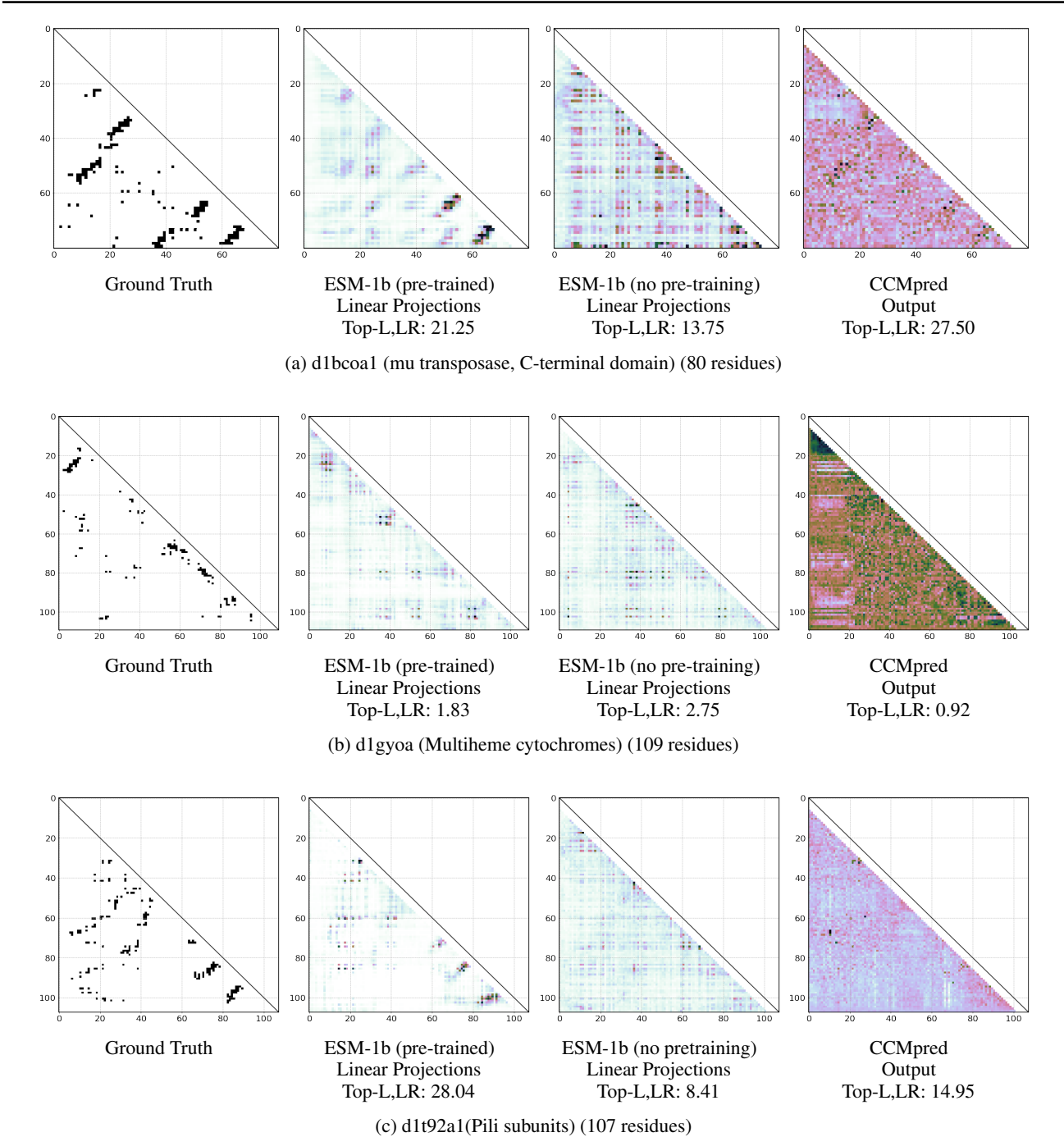


Figure S7. Comparison of ground truth contact map, projections from ESM-1b with and without pre-training, and CCMpred output. Labels indicate SCOPe domain, fold name, and number of residues. Eight domains randomly sampled from fold-level test sets are shown.

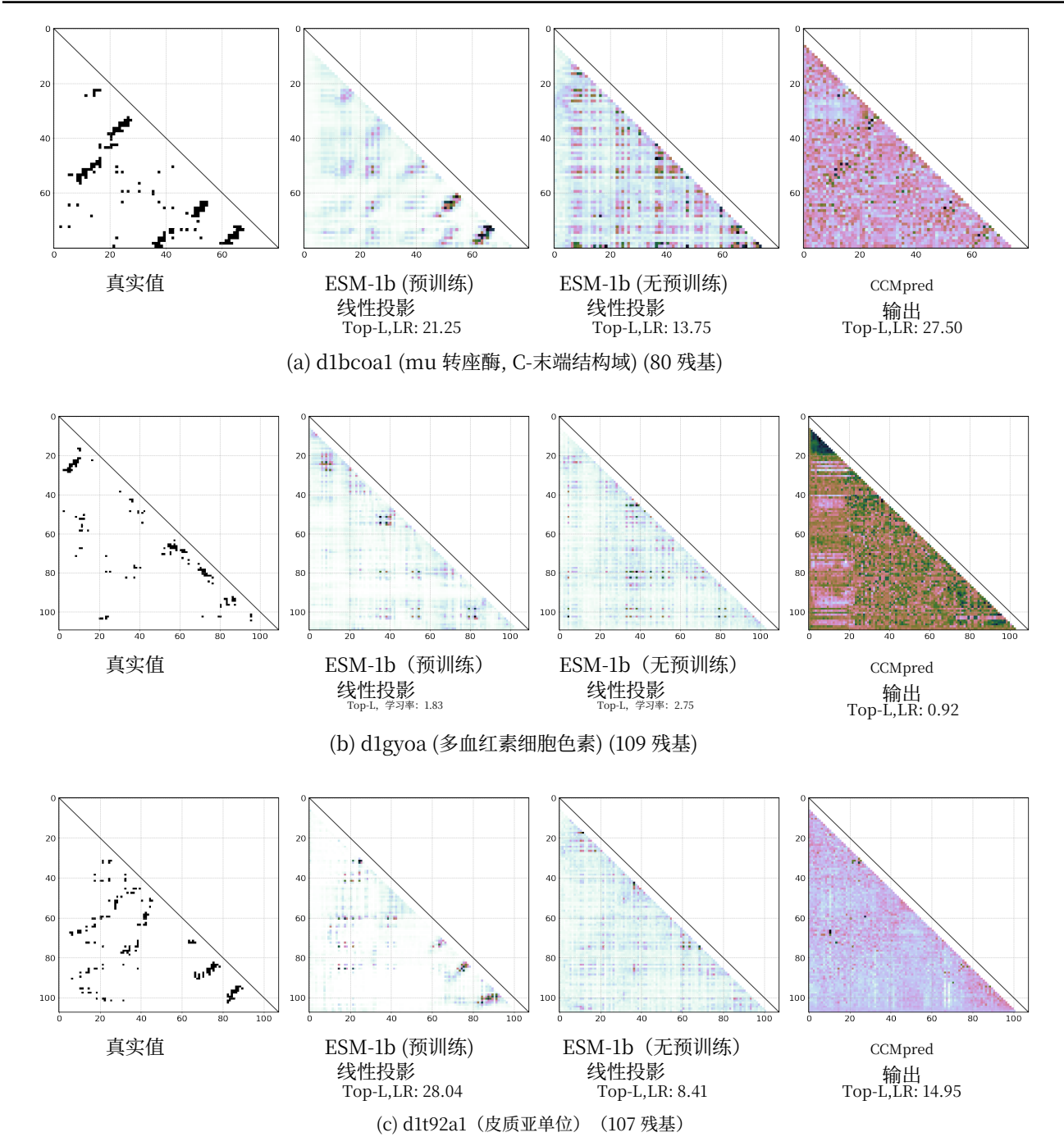


图 S7。真实值接触图与 ESM-1b 在有无预训练下的投影以及 CCMpred 输出的比较。标签显示 SCOPe 域、折叠名称和残基数量。展示了从折叠级测试集随机抽取的八个域。

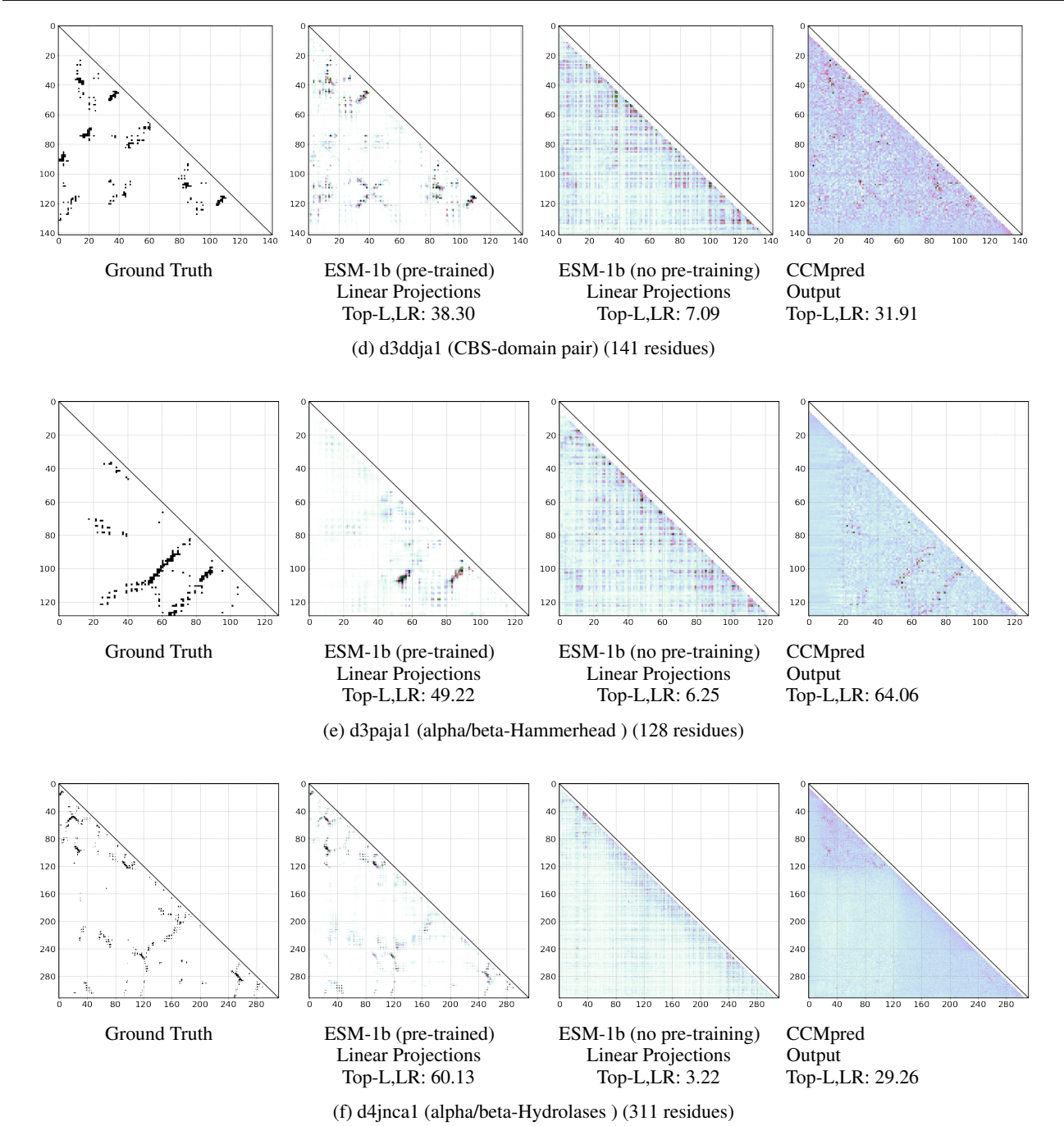


Figure S7. (continued from above)

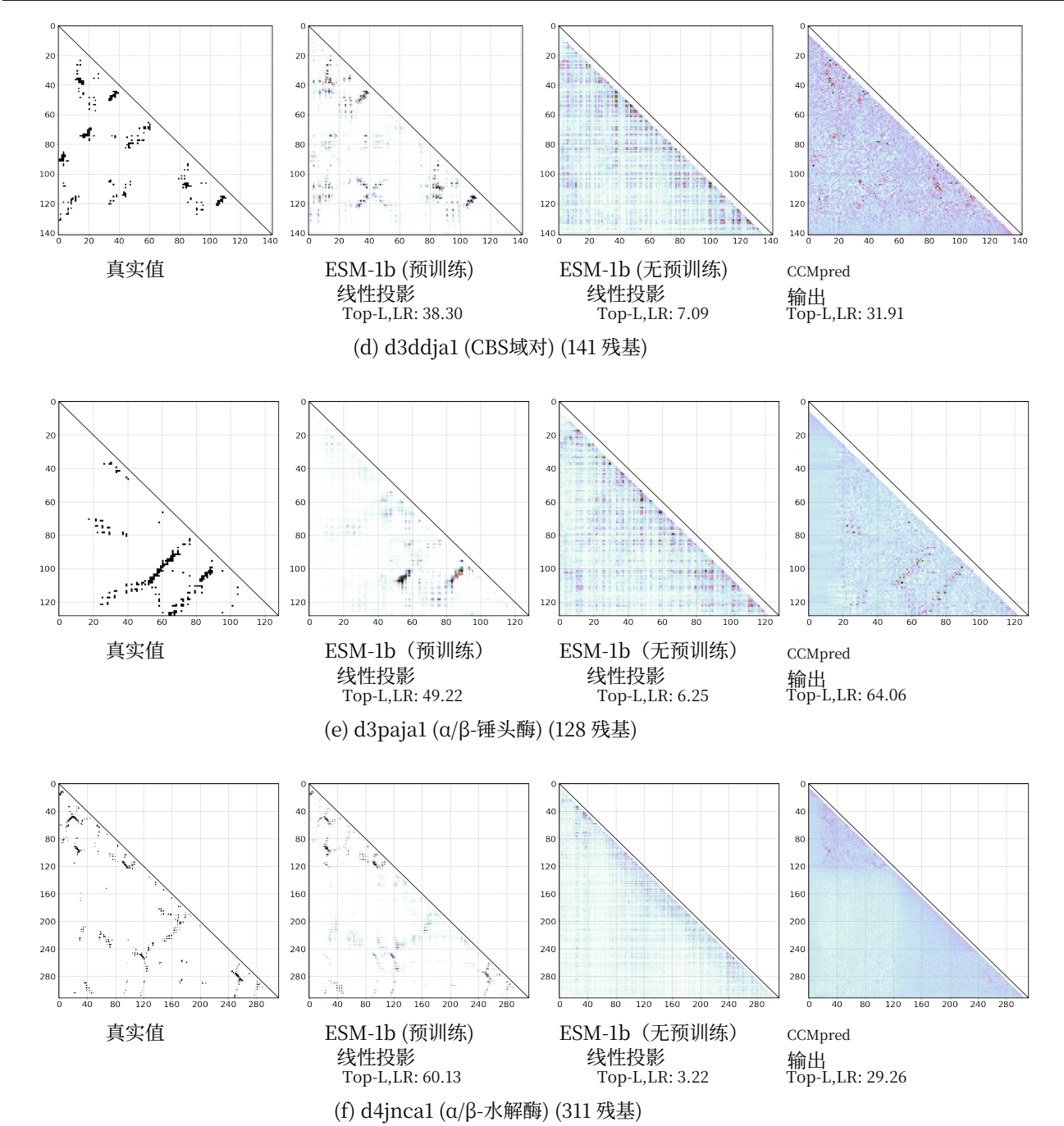


图 S7. (继续从上文)

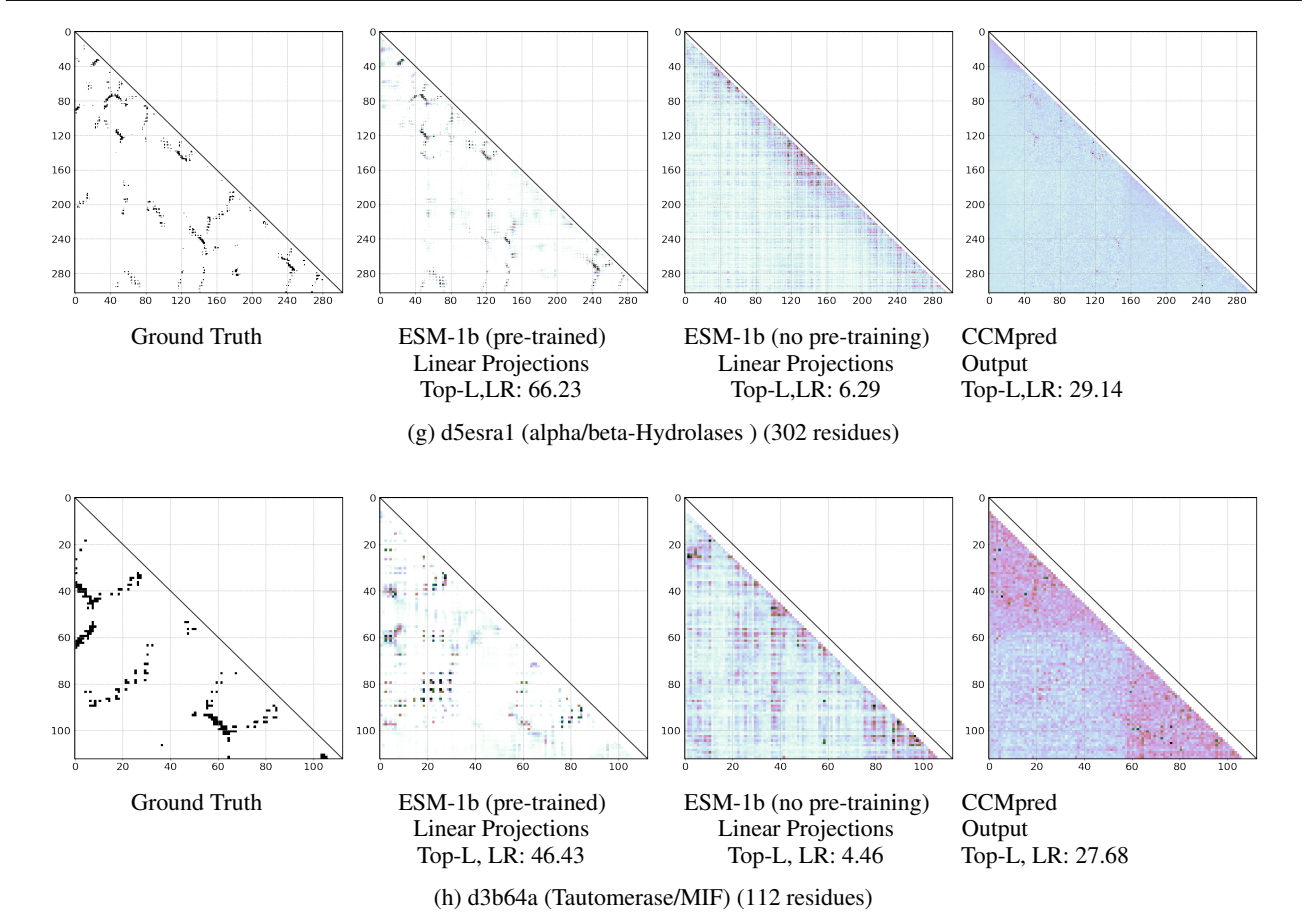


Figure S7. (continued from above)

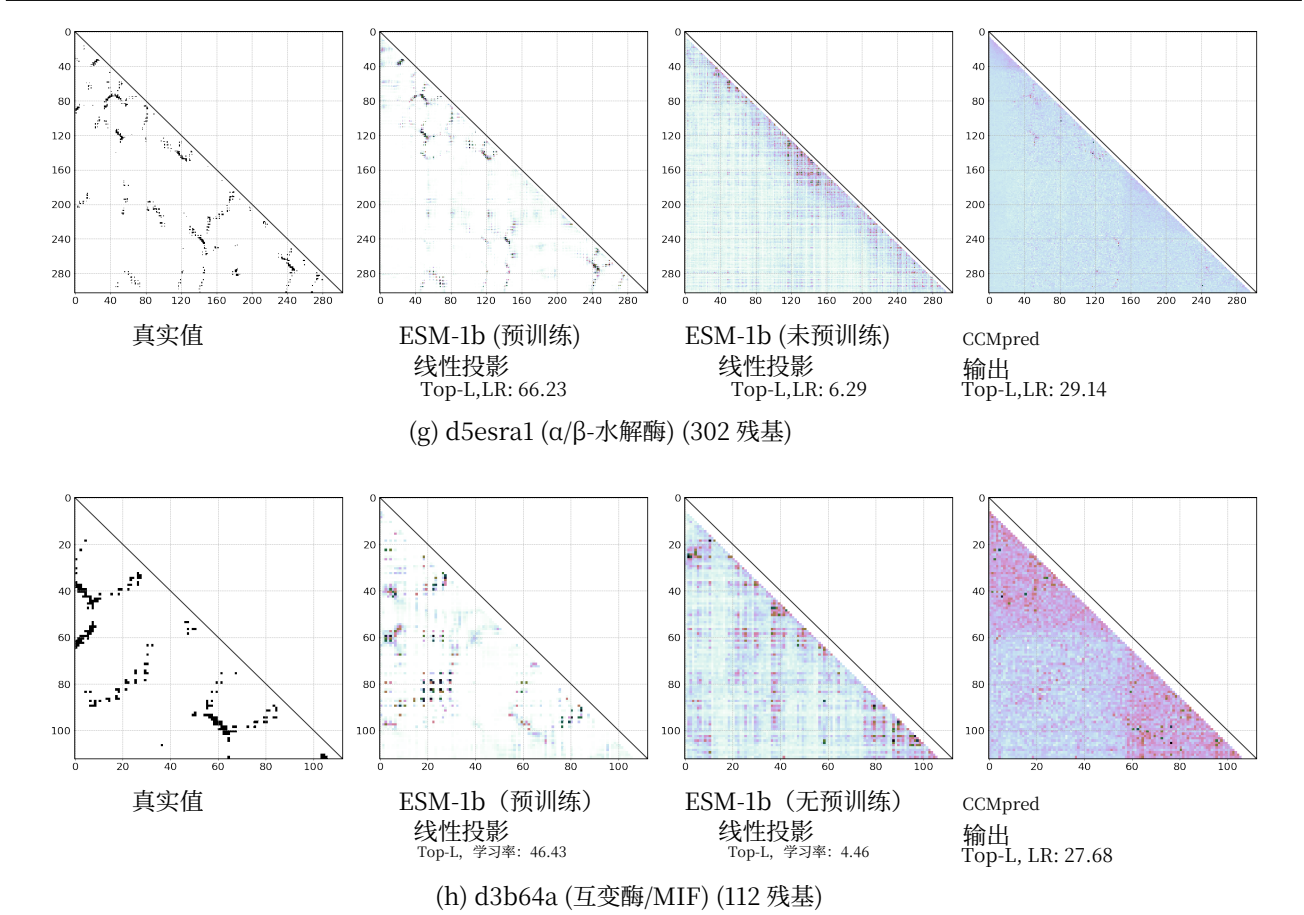


图 S7. (续上文)

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